

Custom Compiler™ analogLib Element Reference Manual

Version W-2024.09, September 2024



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About the analogLib Element Reference Manual

The *analogLib Element Reference Manual* describes the analogLib library, which contains elements used for building testbenches and providing necessary stimulus for devices under test.

This preface includes the following sections:

- [New in This Release](#)
- [Related Products, Publications, and Trademarks](#)
- [Conventions](#)
- [Customer Support](#)
- [Statement on Inclusivity and Diversity](#)

New in This Release

Information about new features, enhancements, and changes, known limitations, and resolved Synopsys Technical Action Requests (STARs) is available in the Custom Compiler Release Notes on the SolvNetPlus site.

Related Products, Publications, and Trademarks

For additional information about the Custom Compiler tool, see the documentation on the Synopsys SolvNetPlus support site at the following address:

<https://solvnetplus.synopsys.com>

You might also want to see the documentation for the following related Synopsys products:

- Synopsys® Custom Compiler™
- PrimeSim™ Continuum
- PrimeSim™
- PrimeSim™ Pro
- PrimeSim™ SPICE
- PrimeSim™ HSPICE®
- PrimeSim™ XA

- PrimeSim™ CCK
- PrimeSim™ MOSRA
- VCS® PrimeSim™ AMS
- Synopsys FineSim®
- Synopsys HSPICE®
- Synopsys VCS®
- Synopsys Verdi®
- Synopsys OptoCompiler™
- Synopsys StarRC™
- Mentor Graphics® Calibre®

Conventions

The following conventions are used in Synopsys documentation.

Convention	Description
Courier	Indicates syntax, such as <code>write_file</code> .
<i>Courier italic</i>	Indicates a user-defined value in syntax, such as <code>write_file design_list</code>
Courier bold	Indicates user input—text you type verbatim—in examples, such as <code>prompt> write_file top</code>
Purple	• Within an example, indicates information of special interest. • Within a command-syntax section, indicates a default, such as <code>include_enclosing = true false</code>
[]	Denotes optional arguments in syntax, such as <code>write_file [-format fmt]</code>
...	Indicates that arguments can be repeated as many times as needed, such as <code>pin1 pin2 ... pinN</code> .
	Indicates a choice among alternatives, such as <code>low medium high</code>
\	Indicates a continuation of a command line.

Convention	Description
/	Indicates levels of directory structure.
Bold	Indicates a graphical user interface (GUI) element that has an action associated with it.
Edit > Copy	Indicates a path to a menu command, such as opening the Edit menu and choosing Copy .
Ctrl+C	Indicates a keyboard combination, such as holding down the Ctrl key and pressing C.

Customer Support

Customer support is available through SolvNetPlus.

Accessing SolvNetPlus

The SolvNetPlus site includes a knowledge base of technical articles and answers to frequently asked questions about Synopsys tools. The SolvNetPlus site also gives you access to a wide range of Synopsys online services including software downloads, documentation, and technical support.

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Synopsys is committed to creating an inclusive environment where every employee, customer, and partner feels welcomed. We are reviewing and removing exclusionary language from our products and supporting customer-facing collateral. Our effort also includes internal initiatives to remove biased language from our engineering and working

environment, including terms that are embedded in our software and IPs. At the same time, we are working to ensure that our web content and software applications are usable to people of varying abilities. You may still find examples of non-inclusive language in our software or documentation as our IPs implement industry-standard specifications that are currently under review to remove exclusionary language.

1

analogLib Elements

Describes cells inside analogLib elements and provides examples of how to use those devices.

The analogLib library contains elements used for building testbenches and providing necessary stimulus for devices under test. These elements include resistors, capacitors, independent and behavioral voltage sources, and more.

Elements from analogLib are used in all testbenches. The simplest examples are power and ground supplies. A corresponding element from the analogLib is added so the simulator can understand which node is the ground node and which one is supply, as well as the supply value.

For information about how to copy symbols from the analogLib into your library or how to copy simulation views from the analogLib into your library, see the *Custom Compiler iPDK Developer Guide*.

You can find a sample model file for devices inside the analogLib library in `<product_installation_directory>/samples`.

The following sections describe the cells of the analogLib library. Each section includes an element description, a CDF parameter table, references to HSPICE documentation, and an example of the netlist with that device.

- [Passive Elements](#)
- [Active Elements](#)
- [Independent Sources](#)
- [Dependent Sources](#)
- [Global Nets](#)

Passive Elements

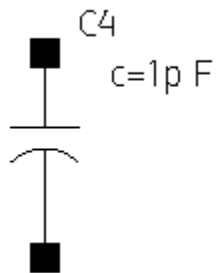
The following topics describe:

- [cap](#)
- [cap3](#)
- [cap4](#)
- [cccap](#)
- [core](#)
- [dcblock](#)
- [dcfeed](#)
- [delay](#)
- [ind](#)
- [Controlling Version Number for Legacy Designs](#)
- [iprobe, iprobe1](#)
- [ixfmr](#)
- [mind](#)
- [msline](#)
- [mtline](#)
- [phyres](#)
- [res](#)
- [res3](#)
- [res4](#)
- [switch](#)
- [tline](#)
- [u1wire - u5wire](#)
- [varactor](#)
- [varactor3](#)
- [vccap](#)

- [vcre](#)
- [winding](#)
- [xfmr](#)

cap

Two-terminal capacitor device.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Capacitance	c	1p	X	X	X
Model Name	model		X	X	-
Initial Condition	ic		X	X	-
Temperature Coefficient 1	tc1		X	X	-
Temperature Coefficient 2	tc2		X	X	-
Width	w		X	X	X
Length	l		X	X	X
Device Area	area		-	X	-
Multiplier	m		X	X	X
Temp Difference	dtemp		X	-	-
Scale Factor	scale		X	X	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Temperature rise from ambient	trise		-	X	-
Capacitor perimeter	perim		-	X	-
Number of Polynomial Coeffs	polyCoef	0	-	X	-
c1-20	c1-20		-	X	-

Example

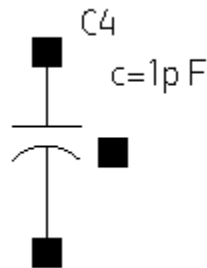
```
c4 plus minus c=1p
```

See Also

- [Passive Elements](#) in the *HSPICE® User Guide: Elements*

cap3

Three-terminal custom macro model capacitor device.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Capacitance	c	1p	X	-	X
Model Name	model	cap3	X	-	-
Initial Condition	ic		X	-	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Temperature Coefficient 1	tc1		X	-	-
Temperature Coefficient 2	tc2		X	-	-
Width	w		X	-	X
Length	l		X	-	X
Multiplier	m		X	-	X
Temp Difference	dtemp		X	-	-
Scale Factor	scale		X	-	-

Example

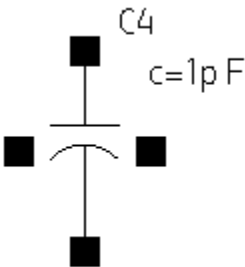
xc5 plus minus b cap3 c=1p

Model Statement

```
.subckt cap3 p m b c=1p
cmain p m c=c
.ends
```

cap4

Four-terminal custom macro model capacitor device.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Capacitance	c	1p	X	-	X
Model Name	model	cap4	X	-	X
Initial Condition	ic		X	-	X
Temperature Coefficeint 1	tc1		X	-	X
Temperature Coefficient 2	tc2		X	-	X
Width	w		X	-	X
Length	l		X	-	X
Multiplier	m		X	-	X
Temp Difference	dtemp		X	-	X
Scale Factor	scale		X	-	X

Example

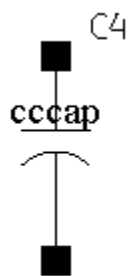
```
xc6 plus minus b t cap4 c=1p
```

Model Statement

```
.subckt cap4 p m b t c=1p
cmain p m c=c
.ends
```

cccap

Two-terminal charge-conserved capacitor. Capacitance is calculated based on provided charge equation.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Charge Equation	q		X	-	X

Example

```
c7 n1 n2 q='(1e-9*v(10))'
```

See Also

- [Charge-conserved Capacitors](#) in the *HSPICE® User Guide: Elements*

core

Magnetic core element.

CORE4



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Model Name	model	core	X	X	-
Magnetization	mag		X	-	-
Number of windings	numOfL	1	X	-	-
Winding 1	l1		X	-	-
Winding 2	l2		X	-	-
Winding 3	l3		X	-	-
Winding 4	l4		X	-	-
Winding 5	l5		X	-	-
Winding 6	l6		X	-	-
Winding 7	l7		X	-	-
Winding 8	l8		X	-	-
Winding 9	l9		X	-	-
Winding 10	l10		X	-	-
Winding 11	l11		X	-	-
Winding 12	l12		X	-	-
Winding 13	l13		X	-	-
Winding 14	l14		X	-	-
Winding 15	l15		X	-	-
Winding 16	l16		X	-	-
Winding 17	l17		X	-	-
Winding 18	l18		X	-	-
Winding 19	l19		X	-	-
Winding 20	l20		X	-	-
Multiplier	m		-	X	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Device area	area		-	X	-
Physical length	len		-	X	-
Inner diameter	idiam		-	X	-
Outer diameter	od		-	X	-
Gap length	gap		-	X	-

Example

```
11 2 0 nt=300
12 3 0 nt=90
13 0 4 nt=90
kcore1 11 12 13 core mag=0
```

Model Statement

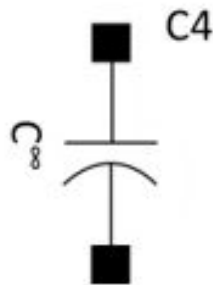
```
.model core l(ac=1e4 lc=100 hc=.1 tc=1u br=6.4k
+ bs=6.75k hs=.6 hcr=0)
```

See Also

- [Inductor Device Model and Equations](#) in the *HSPICE® Reference Manual: Device Models*. You can also check the HSPICE installation directory for a demo netlist:
<HSPICE_INSTALL_DIR>/hspice/demo/hspice/mag/magcore.sp

dcblock

Two-terminal capacitor device, the capacitance is infinite.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Initial Condition	ic		X	X	-
Model name	model		-	X	-
Capacitance used in tran	c	1u	-	X	-

Example

c4 plus minus infinity

dcfeed

Two-terminal inductor device, the inductance is infinite.



CDF Parameters

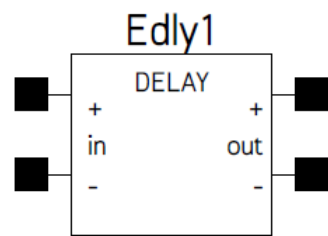
CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Initial Condition	ic		X	X	-
Model name	model		-	X	-
Inductance used in tran	l	1u	-	X	-

Example

l4 plus minus infinity

delay

This element adjusts propagation delay of E-element which is used as a behavioral voltage source where the voltage source can be bias and/or frequency dependent.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Delay Time	td		X	X	-
Gain	gain	1.0	X	X	-
Number of data points	npdelay		X	-	-
Multiplier	m		-	X	-

Example

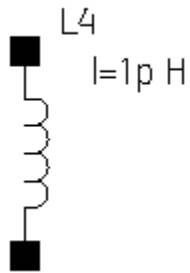
```
edly1 p n vcvs delay ps ns td=1n scale=1.0 npdelay=2
```

See Also

- [Voltage-Dependent Voltage Sources – E-Elements](#) in the *HSPICE® User Guide: Elements*

ind

Inductor element.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Inductance	l	1n	X	X	X
Model Name	model		X	X	-
Initial Condition	ic		X	X	-
Temperature Coefficient 1	tc1		X	-	-
Temperature Coefficient 2	tc2		X	-	-
Resistance	r		X	X	-
Multiplier	m		X	X	X
Temp Difference	dtemp		-	X	-
Scale Factor	scale		X	-	-
Maximum number of coefficients	polyCoef	0	-	X	-
Poly coefficient	c1 – c20	0	-	X	-
Generate noise?	isnoisy		-	X	-
Temperature rise from ambient	trise		-	X	-

Example

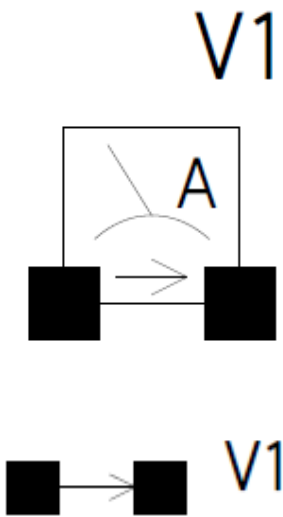
```
l12 plus minus l=1n
```

Controlling Version Number for Legacy Designs

To control the version number for legacy designs, create and use the `dbAnalogLibIndVersion` Tcl preference before the analogLib library is loaded.

iprobe, iprobe1

Current Probe. In the HSPICE netlist, this creates a voltage source with 0 DC voltage. It computes the current flowing from the PLUS to the MINUS terminal.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Source Type	srcType	dc	X	-	-
Zero Voltage	vdummy	0	-	-	-
Ignore	lvIgnore	1	-	-	-
Remove lvs Device	lvsRemoveDevice	(short(P LUS MINUS))	-	-	-
Remove lx Device	lxRemoveDevice	(short(P LUS MINUS))	-	-	-

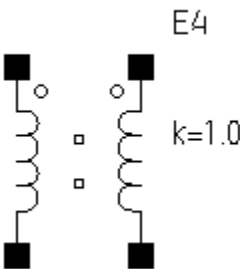
CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Action	nlAction	stop	-	-	-

Example

```
v1 p n dc=0
```

ixfmr

Ideal transformer element.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Coupling Coefficient	k	1	X	-	-

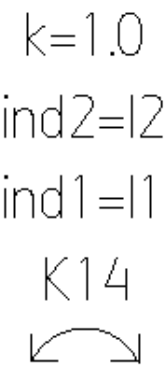
Example

```
e13 n+ n- transformer in+ in- 0.85
```

See Also

- [Ideal Transformer](#) in the *HSPICE® User Guide: Elements*

mind



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
First Coupled Inductor	ind1		X	X	-
Second Coupled Inductor	ind2		X	X	-
Coupling Coefficient	k	1	X	X	-

Example

```
k14 l1 l2 k=1.0
```

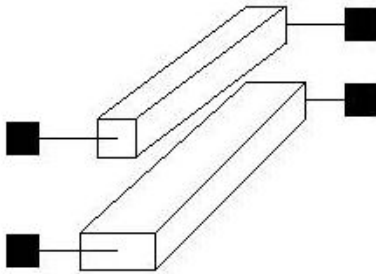
See Also

- [Inductor Device Model and Equations](#) in the *HSPICE® Reference Manual: Device Models*

msline

Microstrip line model.

The following figure illustrates the planar transmission line at the IC-circuits:



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Length	l	-	-	X	-
Width	w	-	-	X	-
Substrate height	h	-	-	X	-
Conductor thickness	t	-	-	X	-
Relative permittivity	eps	-	-	X	-
Max signal frequency	fmax	-	-	X	-
Multiplicity	m	-	-	X	-

mtline

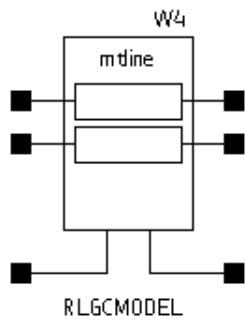
This device models distributed transmission lines using HSPICE W-element.

The W-element supports five formats to specify the transmission line properties:

- Model 1: RLGC-Model specification
 - Internally specified in a .model statement
 - Externally specified in a different file
- Model 2: U-Model specification
 - RLGC input for up to five coupled conductors
 - Geometric input (planer, coax, twin-lead)

- Measured-parameter input
- Skin effect
- Model 3: Built-in field solver model
 - Standard format (using geometric data with the W-element)
 - Tabular format
- Model 4: Frequency-dependent tabular model
- Model 5: S-parameter Model

The following figure illustrates the mtline element with two conductors:



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Number of Conductors	n		X	X	-
Model Type	mtype	RLGCMODEL	X	X	-
Model Name	model		-	X	-
FS Model	fsmodel		X	-	-
RLGC Model	rlgcmode		X	-	-
RLGC file	rlgcfile		X	-	-
RLGC data file	file		-	X	-
U Model	umodel		X	-	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
TABLE Model	tablemodel		X	-	-
S Model	smodel		X	-	-
Layer Stack	layerstack		X	-	-
FS Options	fsoptions		X	-	-
Coordinate Type	coord		X	-	-
Output Format	outputformat	RLGCF ILE	X	-	-
Conductor 1 - 256	conductor1		X	-	-
Imaginary Skin Effect?	includersimag		X	-	-
Complex Dielectric Loss?	includegdimag		X	-	-
Cut-off Frequency	fgd		X	-	-
Delay Optimization	delayopt		X	-	-
DC Accuracy	dcacc		X	-	-
Generate Noise?	noise		X	-	-
Temperature Difference	dtemp		X	-	-
LMG subcircuit file	subcktfile		-	X	-
Maximum signal frequency	fmax		-	X	-
Multiplicity factor	mf	1	-	X	-
Carrier frequency	fc		-	X	-
Specify mtline choice?	mtlineChoice	0	-	X	-
S-parameter File	file1		-	X	-
R matrix per unit length	R		-	X	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
L matrix per unit length	L		-	X	-
G matrix per unit length	G		-	X	-
C matrix per unit length	C		-	X	-
Skin effect res matrix per unit length	rskin		-	X	-
Dielectric loss cond matrix per unit length	gdloss		-	X	-
Frequency scale factor	freqscale		-	X	-
Transmission line type	linetype	microstrip	-	X	-
Number of dielectric layers	numlayer	1	-	X	-
Number of Ground Planes	numgnd	1	-	X	-
Rel dielectric const of layers(er)	er		-	X	-
Dielectric layer thickness (d)	layerthickness		-	X	-
Dielectric loss type	dlosstype		-	X	-
Dielectric layer loss	dloss		-	X	-
Signal line width	linewidth		-	X	-
Signal line thickness	linethickness		-	X	-
Signal line height (h)	lineheight		-	X	-
Signal line spacing	linespace		-	X	-
Signal line conductivity	linesigma		-	X	-
Ground plane thickness	gndthickness		-	X	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Ground plane conductivity	gndsigma		-	X	-
Characteristic impedance	z0		-	X	-
Delay time	tdmt		-	X	-
Frequency	fmt		-	X	-
Normalized length	nlmt		-	X	-
Propagation velocity normalized to c	velmt		-	X	-
Corner frequency	corner		-	X	-
DC series rest/Length	dcr		-	X	-
Loss resistance per unit length	seriesr		-	X	-
Conductor loss at fc	alphac		-	X	-
Conductor loss quality factor	qc		-	X	-
Dielectric loss frequency	fd		-	X	-
Loss conductance per unit length	shuntg		-	X	-
Dielectric loss	alphad		-	X	-
Dielectric loss quality factor	gd		-	X	-
Use LMG subckt	uselmg	0	-	X	-
firelmg	firelmg	false	-	X	-
Enter RLGC etc. matrices	entermatrices	0	-	X	-
Type of Input	modelType		-	X	-
Display Cross-section	firedisplay	nil	-	X	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Conductor loss frequency	fcmt		-	X	-
Physical length	len		X	X	-

Example

The W1 lossy transmission line connects the in node to the out node:

```
W1 in gnd out gnd RLGCfile=cable.rlgc N=1 L=5
```

See Also

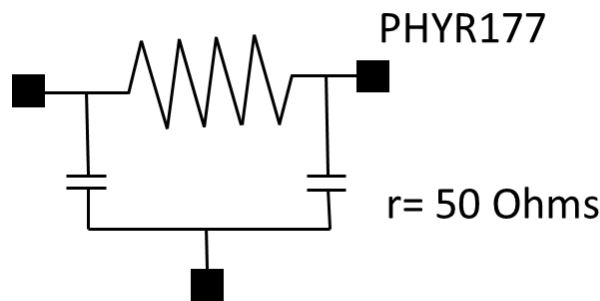
- [W-element](#) in the *HSPICE® User Guide: Elements*

Controlling Version Number for Legacy Designs

To control the version number for legacy designs, create and use the [dbAnalogLibMTlineVersion](#) Tcl preference before the analogLib library is loaded.

phyres

Physical resistor.



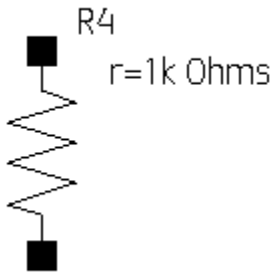
CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Model Name	model	phyres	-	X	-
Temp coefficient 1	tc1		-	X	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Temp coefficient 2	Tc2		-	X	-
Temperature rise from ambient	trise		-	X	-
Multiplier	m		-	X	-
Resistance	r	50	-	X	-
Capacitance	c		-	X	-
Length	l		-	X	-
Width	w		-	X	-
Linear temp coefficient of cap	tc1c		-	X	-
Quadratic temp coefficient of cap	tc2c		-	X	-
Estimated operating region	region		-	X	-

res

Two-terminal resistor element.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Resistance	r	1k	X	X	X

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Temperature Coefficient 1	tc1		X	X	-
Temperature Coefficient 2	tc2		X	X	-
Multiplier	m		X	X	-
Model Name	model		X	X	-
AC Resistance	ac		X	X	-
Temp Difference	dtemp		X	X	-
Width	w		X	X	X
Length	l		X	X	X
Scale Factor	scale		X	X	-
Capacitance	c		X	X	-
Generate Noise?	isnoisy		X	X	-
Skin Effect Coefficient	rs		X	-	-
Temperature rise from ambient	trise		-	X	-
Resistance form	resform		-	X	-
Alias for linear temperature coefficient	tc1r		-	X	-
Alias for quadratic temperature coefficient	tc2r		-	X	-
Linear temperature coefficient of linear cap	tc1c		-	X	-
Quadratic temperature coefficient of linear cap	tc2c		-	X	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Resistance scaling factor	scaler		-	X	-
Capacitance scaling factor	scalec		-	X	-

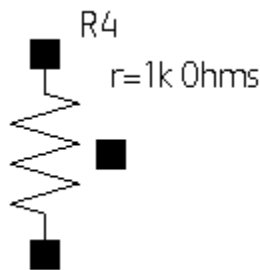
Example

r18 plus minus r=1k

See Also

- [Resistors](#) in the *HSPICE® User Guide: Elements*

res3



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Resistance	r	1k	X	-	X
Temperature Coefficient 1	tc1		X	-	-
Temperature Coefficient 2	tc2		X	-	-
Multiplier	m		X	-	-
Model Name	model	res3	X	-	-
AC Resistance	ac		X	-	-

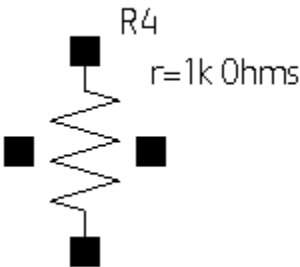
CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Temp Difference	dtemp		X	-	-
Width	w		X	-	X
Length	l		X	-	X
Scale Factor	scale		X	-	-
Capacitance	c		X	-	-
Generate Noise?	isnoisy		X	-	-
Skin Effect Coefficient	rs		X	-	-

Example

```
xr19 plus minus b res3 r=1k
```

res4

Custom four-terminal resistor macro model.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Resistance	r	1k	X	-	X
Temperature Coefficeint 1	tc1		X	-	-

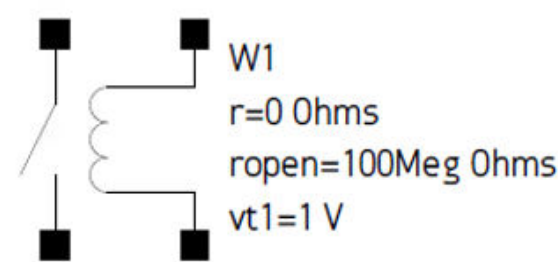
CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Temperature Coefficient 2	tc2		X	-	-
Multiplier	m		X	-	-
Model Name	model	res3	X	-	-
AC Resistance	ac		X	-	-
Temp Difference	dtemp		X	-	-
Width	w		X	-	X
Length	l		X	-	X
Scale Factor	scale		X	-	-
Capacitance	c		X	-	-
Generate Noise?	isnoisy		X	-	-
Skin Effect Coefficient	rs		X	-	-

Example

```
xr24 plus minus b t res4 r=1k
```

switch

Voltage-controlled resistor.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Type	swType	linear	-	-	-
Closed Resistance	r	0	x	-	-
Open Resistance	ropen	100Meg	X	-	-
Open Voltage	vt1		X	X	-
Closed Voltage	vt2		X	X	-
Open switch resistance	ro	1T	-	X	-
Close switch resistance	rc	1	-	X	-
Multiplier	m		-	X	-
Estimated operating region	region		-	X	-
Version	version	3	-	-	-

Note:

- For version 3, the switch element has empty values for the default of vt1 and vt2 parameters. For legacy designs with version 1 and 2, the switch element has old default values of vt1=0.1 and vt2=0.

Example

```
gw0 n+ n- vcr relay nc+ nc- vt1 vt2 ropen r
```

See Also

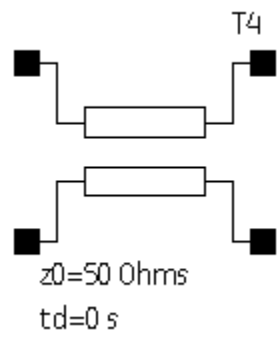
- [Voltage-Controlled Resistor](#) in the *HSPICE® User Guide: Elements*

Controlling Version Number for Legacy Designs

To control the version number for legacy designs, create and use the [dbAnalogLibSwitchVersion](#) Tcl preference before the analogLib library is loaded.

tline

Ideal transmission line model.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Model Name	model		X	X	-
Characteristic Impedance	z0	50	X	-	-
Delay Time	td		X	X	-
Length	l		X	-	-
Frequency	freq		X	X	-
Normalized Length	nl	0.25	X	X	-
Voltage 1	v1	0	X	-	-
Current 1	i1	0	X	-	-
Voltage 2	v2	0	X	-	-
Current 2	i2	0	X	-	-
Multiplier	m		-	X	-
Propagation velocity normalized to c	vel		-	X	-
Resistance	r	1K	-	X	-
Loss resistance per unit length	rs	1K	-	X	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Loss conductance per unit length	g		-	X	-
Physical Length	len		-	X	-
Characteristic impedance	zo	50	-	X	-

Example

```
t25 in+ in- out+ out- z0=50 td=0 l=5u f=5G ic=v1,i1,v2,i2
```

See Also

- [T-element](#) in the *HSPICE® User Guide: Elements*

u1wire - u5wire

Lumped Transmission Lines model.

This can be used for modeling lossy non-ideal transmission lines in terms of lumped-element equivalent circuits.

The U-elements can model effects such as the following:

- Time delay
- Phase shift
- Coupling and Crosstalk
- Distortion

There are several cells which differ by input node counts from one to five.

CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Model Name	model		X	-	-
Length	l		X	-	-
Element Lumps	lumps		X	-	-

Example

```
u1 In10 In20 In30 Vss Out1 Out2 Out3 Vss IcWire L=200um
.model lcWire U LEVEL=3 Dlev=0 NI=3 Nlay=2 Plev=1 Elev=1
+ Llev=0 Ht=2u Wd=5u Sp=15u Th=1u Rho=2.8e-8 Kd=3.9
```

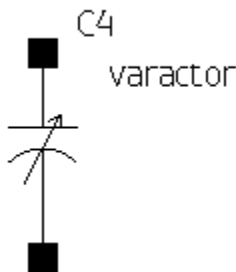
See Also

- [U-element Examples, Models, and Applications](#) in the *HSPICE® User Guide: Elements*

varactor

This macro-model device is intended to model voltage dependent capacitor, and has parameters for model name, width, length and multiplier.

In order to simulate this device, you need to provide model definition using .subckt.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Model Name	model	varactor	X	-	X
Width	w		X	-	X
Length	l		X	-	X
Multiplier	m		X	-	X

Example

```
xc27 n1 n2 varactor w=10u l=3u m=4
```

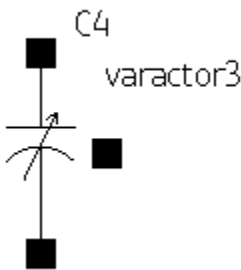
Model Statement

```
.subckt varactor ng nds w=1u l=1u m=1
.param area='l*w'
.param Cmain='20E-11*2*w+1.15E-10*2*l+1.6E-3*area'
.param dC='-1.0E-11*2*w+4E-12*2*l+5.87E-03*area'
.param dVgs='-0.075-7E-05*(temper-27)'
.param Vgn='0.43+4E-04*(temper-27)'
cg ng nds 'm*(Cmain*(1.0+tanh((v(ng,nds)-dVgs)/Vgn)))'
.ends varactor
```

varactor3

This macro-model device is intended to model voltage dependent capacitor, and has parameters for model name, width, length and multiplier.

In order to simulate this device, you need to provide model definition using .subckt.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Model Name	model	varactor	X	-	X
Width	w		X	-	X
Length	l		X	-	X
Multiplier	m		X	-	X

Example

```
xc27 n1 n2 sub varactor3 w=10u l=3u m=4
```

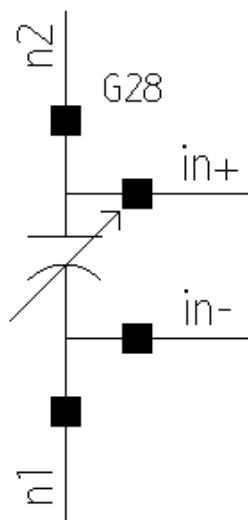
Model Statement

```
.subckt varactor3 ng nds nsub w=1u l=1u m=1
.param area='l*w'
.param Cmain='20E-11*2*w+1.15E-10*2*l+1.6E-3*area'
.param dC='-1.0E-11*2*w+4E-12*2*l+5.87E-03*area'
.param dVgs='-0.075-7E-05*(temper-27)'
.param Vgn='0.43+4E-04*(temper-27)'
cg ng nds 'm*(Cmain*(1.0+tanh((v(ng,nds)-v(nsub,0)-dVgs)/Vgn)))'
.ends varactor3
```

vccap

Voltage-controlled capacitor.

The capacitance value between nodes n1 and n2 depends on the voltage specified within nodes in+ and in-.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Delta	delta		X	X	-
Controlling Pairs	xypairs	2	X	X	-
Control Volt 1–50	x1-x50		X	X	-
Corresp Element1–50	y1-y50		X	X	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Scale Factor	scale		X	X	-
Temperature Coefficient 1	tc1		X	X	-
Temperature Coefficient 2	tc2		X	X	-
Initial Condition	hic		X	-	-
Multiplier	hm	1	X	X	-
Pwl Type	pwlType	pwl	X	-	-
Smoothing Method	smooth	2	X	-	-
Capacitance Calculation Mode	ctype	0	X	-	-
Type	csType	linear	X	X	-

Example

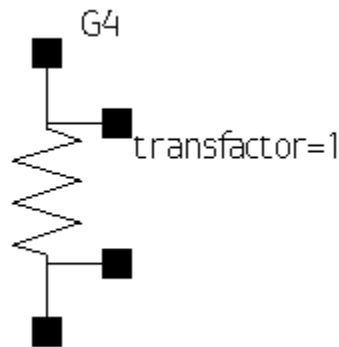
```
g28 n2 n1 vccap pwl(1) in+ in- 0.05, 20f 0.1, 35f 0.35, 85f smooth=2
ctype=0
```

See Also

- [Voltage-Controlled Capacitor](#) in the *HSPICE® User Guide: Elements*

vcres

Voltage-controlled resistor.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Volt/res Conversion factor	transfactor		X	X	-
Type	csType	linear	X	X	-
Controlling Pairs	xypairs	2	X	X	-
Delta	delta		X	-	-
Control Current 1–50	x1-50		X	X	-
Corresp Element 1-50	y1-50		X	X	
Maximum Resistance	maxr		X	-	-
Minimum Resistance	minr		X	-	-
Scale Factor	scale		X	-	-
Temperature Coefficient 1	tc1		X	-	-
Temperature Coefficient 2	tc2		X	-	-
Initial Condition	hic		X	-	-
Multiplier	hm		X	-	-
Pwl Type	pwlType	pwl	X	-	-
Smoothing Method	smooth		X	-	-
Resistance 1–50	y1-y50		X	-	-

Example

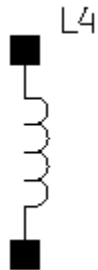
```
g28 n2 n1 vcr in+ in- 1 max=100k min=5k
g28 n2 n1 vcr pwl(1) in+ in- 0.05, 5k 0.1, 7k 0.35, 10k
```

See Also

- [Voltage-Controlled Resistor](#) in the *HSPICE® User Guide: Elements*

winding

Inductor with magnetic winding.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Num of Turns	turn		X	X	-
Model Name	model		X	-	-
Initial Condition	ic		X	X	-
Temperature Coefficient 1	tc1		X	-	-
Temperature Coefficient 2	tc2		X	-	-
Resistance	r		X	X	-
Multiplier	m		X	X	-
Temp Difference	dtemp		X	-	-
Scale Factor	scale		X	-	-
Name of core	core		-	X	-
Res of the winding	resis			X	

Example

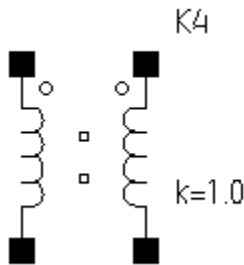
```
l4 n2 n1 nt=100 r=10
```


See Also

- [Inductors](#) in the *HSPICE® User Guide: Elements*

xfmr

Mutual inductors with coupling coefficient k.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Primary Inductance	pi		X	-	-
Secondary Inductance	si		X	-	-
Coupling Coefficient	k	1	X	-	-
Multiplier	m		-	X	-
Number of turns on primary	n1		-	X	-
Number of turns on secondary	n2		-	X	-

Example

```
xk31 p+ p- s+ s- top_xk31
.subckt top_xk31 p1 p2 s1 s2
  l1 p1 p2 l=0.5n
  l2 s1 s2 l=0.75n
  k1 l1 l2 k=0.85
.ends top_xk31
```

See Also

- [Mutual Inductors](#) in the *HSPICE® User Guide: Elements*

Active Elements

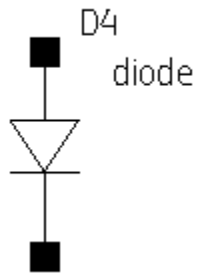
The following topics describe:

- [diode](#)
- [diode3](#)
- [diode4](#)
- [ibis_buffer](#)
- [mport](#)
- [nbsim](#)
- [pbsim](#)
- [nbsim4](#)
- [pbsim4](#)
- [njfet](#)
- [pjfet](#)
- [nmes](#)
- [pmes](#)
- [nmes4](#)
- [pmes4](#)
- [nmos](#)
- [pmos](#)
- [nmos4](#)
- [pmos4](#)
- [nmos4hv](#)
- [pmos4hv](#)
- [nmoshv](#)

- [pmoshv](#)
- [npn](#)
- [pnp](#)
- [npn4](#)
- [nnp4a](#)
- [pnp4](#)
- [pnp4a](#)
- [nsoi](#)
- [psoi](#)
- [nsoi4](#)
- [psoi4](#)
- [nsoifd](#)
- [psoidf](#)
- [nsoifd4](#)
- [psoidf4](#)
- [schottky](#)
- [sp1tswitch](#)
- [sp2tswitch](#)
- [sp3tswitch](#)
- [sp4tswitch](#)
- [u1wire](#), [u2wire](#), [u3wire](#), [u4wire](#), [u5wire](#)

diode

Two-terminal p-n diode.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Model Name	model	diode	X	X	X
Device Area	area		X	X	X
Initial Diode Voltage	Vd		X	-	-
Periphery of Junction	pj		X	X	X
Width of Polycap	wp		X	X	-
Length of Polycap	lp		X	X	-
Width of Metalcap	wm		X	X	-
Length of Metalcap	lm		X	X	-
Width	w		X	X	-
Length	l		X	X	-
Initially Off	off	0	-	X	-
Multiplier	m		X	X	X
Device Temperature	dtemp		X	X	-
Amplitude scale factor	scale		-	X	-

Example

```
d32 n2 n1 diode
```

Model Statement

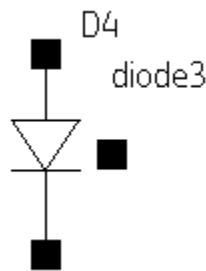
```
.model core 1(ac=1e4 lc=100 hc=.1 tc=1u br=6.4k  
+ bs=6.75k hs=.6 hcr=0)
```

See Also

- [Diode Element](#) in the *HSPICE® User Guide: Elements*

diode3

Three-terminal diode.



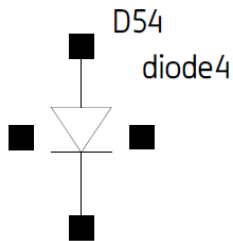
CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Model Name	model	diode3	X	-	X
Device Area	area		X	-	X
Initial Diode Voltage	Vd		X	-	-
Periphery of Junction	pj		X	-	X
Width of Polycap	wp		X	-	-
Length of Polycap	lp		X	-	-
Width of Metalcap	wm		X	-	-
Length of Metalcap	lm		X	-	-
Width	w		X	-	-
Length	l		X	-	-
Initially Off	off	0	-	-	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Multiplier	m		X	-	X
Device Temperature	dtemp		X	-	-

diode4

Four-terminal diode.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Model Name	Model	diode4	-	-	-
Device Area	area		X	-	X
Initial Diode Voltage	Vd		X	-	X
Periphery of Junction	pj		X	-	X
Width of Polycap	wp		X	-	X
Length of Polycap	lp		X	-	X
Width of Metalcap	wm		X	-	X
Length of Metalcap	lm		X	-	X
Width	w		X	-	X
Length	l		X	-	X
Initially Off	off	0	-	-	-
Multiplier	m		X	-	X
Device Temperature	dtemp		X	-	X

Example

```
d1 plus minus b t diode4
```

See Also

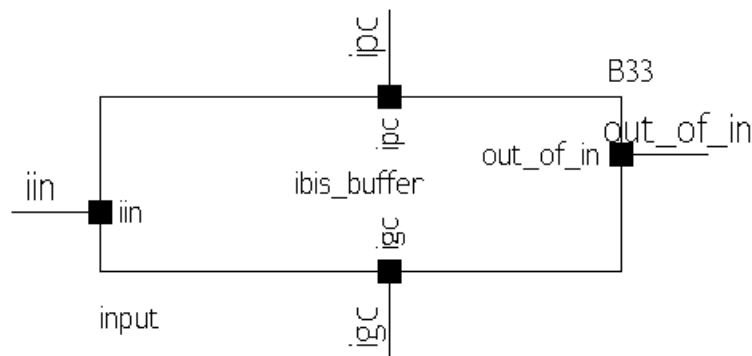
- [Diode Models](#) in the *HSPICE® Reference Manual: Device Models*

ibis_buffer

A behavioral method of modeling input/output buffers based on V/I curve data derived from measurement or full circuit simulation.

There are different IBIS buffer types, including:

- Input buffer
- Output buffer
- Input ECL buffer
- Output ECL buffer
- Tri-state buffer
- Input/Output buffer



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Buffer Type	buffer	input	X	X	X
Power Ground Clamp?	pgclamp	0	X	X	X

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
IBIS File	file		X	-	-
Model Name	model		X	X	X
Type	typ		X	X	-
HSPICE Version	hsp_ver		X	X	-
Power	power		X	X	-
Interpolation	interpol		X	-	-
XV_PU State Node	xv_pu		X	-	-
XV_PD State Node	xv_pd		X	-	-
Ramp Falling Waveform	ramp_fwf		X	-	-
Ramp Rising Waveform	ramp_rwf		X	-	-
Falling Waveform Tune	fwf_tune		X	-	-
Rising Waveform Tune	rwf_tune		X	-	-
Falling Waveform Pullup Delay	fwf_pu_dly		X	-	-
Rising Waveform Pulldown Delay	rwf_pd_dly		X	-	-
Pulldown Scale Factor	pd_scal		X	-	-
Pullup Scale Factor	pu_scal		X	-	-
Power Clamp Scale Factor	pc_scal		X	-	-
Ground Clamp Scale Factor	gc_scal		X	-	-
Falling Waveform Scale Factor	fwf_scal		X	-	-
Rising Waveform Scale Factor	rwf_scal		X	-	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Series Switch State	ss_state		X	-	-
Remove Initial Delay Falling	rm_dly_fwf		X	-	-
Remove Initial Delay Rising	rm_dly_rwf		X	-	-
Remove Flat Tails Falling	rm_tail_fwf		X	-	-
Remove Flat Tails Rising	rm_tail_rwf		X	-	-
Supress Warning Messages	nowarn	9	X	-	-
C_comp Pullup	c_com_pu		X	-	-
C_comp Pulldown	c_com_pd		X	-	-
C_comp Power Clamp	c_com_pc		X	-	-
C_comp Ground Clamp	c_com_gc		X	-	-
Detect OTI MID	detect_oti_mid		X	-	-
Time Control	time_control		X	-	-

Example

```
b33 ipc igc iin out_of_in buffer=input
```

See Also

- [Simulating Circuits with IBIS Models in HSPICE](#) in the *HSPICE® User Guide: Signal Integrity Modeling and Analysis*
- [Using IBIS Files with HSPICE](#) in the *HSPICE® User Guide: Signal Integrity Modeling and Analysis*
- [Modeling Input/Output Buffers Using IBIS Files](#) in the *HSPICE® User Guide: Elements*

mport

To measure mixed mode S-parameters you can use a port element with three terminals. Except for the number of external terminals, the syntax of the port element remains the same. The LIN analysis function internally sets the necessary drive mode (common/differential) of these mixed mode port elements. This symbol specifies mixed mode Port element.

Example

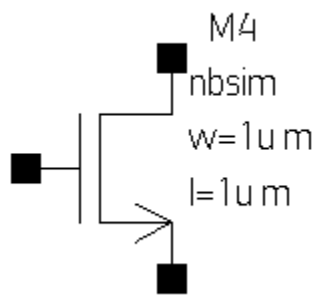
```
p1 plus minus ref dc=0 z0=50
```

See Also

- [Using the Port Element for Mixed-Mode Measurement](#) in the *HSPICE® User Guide: Elements*

nbsim

Three-terminal NMOS device with model name *nbsim*. By default, the bulk terminal of this device is connected to gnd!.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Model Name	model	nbsim	X	X	X
Width	w	1u	X	X	X
Length	l	1u	X	X	X
Drain Diffusion Area	ad		X	X	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Source Diffusion Area	as		X	X	-
Drain Junction Perimeter	pd		X	X	-
Source Junction Perimeter	ps		X	X	-
Drain Diff Resistor Sq	nrd		X	X	-
Source Diff Resistor Sq	nrs		X	X	-
Additional Drain Resistance	rdc		X	X	-
Additional Source Resistance	rsc		X	X	-
Initially Off	off	0	-	-	-
Drain-source Initial Voltage	Vds		X	-	-
Gate-source Initial Voltage	Vgs		X	-	-
Bulk-source Initial Voltage	Vbs		X	-	-
Temp Difference	dtemp		X	-	-
Source/Drain Selector	geo		X	X	-
Threshold Voltage Shift	delvto		X	-	-
Multiplier	m		X	X	X
NQS flag	nqsmod		-	X	-
Drain diffusion length	ld		-	X	-
Source diffusion length	ls		-	X	-
Dist. OD and poly (one side)	sa		-	X	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Dist. OD and poly (other side)	sb		-	X	-
Distance between fingers	sd		-	X	-
Expected operating region	region		-	X	-
Temperature rise from ambient	trise		-	X	-

Example

```
m34 d g s gnd! nbsim w=1u l=1u
```

Model Statement

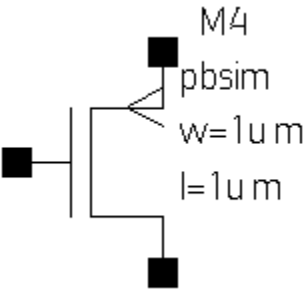
```
.model nbsim nmos level=54
```

See Also

- [MOSFETs](#) in the *HSPICE® User Guide: Elements*

pbsim

Three-terminal PMOS device with model name *pbsim*. By default, the bulk terminal of this device is connected to vdd!.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Model Name	model	nbsim	X	X	X
Width	w	1u	X	X	X
Length	l	1u	X	X	X
Drain Diffusion Area	ad		X	X	-
Source Diffusion Area	as		X	X	-
Drain Junction Perimeter	pd		X	X	-
Source Junction Perimeter	ps		X	X	-
Drain Diff Resistor Sq	nrd		X	X	-
Source Diff Resistor Sq	nrs		X	X	-
Additional Drain Resistance	rdc		X	X	-
Additional Source Resistance	rsc		X	X	-
Initially Off	off	0	-	-	-
Drain-source Initial Voltage	Vds		X	-	-
Gate-source Initial Voltage	Vgs		X	-	-
Bulk-source Initial Voltage	Vbs		X	-	-
Temp Difference	dtemp		X	-	-
Source/Drain Selector	geo		X	X	-
Threshold Voltage Shift	delvto		X	-	-
Multiplier	m		X	X	X

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
NQS flag	nqsmod		-	X	-
Drain diffusion length	ld		-	X	-
Source diffusion length	ls		-	X	-
Dist. OD and poly (one side)	sa		-	X	-
Dist. OD and poly (other side)	sb		-	X	-
Distance between fingers	sd		-	X	-
Expected operating region	region		-	X	-
Temperature rise from ambient	trise		-	X	-

Example

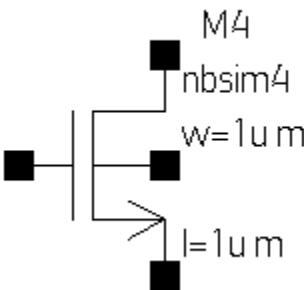
```
m1 s g d vdd! pbsim w=1u l=1u
```

Model Statement

```
.model pbsim pmos level=54
```

nbsim4

Four-terminal MOS device with model name *nbsim4*.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Model Name	model	nbsim4	X	X	X
Width	w	1u	X	X	X
Length	l	1u	X	X	X
Drain Diffusion Area	ad		X	X	-
Source Diffusion Area	as		X	X	-
Drain Junction Perimeter	pd		X	X	-
Source Junction Perimeter	ps		X	X	-
Drain Diff Resistor Sq	nrd		X	X	-
Source Diff Resistor Sq	nrs		X	X	-
Additional Drain Resistance	rdc		X	X	-
Additional Source Resistance	rsc		X	X	-
Initially Off	off	0	-	-	-
Drain-source Initial Voltage	Vds		X	-	-
Gate-source Initial Voltage	Vgs		X	-	-
Bulk-source Initial Voltage	Vbs		X	-	-
Temp Difference	dtemp		X	-	-
Source/Drain Selector	geo		X	X	-
Threshold Voltage Shift	delvto		X	-	-
Multiplier	m		X	X	X

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Expected operating region	region		-	X	-
Temperature rise from ambient	trise		-	X	-
NQS flag	nqsmod		-	X	-
Drain diffusion length	ld		-	X	-
Source diffusion length	ls		-	X	-
Dist. OD and poly (one side)	sa		-	X	-
Dist. OD and poly (other side)	sb		-	X	-
Distance between fingers	sd		-	X	-

Example

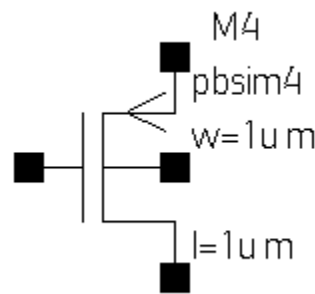
m34 d g s b nbsim4 w=1u l=1u

Model Statement

.model nbsim4 nmos level=54

pbsim4

Four-terminal PMOS device with model name *pbsim4*.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Model Name	model	nbsim4	X	X	X
Width	w	1u	X	X	X
Length	l	1u	X	X	X
Drain Diffusion Area	ad		X	X	-
Source Diffusion Area	as		X	X	-
Drain Junction Perimeter	pd		X	X	-
Source Junction Perimeter	ps		X	X	-
Drain Diff Resistor Sq	nrd		X	X	-
Source Diff Resistor Sq	nrs		X	X	-
Additional Drain Resistance	rdc		X	X	-
Additional Source Resistance	rsc		X	X	-
Initially Off	off	0	-	-	-
Drain-source Initial Voltage	Vds		X	-	-
Gate-source Initial Voltage	Vgs		X	-	-
Bulk-source Initial Voltage	Vbs		X	-	-
Temp Difference	dtemp		X	-	-
Source/Drain Selector	geo		X	X	-
Threshold Voltage Shift	delvto		X	-	-
Multiplier	m		X	X	X

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Expected operating region	region		-	X	-
Temperature rise from ambient	trise		-	X	-
NQS flag	nqsmod		-	X	-
Drain diffusion length	ld		-	X	-
Source diffusion length	ls		-	X	-
Dist. OD and poly (one side)	sa		-	X	-
Dist. OD and poly (other side)	sb		-	X	-
Distance between fingers	sd		-	X	-

Example

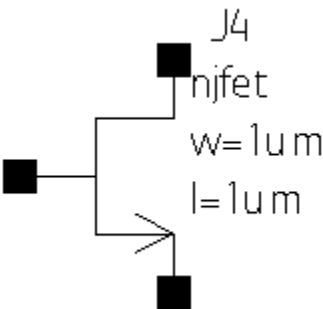
```
M4 s g d vdd! pbsim4 w=1u l=1u
```

Model Statement

```
.model pbsim4 pmos level=54
```

njfet

Three-terminal Junction Gate Field-Effect transistor.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Model Name	model	njfet	X	X	X
Device Area	area		-	X	-
Width	w	1u	X	-	X
Length	l	1u	X	-	X
Initially Off	off	0	X	-	-
Drain-source Initial Voltage	Vds		X	-	-
Gate-source Initial Voltage	Vgs		X	-	-
Multiplier	m		X	X	X
Temp Difference	dtemp		X	-	-
Expected operating region	region		-	X	-

Example

```
j35 d g s gnd! njfet w=1u l=1u
```

Model Statement

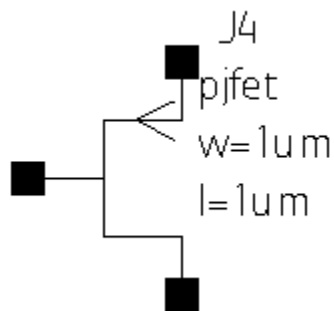
```
.model njfet njf vto=-1.2 beta=.179m lambda=2.2m rs=70 rd=70  
+ is=1.7e-14 cgs=14p cgd=14p
```

See Also

- [JFETs and MESFETs](#) in the *HSPICE® User Guide: Elements*

pjfet

Three-terminal p-type Junction Gate Field-Effect transistor. By default, bulk terminal is connected to global net vdd!.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Model Name	model	pjfet	X	X	X
Device Area	area		-	X	-
Width	w	1u	X	-	X
Length	l	1u	X	-	X
Initially Off	off	0	X	-	-
Drain-source Initial Voltage	Vds		X	-	-
Gate-source Initial Voltage	Vgs		X	-	-
Multiplier	m		X	X	X
Temp Difference	dtemp		X	-	-
Expected operating region	region		-	X	-

Example

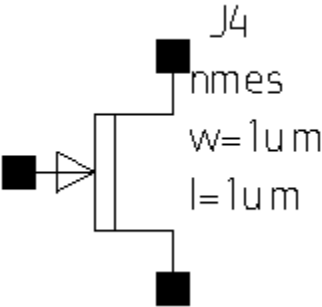
```
j4 s g d vdd! pjfet w=1u l=1u
```

Model Statement

```
.model pjfet pjf vto=-1.2 beta=.179m lambda=2.2m rs=70 rd=70  
+ is=1.7e-14 cgs=14p cgd=14p
```

nmes

Three-terminal Metal-Semiconductor Field-Effect transistor.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Model Name	model	nmes	X	X	4
Device Area	area		-	X	-
Width	w	1u	X	-	4
Length	l	1u	X	-	X
Initially Off	off	0	X	-	-
Drain-source Initial Voltage	Vds		X	-	-
Gate-source Initial Voltage	Vgs		X	-	-
Multiplier	m		X	X	X
Temp Difference	dtemp		X	-	-
Expected operating region	region		-	X	-

Example

```
j36 d g s gnd! nmes w=1u l=1u
```

Model Statement

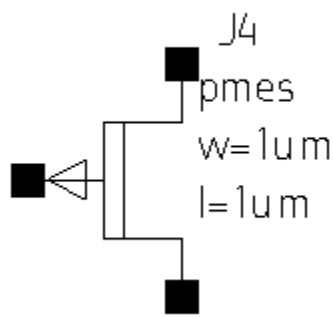
```
.model nmes njf vto=-1.2 beta=.179m lambda=2.2m rs=70 rd=70
+ is=1.7e-14 cgs=14p cgd=14p
```

See Also

- [JFETs and MESFETs](#) in the *HSPICE® User Guide: Elements*

pmes

Three-terminal Metal-Semiconductor Field-Effect transistor.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Model Name	model	nmes	X	X	X
Device Area	area		-	X	-
Width	w	1u	X	-	X
Length	l	1u	X	-	X
Initially Off	off	0	X	-	-
Drain-source Initial Voltage	Vds		X	-	-
Gate-source Initial Voltage	Vgs		X	-	-
Multiplier	m		X	X	X
Temp Difference	dtemp		X	-	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Expected operating region	region		-	X	-

Example

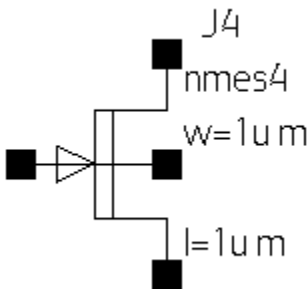
```
jm4 d g s gnd! pmes w=1u l=1u
```

Model Statement

```
.model pmes pjf vto=-1.2 beta=.179m lambda=2.2m rs=70 rd=70  
+ is=1.7e-14 cgs=14p cgd=14p
```

nmes4

Four-terminal Metal-Semiconductor Field-Effect transistor.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Model Name	model	nmes4	X	-	X
Device Area	area		-	-	-
Width	w	1u	X	-	X
Length	l	1u	X	-	X
Initially Off	off	0	X	-	-
Drain-source Initial Voltage	Vds		X	-	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Gate-source Initial Voltage	Vgs		X	-	-
Multiplier	m		X	-	X
Temp Difference	dtemp		X	-	-

Example

```
j36 d g s b nmes4 w=1u l=1u
```

Model Statement

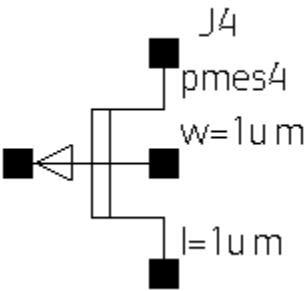
```
.model nmes4 njf vto=-1.2 beta=.179m lambda=2.2m rs=70 rd=70  
+ is=1.7e-14 cgs=14p cgd=14p
```

See Also

- [JFETs and MESFETs](#) in the *HSPICE® User Guide: Elements*

pmes4

Four-terminal Metal-Semiconductor Field-Effect transistor.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Model Name	model	nmes4	X	-	X
Device Area	area		-	-	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Width	w	1u	X	-	X
Length	l	1u	X	-	X
Initially Off	off	0	X	-	-
Drain-source Initial Voltage	Vds		X	-	-
Gate-source Initial Voltage	Vgs		X	-	-
Multiplier	m		X	-	X
Temp Difference	dtemp		X	-	-

Example

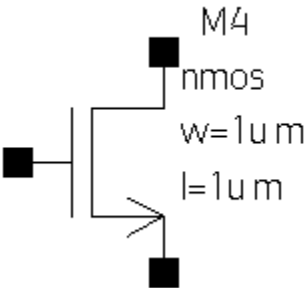
```
j m4 d g s b pmes4 w=1u l=1u
```

Model Statement

```
.model pmes4 pjf vto=-1.2 beta=.179m lambda=2.2m rs=70 rd=70  
+ is=1.7e-14 cgs=14p cgd=14p
```

nmos

Three-terminal N-MOS device. By default, bulk terminal is connected to gnd! net.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Model Name	model	nmos	X	X	X
Width	w	1u	X	X	X
Length	l	1u	X	X	X
Drain Diffusion Area	ad		X	X	-
Source Diffusion Area	as		X	X	-
Drain Junction Perimeter	pd		X	X	-
Source Junction Perimeter	ps		X	X	-
Drain Diff Resistor Sq	nrd		X	X	-
Source Diff Resistor Sq	nrs		X	X	-
Additional Drain Resistance	rdc		X	X	-
Additional Source Resistance	rsc		X	X	-
Initially Off	off	0	-	-	-
Drain-source Initial Voltage	Vds		X	-	-
Gate-source Initial Voltage	Vgs		X	-	-
Bulk-source Initial Voltage	Vbs		X	-	-
Temp Difference	dtemp		X	-	-
Source/Drain Selector	geo		X	X	-
Threshold Voltage Shift	delvto		X	-	-
Multiplier	m		X	X	X

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Drain diffusion length	ld	-	-	X	-
Source diffusion length	ls	-	-	X	-
Dist. OD and poly (one side)	sa	-	-	X	-
Dist. OD and poly (other side)	sb	-	-	X	-
Distance between fingers	sd	-	-	X	-
Hot-electron degradation	degradation	-	-	X	-
Expected operating region	region	-	-	X	-
Temperature rise from ambient	trise	-	-	X	-

Example

```
mj36 d g s gnd! nmos w=1u l=1u
```

Model Statement

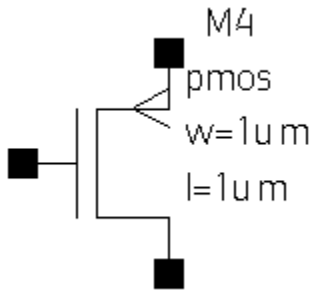
```
.model nmos nmos level=54
```

See Also

- [MOSFETs](#) in the *HSPICE® User Guide: Elements*

pmos

Three-terminal P-MOS device. By default, bulk terminal is connected to vdd! net.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Model Name	model	nmos	X	X	X
Width	w	1u	X	X	X
Length	l	1u	X	X	X
Drain Diffusion Area	ad		X	X	-
Source Diffusion Area	as		X	X	-
Drain Junction Perimeter	pd		X	X	-
Source Junction Perimeter	ps		X	X	-
Drain Diff Resistor Sq	nrd		X	X	-
Source Diff Resistor Sq	nrs		X	X	-
Additional Drain Resistance	rdc		X	X	-
Additional Source Resistance	rsc		X	X	-
Initially Off	off	0	-	-	-
Drain-source Initial Voltage	Vds		X	-	-
Gate-source Initial Voltage	Vgs		X	-	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Bulk-source Initial Voltage	Vbs		X	-	-
Temp Difference	dtemp		X	-	-
Source/Drain Selector	geo		X	X	-
Threshold Voltage Shift	delvto		X	-	-
Multiplier	m		X	X	X
Drain diffusion length	ld		-	X	-
Source diffusion length	ls		-	X	-
Dist. OD and poly (one side)	sa		-	X	-
Dist. OD and poly (other side)	sb		-	X	-
Distance between fingers	sd		-	X	-
Hot-electron degradation	degradation		-	X	-
Expected operating region	region		-	X	-
Temperature rise from ambient	trise		-	X	-

Example

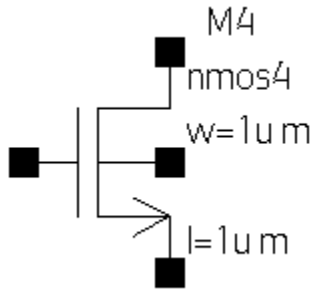
```
m4 d g s vdd! pmos w=1u l=1u
```

Model Statement

```
.model pmos pmos level=54
```

nmos4

Four-terminal N-MOS device.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Model Name	model	nmos	X	X	X
Width	w	1u	X	X	X
Length	l	1u	X	X	X
Drain Diffusion Area	ad		X	X	-
Source Diffusion Area	as		X	X	-
Drain Junction Perimeter	pd		X	X	-
Source Junction Perimeter	ps		X	X	-
Drain Diff Resistor Sq	nrd		X	X	-
Source Diff Resistor Sq	nrs		X	X	-
Additional Drain Resistance	rdc		X	X	-
Additional Source Resistance	rsc		X	X	-
Initially Off	off	0	-	-	-
Drain-source Initial Voltage	Vds		X	-	-
Gate-source Initial Voltage	Vgs		X	-	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Bulk-source Initial Voltage	Vbs		X	-	-
Temp Difference	dtemp		X	-	-
Source/Drain Selector	geo		X	X	-
Threshold Voltage Shift	delvto		X	-	-
Multiplier	m		X	X	X
Drain diffusion length	ld		-	X	-
Source diffusion length	ls		-	X	-
Dist. OD and poly (one side)	sa		-	X	-
Dist. OD and poly (other side)	sb		-	X	-
Distance between fingers	sd		-	X	-
Hot-electron degradation	degradation		-	X	-
Expected operating region	region		-	X	-
Temperature rise from ambient	trise		-	X	-

Example

```
m4 d g s b nmos4 w=1u l=1u
```

Model Statement

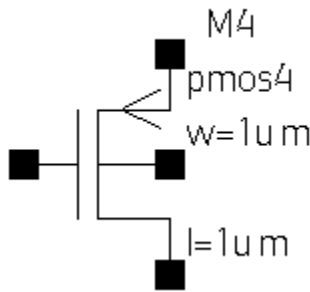
```
.model nmos4 nmos level=54
```

See Also

- [MOSFETs](#) in the *HSPICE® User Guide: Elements*

pmos4

Four-terminal P-MOS device.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Model Name	model	nmos	X	X	X
Width	w	1u	X	X	X
Length	l	1u	X	X	X
Drain Diffusion Area	ad		X	X	-
Source Diffusion Area	as		X	X	-
Drain Junction Perimeter	pd		X	X	-
Source Junction Perimeter	ps		X	X	-
Drain Diff Resistor Sq	nrd		X	X	-
Source Diff Resistor Sq	nrs		X	X	-
Additional Drain Resistance	rdc		X	X	-
Additional Source Resistance	rsc		X	X	-
Initially Off	off	0	-	-	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Drain-source Initial Voltage	Vds		X	-	-
Gate-source Initial Voltage	Vgs		X	-	-
Bulk-source Initial Voltage	Vbs		X	-	-
Temp Difference	dtemp		X	-	-
Source/Drain Selector	geo		X	X	-
Threshold Voltage Shift	delvto		X	-	-
Multiplier	m		X	X	X
Drain diffusion length	ld		-	X	-
Source diffusion length	ls		-	X	-
Dist. OD and poly (one side)	sa		-	X	-
Dist. OD and poly (other side)	sb		-	X	-
Distance between fingers	sd		-	X	-
Hot-electron degradation	degradation		-	X	-
Expected operating region	region		-	X	-
Temperature rise from ambient	trise		-	X	-

Example

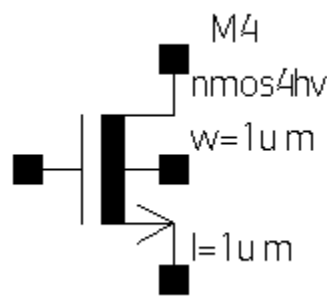
```
m4 d g s b pmos4 w=1u l=1u
```

Model Statement

```
.model pmos4 pmos level=54
```

nmos4hv

Four-terminal high-voltage N-MOS device.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Model Name	model	nmos4hv	X	-	X
Width	w	1u	X	-	X
Length	l	1u	X	-	X
Drain Diffusion Area	ad		X	-	-
Source Diffusion Area	as		X	-	-
Drain Junction Perimeter	pd		X	-	-
Source Junction Perimeter	ps		X	-	-
Drain Diff Resistor Sq	nrd		X	-	-
Source Diff Resistor Sq	nrs		X	-	-
Additional Drain Resistance	rdc		X	-	-
Additional Source Resistance	rsc		X	-	-
Initially Off	off	0	-	-	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Drain-source Initial Voltage	Vds		X	-	-
Gate-source Initial Voltage	Vgs		X	-	-
Bulk-source Initial Voltage	Vbs		X	-	-
Temp Difference	dtemp		X	-	-
Source/Drain Selector	geo		X	-	-
Threshold Voltage Shift	delvto		X	-	-
Multiplier	m		X	-	X

Example

```
mj36 d g s b nmos4hv w=1u l=1u
```

Model Statement

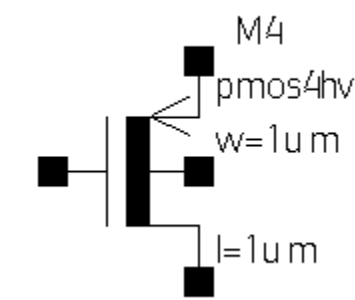
```
.model nmos4hv nmos level=54
```

See Also

- [MOSFETs](#) in the *HSPICE® User Guide: Elements*

pmos4hv

Four-terminal high-voltage P-MOS device.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Model Name	model	nmos4hv	X	-	X
Width	w	1u	X	-	X
Length	l	1u	X	-	X
Drain Diffusion Area	ad		X	-	-
Source Diffusion Area	as		X	-	-
Drain Junction Perimeter	pd		X	-	-
Source Junction Perimeter	ps		X	-	-
Drain Diff Resistor Sq	nrd		X	-	-
Source Diff Resistor Sq	nrs		X	-	-
Additional Drain Resistance	rdc		X	-	-
Additional Source Resistance	rsc		X	-	-
Initially Off	off	0	-	-	-
Drain-source Initial Voltage	Vds		X	-	-
Gate-source Initial Voltage	Vgs		X	-	-
Bulk-source Initial Voltage	Vbs		X	-	-
Temp Difference	dtemp		X	-	-
Source/Drain Selector	geo		X	-	-
Threshold Voltage Shift	delvto		X	-	-
Multiplier	m		X	-	X

Example

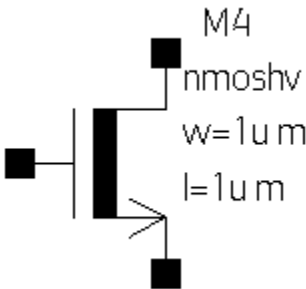
```
m4 d g s b pmos4hv w=1u l=1u
```

Model Statement

```
.model pmos4hv pmos level=54
```

nmoshv

Three-terminal high-voltage N-MOS device.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Model Name	model	nmos4hv	X	-	X
Width	w	1u	X	-	X
Length	l	1u	X	-	X
Drain Diffusion Area	ad		X	-	-
Source Diffusion Area	as		X	-	-
Drain Junction Perimeter	pd		X	-	-
Source Junction Perimeter	ps		X	-	-
Drain Diff Resistor Sq	nrd		X	-	-
Source Diff Resistor Sq	nrs		X	-	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Additional Drain Resistance	rdc		X	-	-
Additional Source Resistance	rsc		X	-	-
Initially Off	off	0	-	-	-
Drain-source Initial Voltage	Vds		X	-	-
Gate-source Initial Voltage	Vgs		X	-	-
Bulk-source Initial Voltage	Vbs		X	-	-
Temp Difference	dtemp		X	-	-
Source/Drain Selector	geo		X	-	-
Threshold Voltage Shift	delvto		X	-	-
Multiplier	m		X	-	X

Example

```
mj36 d g s gnd! nmoshv w=1u l=1u
```

Model Statement

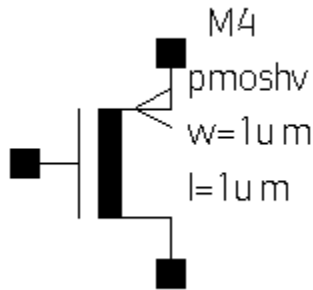
```
.model nmoshv nmos level=54
```

See Also

- [MOSFETs](#) in the *HSPICE® User Guide: Elements*

pmoshv

Three-terminal high-voltage P-MOS device. By default, bulk terminal is connected to vdd!.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Model Name	model	nmos4hv	X	-	X
Width	w	1u	X	-	X
Length	l	1u	X	-	X
Drain Diffusion Area	ad		X	-	-
Source Diffusion Area	as		X	-	-
Drain Junction Perimeter	pd		X	-	-
Source Junction Perimeter	ps		X	-	-
Drain Diff Resistor Sq	nrd		X	-	-
Source Diff Resistor Sq	nrs		X	-	-
Additional Drain Resistance	rdc		X	-	-
Additional Source Resistance	rsc		X	-	-
Initially Off	off	0	-	-	-
Drain-source Initial Voltage	Vds		X	-	-
Gate-source Initial Voltage	Vgs		X	-	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Bulk-source Initial Voltage	Vbs		X	-	-
Temp Difference	dtemp		X	-	-
Source/Drain Selector	geo		X	-	-
Threshold Voltage Shift	delvto		X	-	-
Multiplier	m		X	-	X

Example

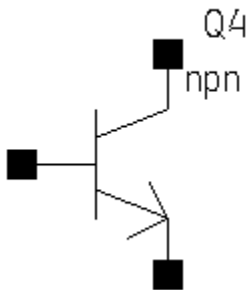
```
m4 d g s vdd! pmoshv w=1u l=1u
```

Model Statement

```
.model pmoshv pmos level=54
```

npn

Three-terminal N-P-N bipolar junction transistor.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Model Name	model	npn	X	X	X
Device Area	area		X	X	X

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Base-emitter Voltage	Vbe		X	-	-
Collector-emitter Voltage	Vce		X	-	-
Initially Off	off	0	-	-	-
Multiplier	m		X	X	X
Temp Difference	dtemp		X	X	-
Base Area	areab		X	-	-
Collector Area	areac		X	-	-
Temperature rise	dtmp		-	-	-
Temperature rise	dtempn		-	-	-
Expected operating region	region		-	X	-

Example

```
q37 c b e npn area=10u
```

Model Statement

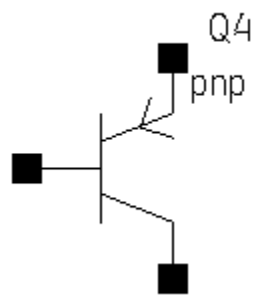
```
.model npn npn (bf=147 is=1.26e-17 br=0.7 ise=0 isc=0
+ ikf=2.7e-3 ikr=7.2e-3 ne=1.5 nc=1.5 tf=2.27e-10 tr=2.27e-9
+ cje=7.71e-14 cjc=5.74e-14 vje=0.88 vjc=0.74 mje=0.42 mjc=0.41
+ cjs=2.82e-13 vjs=0.67 mjs=0.44 vaf=17 var=4.25 rc=87.34
+ rb=1040 re=1.578 xtb=1.58936 eg=1.17 fc=0.5 xti=3.5)
```

See Also

- [Bipolar Junction Transistor](#) in the *HSPICE® User Guide: Elements*

pnp

Three-terminal P-N-P bipolar junction transistor.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Model Name	model	npn	X	X	X
Device Area	area		X	X	X
Base-emitter Voltage	Vbe		X	-	-
Collector-emitter Voltage	Vce		X	-	-
Initially Off	off	0	-	-	-
Multiplier	m		X	X	X
Temp Difference	dtemp		X	X	-
Base Area	areab		X	-	-
Collector Area	areac		X	-	-
Temperature rise	dtmp		-	-	-
Temperature rise	dtempn		-	-	-
Expected operating region	region		-	X	-

Example

q4 c b e pnp area=10u

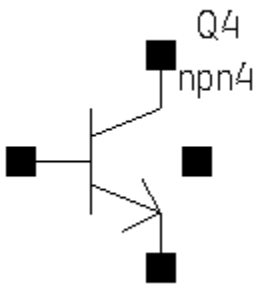
Model Statement

```
.model pnp pnp (bf=92 is=2.8e-16 br=0.5 ise=0 isc=0 ikf=.0003
+ ikr=9e-4 ne=1.5 nc=1.5 tf=3.54e-8 tr=1.06e-7
```

```
+ cje=1.14e-13 cjc=5.8e-13 vje=.74 vjc=.74
+ mje=.41 mjc=.41 cjs=0
+ vaf=16.1 var=5.38 rc=100 rb=50 re=5
+ xtb=1.3 eg=1.17 fc=.5 xti=3.5)
```

npn4

N-P-N bipolar junction transistor with substrate terminal.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Model Name	model	npn4	X	-	X
Device Area	area		X	-	X
Base-emitter Voltage	Vbe		X	-	-
Collector-emitter Voltage	Vce		X	-	-
Initially Off	off	0	-	-	-
Multiplier	m		X	-	X
Temp Difference	dtemp		X	-	-
Base Area	areab		X	-	-
Collector Area	areac		X	-	-

Example

```
q37 c b e s npn area=10u
```

Model Statement

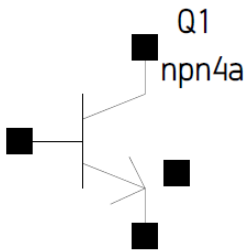
```
.model npn4 npn (bf=147 is=1.26e-17 br=0.7 ise=0 isc=0
+ cje=7.71e-14 cjc=5.74e-14 vje=0.88 vjc=0.74 mje=0.42 mjc=0.41
+ cjs=2.82e-13 vjs=0.67 mjs=0.44 vaf=17 var=4.25 rc=87.34
+ rb=1040 re=1.578 xtb=1.58936 eg=1.17 fc=0.5 xti=3.5)
+ ikf=2.7e-3 ikr=7.2e-3 ne=1.5 nc=1.5 tf=2.27e-10 tr=2.27e-9
```

See Also

- [Bipolar Junction Transistor](#) in the *HSPICE® User Guide: Elements*

npn4a

N-P-N bipolar junction transistor with substrate terminal.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Model Name	model	npn4a	X	-	X
Device Area	area		X	-	-
Base-emitter Voltage	Vbe		X	-	-
Collector-emitter Voltage	Vce		X	-	-
Initially Off	off	0	-	-	-
Multiplier	m		X	-	X
Temp Difference	dtemp		X	-	-
Base Area	areab		X	-	-
Collector Area	areac		X	-	-

Example

```
q1 c b e s npn4a area=10u
```

Model Statement

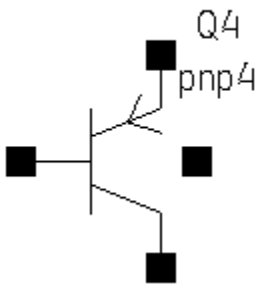
```
.model npn4a npn (bf=147 is=1.26e-17 br=0.7 ise=0 isc=0
+ ikf=2.7e-3 ikr=7.2e-3 ne=1.5 nc=1.5 tf=2.27e-10 tr=2.27e-9
+ cje=7.71e-14 cjc=5.74e-14 vje=0.88 vjc=0.74 mje=0.42 mjc=0.41
+ cjs=2.82e-13 vjs=0.67 mjs=0.44 vaf=17 var=4.25 rc=87.34
+ rb=1040 re=1.578 xtb=1.58936 eg=1.17 fc=0.5 xti=3.5)
```

See Also

- [BJT Model Equations: \(NPN and PNP\)](#) in the *HSPICE® Reference Manual: Device Models*

npn4

P-N-P bipolar junction transistor with substrate terminal.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Model Name	model	npn4	X	-	X
Device Area	area		X	-	X
Base-emitter Voltage	Vbe		X	-	-
Collector-emitter Voltage	Vce		X	-	-
Initially Off	off	0	-	-	-
Multiplier	m		X	-	X

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Temp Difference	dtemp		X	-	-
Base Area	areab		X	-	-
Collector Area	areac		X	-	-

Example

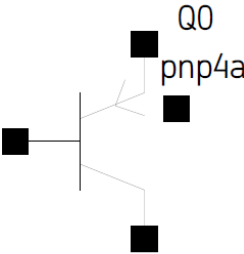
q4 c b e s pnp4 area=10u

Model Statement

```
.model pnp4 pnp (bf=92 is=2.8e-16 br=0.5 ise=0 isc=0 ikf=.0003
+ ikr=9e-4 ne=1.5 nc=1.5 tf=3.54e-8 tr=1.06e-7
+ cje=1.14e-13 cjc=5.8e-13 vje=.74 vjc=.74
+ mje=.41 mjc=.41 cjs=0
+ vaf=16.1 var=5.38 rc=100 rb=50 re=5
+ xtb=1.3 eg=1.17 fc=.5 xti=3.5)
```

pnp4a

P-N-P bipolar junction transistor with substrate terminal.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Model Name	model	pnp4a	X	-	X
Device Area	area		X	-	X
Base-emitter Voltage	Vbe		X	-	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Collector-emitter Voltage	Vce		X	-	-
Initially Off	off	0	X	-	-
Drain-source Initial Voltage	Vds		X	-	-
Multiplier	m		X	X	X
Temp Difference	dtemp		X	-	-
Expected operating region	region		-	X	-
Base Area	areab		-	-	-
Collector Area	areac		-	-	-

Example

q0 c b e s pnp4a

Model Statement

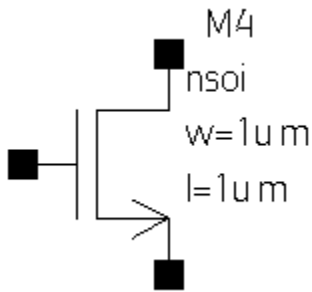
```
.model pnp4a pnp (bf=92 is=2.8e-16 br=0.5 ise=0 isc=0 ikf=.0003
+ ikr=9e-4 ne=1.5 nc=1.5 tf=3.54e-8 tr=1.06e-7
+ cje=1.14e-13 cjc=5.8e-13 vje=.74 vjc=.74
+ mje=.41 mjc=.41 cjs=0
+ vaf=16.1 var=5.38 rc=100 rb=50 re=5
+ xtb=1.3 eg=1.17 fc=.5 xti=3.5)
```

See Also

- [BJT Model Equations: \(NPN and PNP\)](#) in the *HSPICE® Reference Manual: Device Models*

nsoi

N-type three-terminal Silicon on Insulator (SOI) device. By default, bulk terminal is connected to gnd! node.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Model Name	model	nsoi	X	-	X
Width	w	1u	X	-	X
Length	l	1u	X	-	X
Drain Diffusion Area	ad		X	-	-
Source Diffusion Area	as		X	-	-
Drain Junction Perimeter	pd		X	-	-
Source Junction Perimeter	ps		X	-	-
Drain Diff Resistor Sq	nrd		X	-	-
Source Diff Resistor Sq	nrs		X	-	-
Additional Drain Resistance	rdc		X	-	-
Additional Source Resistance	rsc		X	-	-
Initially Off	off	0	-	-	-
Drain-source Initial Voltage	Vds		X	-	-
Gate-source Initial Voltage	Vgs		X	-	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Bulk-source Initial Voltage	Vbs		X	-	-
Temp Difference	dtemp		X	-	-
Source/Drain Selector	geo		X	-	-
Threshold Voltage Shift	delvto		X	-	-
Multiplier	m		X	-	X

Example

```
m38 d g s gdn! nsoi w=1u l=1u
```

Model Statement

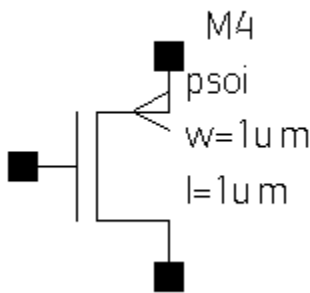
```
.model nsoi nmos level=57
```

See Also

- [LEVEL 57: BSIM3 SOI Model](#) in the *HSPICE® Reference Manual: MOSFET Models*

psoi

P-type three-terminal Silicon on Insulator (SOI) device. By default, bulk terminal is connected to vdd! node.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Model Name	model	psoi	X	-	X
Width	w	1u	X	-	X
Length	l	1u	X	-	X
Drain Diffusion Area	ad		X	-	-
Source Diffusion Area	as		X	-	-
Drain Junction Perimeter	pd		X	-	-
Source Junction Perimeter	ps		X	-	-
Drain Diff Resistor Sq	nrd		X	-	-
Source Diff Resistor Sq	nrs		X	-	-
Additional Drain Resistance	rdc		X	-	-
Additional Source Resistance	rsc		X	-	-
Initially Off	off	0	-	-	-
Drain-source Initial Voltage	Vds		X	-	-
Gate-source Initial Voltage	Vgs		X	-	-
Bulk-source Initial Voltage	Vbs		X	-	-
Temp Difference	dtemp		X	-	-
Source/Drain Selector	geo		X	-	-
Threshold Voltage Shift	delvto		X	-	-
Multiplier	m		X	-	X

Example

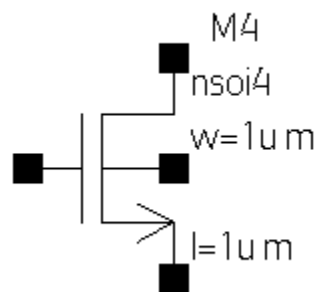
```
m38 d g s vdd! psoi w=1u l=1u
```

Model Statement

```
.model psoi pmos level=57
```

nsoi4

N-type four-terminal Silicon on Insulator (SOI) device.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Model Name	model	nsoi4	X	-	X
Width	w	1u	X	-	X
Length	l	1u	X	-	X
Drain Diffusion Area	ad		X	-	-
Source Diffusion Area	as		X	-	-
Drain Junction Perimeter	pd		X	-	-
Source Junction Perimeter	ps		X	-	-
Drain Diff Resistor Sq	nrd		X	-	-
Source Diff Resistor Sq	nrs		X	-	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Additional Drain Resistance	rdc		X	-	-
Additional Source Resistance	rsc		X	-	-
Initially Off	off	0	-	-	-
Drain-source Initial Voltage	Vds		X	-	-
Gate-source Initial Voltage	Vgs		X	-	-
Bulk-source Initial Voltage	Vbs		X	-	-
Temp Difference	dtemp		X	-	-
Source/Drain Selector	geo		X	-	-
Threshold Voltage Shift	delvto		X	-	-
Multiplier	m		X	-	X

Example

```
m38 d g s b nsoi4 w=1u l=1u
```

Model Statement

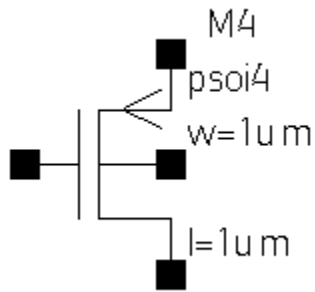
```
.model nsoi4 nmos level=57
```

See Also

- [LEVEL 57: BSIM3 SOI Model](#) in the *HSPICE® Reference Manual: MOSFET Models*

psoi4

N-type four-terminal Silicon on Insulator (SOI) device.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Model Name	model	nsoi4	X	-	X
Width	w	1u	X	-	X
Length	l	1u	X	-	X
Drain Diffusion Area	ad		X	-	-
Source Diffusion Area	as		X	-	-
Drain Junction Perimeter	pd		X	-	-
Source Junction Perimeter	ps		X	-	-
Drain Diff Resistor Sq	nrd		X	-	-
Source Diff Resistor Sq	nrs		X	-	-
Additional Drain Resistance	rdc		X	-	-
Additional Source Resistance	rsc		X	-	-
Initially Off	off	0	-	-	-
Drain-source Initial Voltage	Vds		X	-	-
Gate-source Initial Voltage	Vgs		X	-	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Bulk-source Initial Voltage	Vbs		X	-	-
Temp Difference	dtemp		X	-	-
Source/Drain Selector	geo		X	-	-
Threshold Voltage Shift	delvto		X	-	-
Multiplier	m		X	-	X

Example

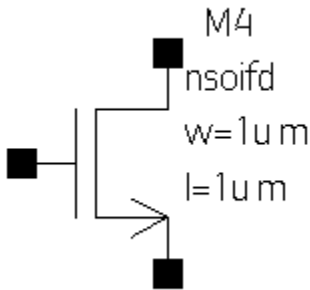
```
m38 d g s b psoi4 w=1u l=1u
```

Model Statement

```
.model psoi4 pmos level=57
```

nsoifd

Three-terminal Fully Depleted Silicon On Insulator, or FD-SOI device. By default, bulk terminal is connected to gnd!.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Model Name	model	nsoifd	X	-	X

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Width	w	1u	X	-	X
Length	l	1u	X	-	X
Drain Diffusion Area	ad		X	-	-
Source Diffusion Area	as		X	-	-
Drain Junction Perimeter	pd		X	-	-
Source Junction Perimeter	ps		X	-	-
Drain Diff Resistor Sq	nrd		X	-	-
Source Diff Resistor Sq	nrs		X	-	-
Additional Drain Resistance	rdc		X	-	-
Additional Source Resistance	rsc		X	-	-
Initially Off	off	0	-	-	-
Drain-source Initial Voltage	Vds		X	-	-
Gate-source Initial Voltage	Vgs		X	-	-
Bulk-source Initial Voltage	Vbs		X	-	-
Temp Difference	dtemp		X	-	-
Source/Drain Selector	geo		X	-	-
Threshold Voltage Shift	delvto		X	-	-
Multiplier	m		X	-	X

Example

```
m38 d g s gnd! nsoifd w=1u l=1u
```

Model Statement

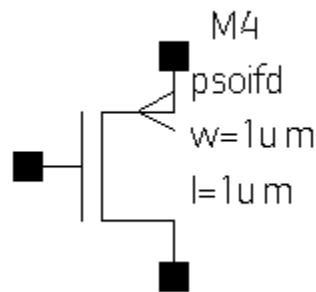
```
.model nsoifd nmos level=59
```

See Also

- [LEVEL 59: BSIM3 SOI FD Model](#) in the *HSPICE® Reference Manual: MOSFET Models*

psoifd

Three-terminal Fully Depleted Silicon On Insulator, or FD-SOI device. By default, bulk terminal is connected to vdd!.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Model Name	model	nsoifd	X	-	X
Width	w	1u	X	-	X
Length	l	1u	X	-	X
Drain Diffusion Area	ad		X	-	-
Source Diffusion Area	as		X	-	-
Drain Junction Perimeter	pd		X	-	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Source Junction Perimeter	ps		X	-	-
Drain Diff Resistor Sq	nrd		X	-	-
Source Diff Resistor Sq	nrs		X	-	-
Additional Drain Resistance	rdc		X	-	-
Additional Source Resistance	rsc		X	-	-
Initially Off	off	0	-	-	-
Drain-source Initial Voltage	Vds		X	-	-
Gate-source Initial Voltage	Vgs		X	-	-
Bulk-source Initial Voltage	Vbs		X	-	-
Temp Difference	dtemp		X	-	-
Source/Drain Selector	geo		X	-	-
Threshold Voltage Shift	delvto		X	-	-
Multiplier	m		X	-	X

Example

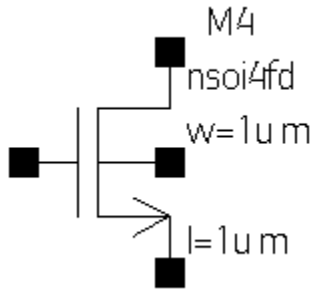
```
m38 d g s vdd! nsoifd w=1u l=1u
```

Model Statement

```
.model psoifd pmos level=59
```

nsoifd4

Four-terminal Fully Depleted Silicon On Insulator, or FD-SOI device.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Model Name	model	nsoifd	X	-	X
Width	w	1u	X	-	X
Length	l	1u	X	-	X
Drain Diffusion Area	ad		X	-	-
Source Diffusion Area	as		X	-	-
Drain Junction Perimeter	pd		X	-	-
Source Junction Perimeter	ps		X	-	-
Drain Diff Resistor Sq	nrd		X	-	-
Source Diff Resistor Sq	nrs		X	-	-
Additional Drain Resistance	rdc		X	-	-
Additional Source Resistance	rsc		X	-	-
Initially Off	off	0	-	-	-
Drain-source Initial Voltage	Vds		X	-	-
Gate-source Initial Voltage	Vgs		X	-	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Bulk-source Initial Voltage	Vbs		X	-	-
Temp Difference	dtemp		X	-	-
Source/Drain Selector	geo		X	-	-
Threshold Voltage Shift	delvto		X	-	-
Multiplier	m		X	-	X

Example

```
m38 d g s b nsoi4fd w=1u l=1u
```

Model Statement

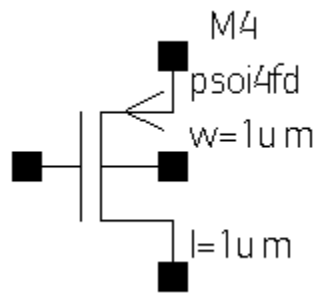
```
.model nsoi4fd nmos level=59
```

See Also

- [LEVEL 59: BSIM3 SOI FD Model](#) in the *HSPICE® Reference Manual: MOSFET Models*

psoifd4

Four-terminal Fully Depleted Silicon On Insulator, or FD-SOI device.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Model Name	model	nsoifd	X	-	X
Width	w	1u	X	-	X
Length	l	1u	X	-	X
Drain Diffusion Area	ad		X	-	-
Source Diffusion Area	as		X	-	-
Drain Junction Perimeter	pd		X	-	-
Source Junction Perimeter	ps		X	-	-
Drain Diff Resistor Sq	nrd		X	-	-
Source Diff Resistor Sq	nrs		X	-	-
Additional Drain Resistance	rdc		X	-	-
Additional Source Resistance	rsc		X	-	-
Initially Off	off	0	-	-	-
Drain-source Initial Voltage	Vds		X	-	-
Gate-source Initial Voltage	Vgs		X	-	-
Bulk-source Initial Voltage	Vbs		X	-	-
Temp Difference	dtemp		X	-	-
Source/Drain Selector	geo		X	-	-
Threshold Voltage Shift	delvto		X	-	-
Multiplier	m		X	-	X

Example

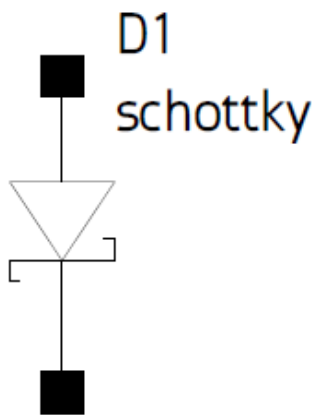
```
m38 d g s b psoi4fd w=1u l=1u
```

Model Statement

```
.model psoi4fd pmos level=59
```

schottky

Schottky barrier diode element.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Model Name	model	schottky	X	X	-
Device Area	area		X	X	X
Initial Diode Voltage	Vd		X	-	-
Periphery of Junction	pj		X	X	X
Width of Polycap	wp		X	X	-
Length of Polycap	lp		X	X	-
Width of Metalcap	wm		X	X	-
Length of Metalcap	lm		X	X	-
Width	w		X	X	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Length	l	0	X	X	-
Initially Off	off		-	-	-
Multiplier	m		X	X	X
Device Temperature	dtemp		X	X	-
Amplitude scale factor	scale		-	X	-

Example

```
dl p n schottky
```

Model Statement

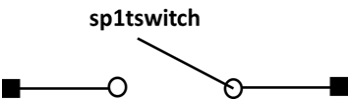
```
.MODEL schottky D
+ LEVEL=1 XP =0.0 EG =1.1
+ XO1 =0.0 XOM =0.0 XM =0.0
+ WP =0.0 WM =0.0 LP =0.0
+ LM =0.0 AF =1.0 JSW =0.0
+ PB =0.65 PHP =0.8 M =0.2994
+ EG =0.69
+ FC =0.95 FCS =0.4 MJSW=0.5
+ TT =2.446e-9 BV =4.65 RS =19
+ IS =1.485e-11 CJO =1.09e-9 CJP =0.0
+ PJ =0.0 N =1.615 IK =0.0
+ IKR =1.100e-2 IBV =2.00e-2
```

See Also

- [Diode Types](#) in the *HSPICE® Reference Manual: Device Models*

sp1tswitch

Ideal switch with 1 and 2 positions.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Offset voltage	offset		X	X	-
Switch position	position	0	X	X	-
dc_position	dc_position	-1	X	-	-
ac_position	ac_position	-1	X	-	-
ic_position	ic_position	-1	X	-	-
tran_position	tran_position	-1	X	-	-
DC position	dcPosition	0	-	X	-
AC position	acPosition	0	-	X	-
IC position	icPosition	0	-	X	-
Tran position	tranPosition	0	-	X	-
Switch position	position_str	0	-	X	-
DC position	dcPosition_str	0	-	X	-
AC position	acPosition_str	0	-	X	-
IC position	icPosition_str	0	-	X	-
Tran position	tranPosition_str	0	-	X	-
Multiplier	m		X	X	-
Parameter Type	paramType	cyclic	-	X	-

Example

```

xsl t0 t1 t2 sp2tswitch offset=0
x2 t1 t0 spltswitch offset=0
.hdl
'<installation_dir>/auxx/samples/analogLib/sp2tswitch/veriloga/veriloga.
va'
.hdl
'<installation_dir>/auxx/samples/analogLib/spltswitch/veriloga/veriloga.
va'

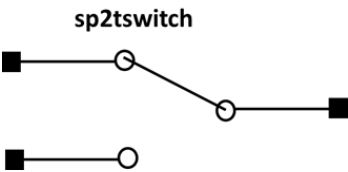
```

Controlling Version Number for Legacy Designs

To control the version number for legacy designs, create and use the `dbAnalogLibSpltswitchVersion` Tcl preference before the analogLib library is loaded.

sp2tswitch

Ideal switch with 1 and 2 positions.



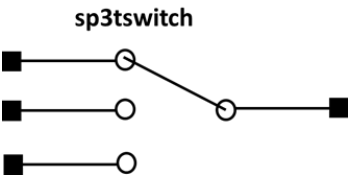
CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Offset voltage	offset		X	X	-
Switch position	position	0	X	X	-
dc_position	dc_position	-1	X	-	-
ac_position	ac_position	-1	X	-	-
ic_position	ic_position	-1	X	-	-
tran_position	tran_position	-1	X	-	-
DC position	dcPosition	0	-	X	-
AC position	acPosition	0	-	X	-
IC position	icPosition	0	-	X	-
Tran position	tranPosition	0	-	X	-
Switch position	position_str	0	-	X	-
DC position	dcPosition_str	0	-	X	-
AC position	acPosition_str	0	-	X	-
IC position	icPosition_str	0	-	X	-
Tran position	tranPosition_str	0	-	X	-

	m		-	X	-
	paramType	cyclic	-	X	-

sp3tswitch

Voltage-controlled switch.



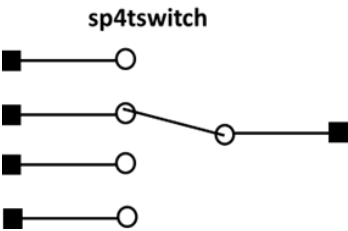
CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Offset voltage	offset		X	X	-
Switch position	position	0	X	X	-
dc_position	dc_position	-1	X	-	-
ac_position	ac_position	-1	X	-	-
ic_position	ic_position	-1	X	-	-
tran_position	tran_position	-1	X	-	-
DC position	dcPosition	0	-	X	-
AC position	acPosition	0	-	X	-
IC position	icPosition	0	-	X	-
Tran position	tranPosition	0	-	X	-
Switch position	position_str	0	-	X	-
DC position	dcPosition_str	0	-	X	-
AC position	acPosition_str	0	-	X	-
IC position	icPosition_str	0	-	X	-
Tran position	tranPosition_str	0	-	X	-

	m		-	X	-
	paramType	cyclic	-	X	-

sp4tswitch

Voltage-controlled switch.



CDF Parameters

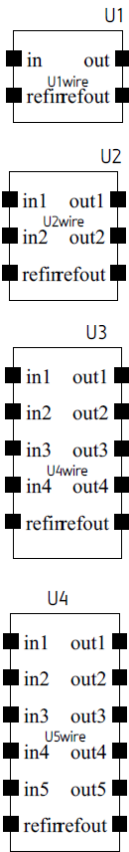
CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Offset voltage	offset		X	X	-
Switch position	position	0	X	X	-
dc_position	dc_position	-1	X	-	-
ac_position	ac_position	-1	X	-	-
ic_position	ic_position	-1	X	-	-
tran_position	tran_position	-1	X	-	-
DC position	dcPosition	0	-	X	-
AC position	acPosition	0	-	X	-
IC position	icPosition	0	-	X	-
Tran position	tranPosition	0	-	X	-
Switch position	position_str	0	-	X	-
DC position	dcPosition_str	0	-	X	-
AC position	acPosition_str	0	-	X	-
IC position	icPosition_str	0	-	X	-

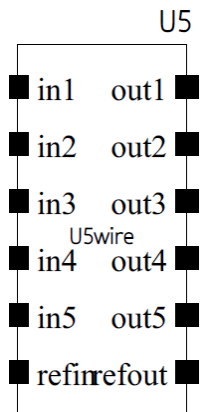
Tran position	tranPosition_str	0	-	X	-
	m		-	X	-
	paramType	cyclic	-	X	-

u1wire, u2wire, u3wire, u4wire, u5wire

U-element is useful for modeling lossy non-ideal transmission lines in terms of lumped-element equivalent circuits.

The U-element models single and coupled lossy transmission lines for various planar, coaxial, and twinlead structures. When a U-element is included in your netlist, the simulation creates an internal network of R, L, C, and G-elements to represent up to five lines and their coupling capacitances and inductances.





CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Model Name	model		-	-	-
Length	l		X	-	-
Element Lumps	lumps		X	-	-

Example

```
u1 in refin out refout ustrip
u2 in1 in2 refin out1 out2 refout ustrip
u3 in1 in2 in3 in4 refin out1 out2 out3 out4 refout ustrip
u4 in1 in2 in3 in4 in5 refin out1 out2 out3 out4 out5 refout ustrip
u5 in1 in2 in3 in4 in5 refin out1 out2 out3 out4 out5 refout usrip
l=0.178 lumps=3
```

Model Statement

```
.Model USTRIP U LEVEL=3 PLev=1 Elev=1 Dlev=2 Nl=3 Ht=381u
+ Wd=305u Th=25u Sp=102u Ts=838u Kd=4.7
.Probe v(13) v(7) v(8) v(6)
.End
```

See Also

- [Lumped Transmission Line -- UModel](#) in the *HSPICE® User Guide: Elements*

Independent Sources

The following topics describe:

- [bcs](#)
- [bvs](#)
- [iam](#)
- [ibit](#)
- [idc](#)
- [iexp](#)
- [ilfsr](#)
- [ipat](#)
- [ipulse](#)
- [ipwl](#)
- [ipwlf](#)
- [ipwlz](#)
- [iprbs](#)
- [isffm](#)
- [isin](#)
- [isource](#)
- [ivmrf](#)
- [multibit](#)
- [n1port](#)
- [n2port](#)
- [n3port](#)
- [n4port](#)
- [ncs](#)
- [nport](#)
- [nport_ref](#)

Chapter 1: analogLib Elements

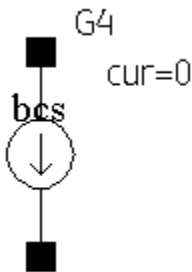
Independent Sources

- [nvs](#)
- [pam](#)
- [pdc](#)
- [pexp](#)
- [plfsr](#)
- [port](#)
- [port3t](#)
- [pprbs](#)
- [ppulse](#)
- [ppwl](#)
- [ppwlf](#)
- [psffm](#)
- [psin](#)
- [vam](#)
- [vbit](#)
- [vcos](#)
- [vdc](#)
- [vexp](#)
- [vgpulse](#)
- [vlfsr](#)
- [vpat](#)
- [vprbs](#)
- [vpulse](#)
- [vpwl](#)
- [vpwlf](#)
- [vpwlz](#)
- [vsffm](#)

- [vsin](#)
- [vsource](#)
- [vwmrf](#)

bcS

Behavioral current source. Useful if you want to model current source, which can be dependent on bias and/or frequency.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Current Equation	cur	0	X	X	-
Min Current	min		X	X	-
Max Current	max		X	X	-
Scale Factor	scale		X	-	-
Multiplier	m		X	X	-

Example

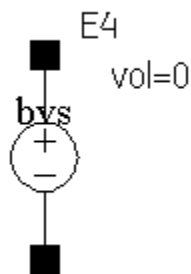
```
v1 1 0 1
r1 1 2 1k
g4 1 2 cur='v(1,2)*0.001'
rout 2 0 1meg
```

See Also

- [Behavioral Current Source](#) in the *HSPICE® User Guide: Elements*

bvs

Behavioral voltage source. Useful if you want to model voltage source, which can be dependent on bias and/or frequency.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Voltage Equation	vol	0	X	X	-
Min Voltage	min		X	X	-
Max Voltage	max		X	X	-
Scale Factor	scale		X	-	-

Example

```
r1 n1 gnd 1
r2 n2 gnd 1
e1 n3 gnd vol='v(n1)+v(n2)'
```

See Also

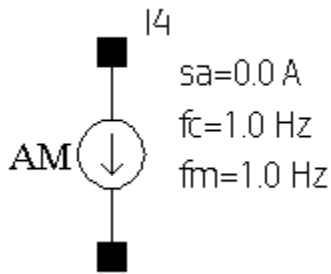
- [Behavioral Voltage Source](#) in the *HSPICE® User Guide: Elements*

iam

Single-frequency AM current source.

The following expression defines the waveform shape:

sourcevalue = *sa* · {*oc* + *SIN*[2 · π · *fm* · (*Time* – *td*)]} · *SIN*[2 · π · *fc* · (*Time* – *td*)]



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
DC Current	idc	0	X	-	-
Source Type	srcType	am	-	-	-
Signal Amplitude	sa	0	X	-	-
Carrier Frequency	fc	1	X	-	-
Modulation Frequency	fm	1	X	-	-
Offset Constant	oc	0	X	-	-
Delay Time	td	0	X	-	-
AC Magnitude	acm		X	-	-
AC Phase	acp		X	-	-
# HB Tones	hbtones	0	X	-	-
HB Magnitude 1-7	hbmag1-7		X	-	-
HB Phase1-7	hbphase1-7	0	X	-	-
HB Harmonic1-7	hbharm1-7	1	X	-	-
HB Tone1-7	hbtone1-7	1-7	X	-	-
HB ModHarm1-7	hbmodharm1		X	-	-
HB ModTone1-7	hbmodtone1		X	-	-
Open DC Connection	dcopen		-	-	-
DC Resistance	rdc		X	-	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
AC Resistance	rac		X	-	-
HB Resistance	rhb		X	-	-
HBAC Resistance	rhbac		X	-	-
TRAN Resistance	rtran		X	-	-
HBAC Magnitude	hbacm		X	-	-
HBAC Phase	hbacp		X	-	-
SNAC Magnitude	snacm		X	-	-
SNAC Phase	snacp		X	-	-
Multiplier	m		X	-	-
Impedance	z0		X	-	-

Example

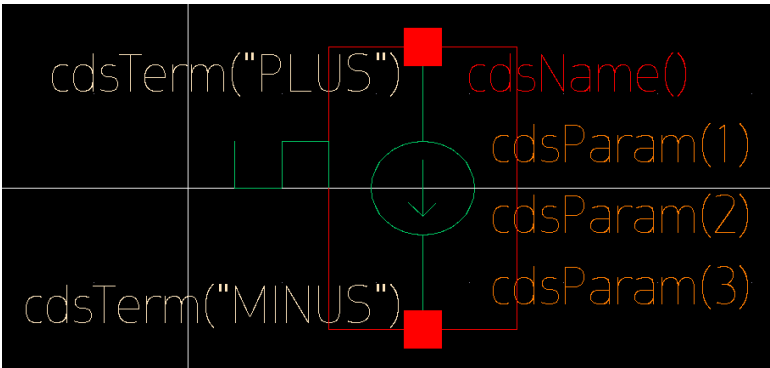
```
i44 plus minus dc=0 am ( 0.85 0.0 10k 1k 0 )
```

See Also

- [Single-frequency AM Source](#) in the *HSPICE® User Guide: Elements*

ibit

Current source that is a pattern source type with various parameters settings.

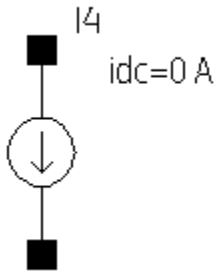


CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Pattern Parameter data	data		-	X	-
Pattern Parameter data is variable	data_as_var	0	-	X	-
Pattern Parameter rptstart	rptstart	1	-	X	-
Pattern Parameter rpttimes	rpttimes	0	-	X	-
Trigger	trigger		-	X	-
RJ(rms)	rjrms		-	X	-
RJ(seed)	rjseed		-	X	-
Number of periodic jitters	nPeriodicJitter	0	-	X	-
PJ%d(amplitude)	pjump1-3		-	X	-
PJ%d(frequency)	pjfreq1-3		-	X	-
PJ%d(type)	pjtype1-3	sine	-	X	-
One Value	val1		-	X	-
Zero Value	val0		-	X	-
Delay Time	td		-	X	-
Rise Time	tr	10f	-	X	-
Fall Time	tf	10f	-	X	-
Period	per		-	X	-
Source Type	srcType	bit	-	X	-

idc

Single-frequency AM current source.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
DC Current	idc	0	X	X	-
AC Magnitude	acm		X	X	-
AC Phase	acp		X	X	-
# HB Tones	hbtones	0	X	-	-
HB Magnitude 1-7	hbmag1-7		X	-	-
HB Phase1-7	hbphase1-7	0	X	-	-
HB Harmonic1-7	hbharm1-7	1	X	-	-
HB Tone1-7	hbtone1-7	1-7	X	-	-
HB ModHarm1-7	hbmodharm1-7		X	-	-
HB ModTone1-7	hbmodtone1-7		X	-	-
Open DC Connection	dcopen		-	-	-
DC Resistance	rdc		X	-	-
AC Resistance	rac		X	-	-
HB Resistance	rhb		X	-	-
HBAC Resistance	rhbac		X	-	-
TRAN Resistance	rtran		X	-	-
HBAC Magnitude	hbacm		X	-	-
HBAC Phase	hbacp		X	-	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
SNAC Magnitude	snacm		X	-	-
SNAC Phase	snacp		X	-	-
Multiplier	m		X	X	-
Source Type	srcType	dc	-	X	-
PAC magnitude	pacm		-	X	-
PAC phase	pacp		-	X	-
Temp coefficient 1	tc1		-	X	-
Temp coefficient 2	tc2		-	X	-
XF magnitude	xfm		-	X	-
Number of noise/frequency pairs	FNpairs	0	-	X	-
Noise file name	noisefile		-	X	-
Noise file name is variable	noise_file_as_var	0	X	X	-
Frequency1-50	F1-50		-	X	-
Noise1-50	N1-50		-	X	-
Nominal temperature	tnom		-	X	-
Impedance	z0		X	-	-

Note:

- When frequency/noise pairs have empty values or the number of frequency/noise pairs does not match with the FNpairs parameter, the Spectre netlister shows an error in the console.
- The default of noise_file_as_var parameter is *false*. If the noisefile parameter is set as *NOISEFILE*, the netlisting behavior for noisefile and noise_file_as_var parameters is as follows:

CDF Parameter	hspiceD Output	spectre Output
noise_file_as_var		

CDF Parameter	hspiceD Output	spectre Output
false (default)	noisefile='NOISEFILE'	noisefile="NOISEFILE"
true	noisefile=str(NOISEFILE)	noisefile=NOISEFILE

- For Hspice-format netlisting, a pair of single quotes is added to noisefile parameter when noise_file_as_var is *false*. When noise_file_as_var parameter is *true*, str(NOISEFILE) is used as the value format.
- For Spectre-format netlisting, a pair of double quotes is added when noise_file_as_var parameter is *false*. No single/double quotes are added when noise_file_as_var parameter is *true*.

Example

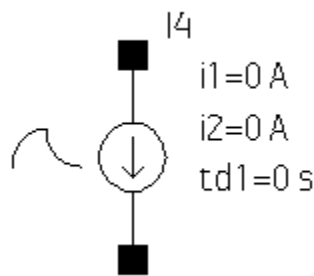
```
i4 plus minus dc=0
```

See Also

- [Independent Source Elements](#) in the *HSPICE® User Guide: Elements*

iexp

Exponential current source.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
DC Current	idc	0	X	X	-
Source Type	srcType	exp		X	-
Current1	i1	0	X	X	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Current2	i2	0	X	X	-
Rise Delay Time	td1	0	X	X	-
Fall Delay Time	td2		X	X	-
Rise Time Constant	tau1		X	X	-
Fall Time Constant	tau2	0	X	X	-
AC Magnitude	acm		X	X	-
AC Phase	acp		X	X	-
# HB Tones	hbtones	0	X	-	-
HB Magnitude 1-7	hbmag1-7		X	-	-
HB Phase1-7	hbphase1-7	0	X	-	-
HB Harmonic1-7	hbharm1-7	1	X	-	-
HB Tone1-7	hbtone1-7	1-7	X	-	-
HB ModHarm1-7	hbmodharm1-7		X	-	-
HB ModTone1-7	hbmodtone1-7		X	-	-
Open DC Connection	dcopen		-	-	-
DC Resistance	rdc		X	-	-
AC Resistance	rac		X	-	-
HB Resistance	rhb		X	-	-
HBAC Resistance	rhbac		X	-	-
TRAN Resistance	rtran		X	-	-
HBAC Magnitude	hbacm		X	-	-
HBAC Phase	hbacp		X	-	-
SNAC Magnitude	snacm		X	-	-
SNAC Phase	snacp		X	-	-
Multiplier	m		X	X	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
PAC magnitude	pacm		-	X	-
PAC phase	pacp		-	X	-
Temp coefficient 1	tc1		-	X	-
Temp coefficient 2	tc2		-	X	-
XF magnitude	xfm	-	-	X	-
Number of noise/frequency pairs	FNpairs	0	-	X	-
Noise file name	noisefile		-	X	-
Noise file name is variable	noise_file_as_var	0	-	X	-
Frequency1-50	F1-50		-	X	-
Noise1-50	N1-50		-	X	-
Nominal temperature	tnom		-	X	-
Delay time	td			X	-
Impedance	z0		X	-	-

Note:

- When frequency/noise pairs have empty values or the number of frequency/noise pairs does not match with the FNpairs parameter, the Spectre netlister shows an error in the console.
- The default of noise_file_as_var parameter is *false*. If the noisefile parameter is set as *NOISEFILE*, the netlisting behavior for noisefile and noise_file_as_var parameters is as follows:

CDF Parameter	hspiceD Output	spectre Output
noise_file_as_var		
false (default)	noisefile='NOISEFILE'	noisefile="NOISEFILE"
true	noisefile=str(NOISEFILE)	noisefile=NOISEFILE

- For Hspice-format netlisting, a pair of single quotes is added to noisefile parameter when noise_file_as_var is *false*. When noise_file_as_var parameter is *true*, `str(NoiseFile)` is used as the value format.
- For Spectre-format netlisting, a pair of double quotes is added when noise_file_as_var parameter is *false*. No single/double quotes are added when noise_file_as_var parameter is *true*.

Example

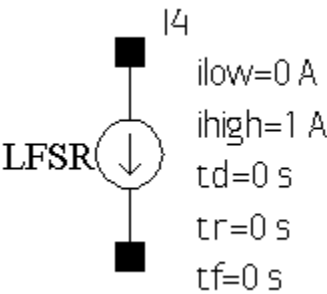
```
i4 n1 n2 dc=0 exp ( 0 50u 0 20n 0 30n )
```

See Also

- [Exponential Source Function](#) in the *HSPICE® User Guide: Elements*

lfsr

Pseudo Random Bit Generator Source (PRBS) function as an independent current source. The PRBS source uses a Linear Feedback Shift Register (LFSR) to generate a pseudo random bit sequence. HSPICE supports eight-bit data byte conversion to a 10-bit transmission character-8B/10B encoding for the LFSR keyword.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
DC Current	idc	0	X	X	-
Source Type	srcType	lfsr		X	-
Current Low	ilow	0	X	X	-
Current High	ihigh	1	X	X	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Delay Time	td	0	X	X	-
Rise Time	tr	0	X	X	-
Fall Time	tf	0	X	X	-
Bit Rate	rate	1M	X	-	-
Seed	seed	1	X	-	-
Taps	taps	[1,2]	X	-	-
Output Resistance	rout		X	-	-
AC Magnitude	acm		X	-	-
AC Phase	acp		X	-	-
# HB Tones	hbtones	0	X	-	-
HB Magnitude 1-7	hbmag1-7		X	-	-
HB Phase1-7	hbphase1-7	0	X	-	-
HB Harmonic1-7	hbharm1-7	1	X	-	-
HB Tone1-7	hbtone1-7	1-7	X	-	-
HB ModHarm1-7	hbmodharm1-7		X	-	-
HB ModTone1-7	hbmodtone1-7		X	-	-
Open DC Connection	dcopen		-	-	-
DC Resistance	rdc		X	-	-
AC Resistance	rac		X	-	-
HB Resistance	rhb		X	-	-
HBAC Resistance	rhbac		X	-	-
TRAN Resistance	rtran		X	-	-
HBAC Magnitude	hbacm		X	-	-
HBAC Phase	hbacp		X	-	-
SNAC Magnitude	snacm		X	-	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
SNAC Phase	snacp		X	-	-
PAM4 modulation	pam4_modulation	none	X	-	-
PAM4 mapping	pam4_mapping		X	-	-
Impedance	z0		X	-	-

Example

The following example shows the pattern source connected between nodes n1 and n2.

Example of LFSR, output is 1100011111001101:

```
i4 n1 n2 dc=0 lfsr ( 0 50u 0 1n 1n 10meg 3 [2,5] rout=10 )
```

Where:

- The output low voltage is 0, and the output high voltage is 1 v.
- The delay time is 0 ms.
- The rise and fall times are each 1 ns.
- The bit rate is 10 meg bits/s.
- The seed is 3 (bits: 00011).
- The taps are at the output of the 2nd and 5th shift registers and are xor'd together as the input to the first shift register.
- The output resistance is 10 ohms.

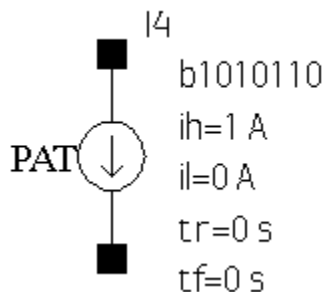
See Also

- [Pseudo Random Bit Generator Source](#) in the *HSPICE® User Guide: Elements*

ipat

Pattern source function. It uses four states, '1','0','m', and 'z', which represent the high, low, middle voltage, or current and high impedance state respectively. The series of these four states is a "b-string." HSPICE supports 8-bit data byte conversion to a 10-bit transmission character-8B/10B encoding, 64-bit data conversion to a 66-bit transmission

character-64B/66B encoding, 128-bit data byte conversion to a 130-bit or 132-bit transmission character-128B/130B or 128B/132B encoding, for the PAT sources.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
DC Voltage	vdc	0	-	-	-
Source Type	srcType	pat	-	-	-
Current High	ih	1	X	-	-
Current Low	il	0	X	-	-
Delay Time	td	0	X	-	-
Rise Time	tr	0	X	-	-
Fall Time	tf	0	X	-	-
Sample Time	tsample	1n	X	-	-
Data	data	b1010 110	X	-	-
Repeat	r		X	-	-
Repeat Bit	rb		X	-	-
AC Magnitude	acm		X	-	-
AC Phase	acp		X	-	-
# HB Tones	hbtones	0	X	-	-
HB Magnitude 1-7	hbmag1-7		X	-	-
HB Phase1-7	hbphase1-7	0	X	-	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
HB Harmonic1-7	hbharm1-7	1	X	-	-
HB Tone1-7	hbtone1-7	1-7	X	-	-
HB ModHarm1-7	hbmodharm1-7		X	-	-
HB ModTone1-7	hbmodtone1-7		X	-	-
Open DC Connection	dcopen		-	-	-
DC Resistance	rdc		X	-	-
AC Resistance	rac		X	-	-
HB Resistance	rhb		X	-	-
HBAC Resistance	rhbac		X	-	-
TRAN Resistance	rtran		X	-	-
HBAC Magnitude	hbacm		X	-	-
HBAC Phase	hbacp		X	-	-
SNAC Magnitude	snacm		X	-	-
SNAC Phase	snacp		X	-	-
PAM4 modulation	pam4_modulation	none	X	-	-
PAM4 mapping	pam4_mapping		X	-	-
Impedance	z0		X	-	-

Example

```
i4 n1 n2 pat ( 50u 10u 1n 1n 5n b1011 r=5 rb=2 )
```

In this example:

- High current is 50 uA
- Low current is 10 uA
- Time delay is 0 n
- Rise time is 1 n

- Fall time is 1 n
- Sample time is 5 n

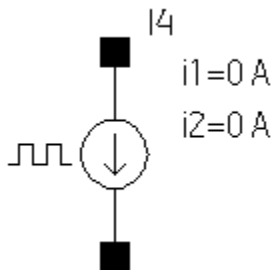
The first b-string is 1011, which repeats once and then repeats from the second bit, which is 0. The pattern is repeated 5 times.

See Also

- [Pattern Source](#) in the *HSPICE® User Guide: Elements*

ipulse

Trapezoidal pulse current source that starts with an initial delay from the beginning of the transient simulation interval to an onset ramp. During the onset ramp, the current changes linearly from its initial value to the pulse plateau value. After the pulse plateau, the current moves linearly along a recovery ramp back to its initial value. The entire pulse repeats, with a period named per, from onset to onset.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
DC Current	idc	0	X	X	-
Source Type	srcType	pulse	-	X	-
Current1	i1	0	X	X	-
Current2	i2	0	X	X	-
Delay Time	td	0	X	X	-
Rise Time	tr		X	X	-
Fall Time	tf		X	X	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Pulse Width	pw		X	X	-
Period	per		X	X	-
Period Jitter	perjitter		X	-	-
Seed	seed		X	-	-
AC Magnitude	acm		X	X	-
AC Phase	acp		X	X	-
# HB Tones	hbtones	0	X	-	-
HB Magnitude1-7	hbmag1-7		X	-	-
HB Phase1-7	hbphase1-7	0	X	-	-
HB Harmonic1-7	hbharm1-7	1	X	-	-
HB Tone1-7	hbtone1-7	1-7	X	-	-
HB ModHarm1-7	hbmodharm1-7		X	-	-
HB ModTone1-7	hbmodtone1-7		X	-	-
Open DC Connection	dcopen		X	-	-
DC Resistance	rdc		X	-	-
AC Resistance	rac		X	-	-
HB Resistance	rhb		X	-	-
HBAC Resistance	rhbac		X	-	-
TRAN Resistance	rtran		X	-	-
HBAC Magnitude	hbacm		X	-	-
HBAC Phase	hbacp		X	-	-
SNAC Magnitude	snacm		X	-	-
SNAC Phase	snacp		X	-	-
Multiplier	m		X	X	-
PAC magnitude	pacm		-	X	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
PAC phase	pacp		-	X	-
Edge type	edgetype	linear	-	X	-
Temp coefficient 1	tc1		-	X	-
Temp coefficient 2	tc2		-	X	-
XF magnitude	xfm		-	X	-
Number of noise/frequency pairs	FNpairs	0	-	X	-
Noise file name	noisefile		-	X	-
Noise file name is variable	noise_file_as_var	0	-	X	-
Frequency1-50	F1-50		-	X	-
Noise1-50	N1-50		-	X	-
Frequency name for fundamental frequency	fundname		-	X	-
Nominal temperature	tnom		-	X	-
Impedance	z0		X	-	-

Note:

- When frequency/noise pairs have empty values or the number of frequency/noise pairs does not match with the FNpairs parameter, the Spectre netlister shows an error in the console.
- The default of noise_file_as_var parameter is *false*. If the noisefile parameter is set as *NOISEFILE*, the netlisting behavior for noisefile and noise_file_as_var parameters is as follows:

CDF Parameter	hspiceD Output	spectre Output
noise_file_as_var		
false (default)	noisefile='NOISEFILE'	noisefile="NOISEFILE"
true	noisefile=str(NOISEFILE)	noisefile=NOISEFILE

- For Hspice-format netlisting, a pair of single quotes is added to noisefile parameter when noise_file_as_var is *false*. When noise_file_as_var parameter is *true*, `str(NOISEFILE)` is used as the value format.
- For Spectre-format netlisting, a pair of double quotes is added when noise_file_as_var parameter is *false*. No single/double quotes are added when noise_file_as_var parameter is *true*.

Example

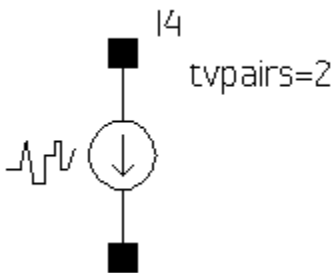
```
i4 plus gate dc=0 pulse ( 0 50u 0 1n 1n 10n 20n )
```

See Also

- [Trapezoidal Pulse Source](#) in the *HSPICE® User Guide: Elements*

ipwl

PWL independent current source.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
DC Current	idc	0	X	X	-
Source Type	srcType	pwl	-	X	-
Time Value Pairs	tvpairs	2	X	X	-
Display Pairs count	dpcount	5	X	-	-
Time1-50	t1-50		X	X	-
Current1-50	i1-50	0	X	X	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Repeat	r		X	-	-
Repeat Stop Time	rstop		X	-	-
Repeat Stop Value	stopvalue		X	-	-
Repeat Stop Slope	stopslope		X	-	-
Delay Time	td	0	X	X	-
AC Magnitude	acm		X	X	-
AC Phase	acp		X	X	-
# HB Tones	hbtones	0	X	-	-
HB Magnitude 1-7	hbmag1-7		X	-	-
HB Phase1-7	hbphase1-7	0	X	-	-
HB Harmonic1-7	hbharm1-7	1	X	-	-
HB Tone1-7	hbtone1-7	1-7	X	-	-
HB ModHarm1-7	hbmodharm1-7		X	-	-
HB ModTone1-7	hbmodtone1-7		X	-	-
Open DC Connection	dcopen		-	-	-
DC Resistance	rdc		X	-	-
AC Resistance	rac		X	-	-
HB Resistance	rhb		X	-	-
HBAC Resistance	rhbac		X	-	-
TRAN Resistance	rtran		X	-	-
HBAC Magnitude	hbacm		X	-	-
HBAC Phase	hbacp		X	-	-
SNAC Magnitude	snacm		X	-	-
SNAC Phase	snacp		X	-	-
Multiplier	m		X	X	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Delay Time	delay	0	-	X	-
Offset current	io		-	X	-
Amplitude scale factor	scale		-	X	-
Number of noise/frequency pairs	FNpairs	0	-	X	-
Noise file name	noisefile		-	X	-
Noise file name is variable	noise_file_as_var	0	-	X	-
Frequency1-50	F1-50		-	X	-
Noise1-50	N1-50		-	X	-
Frequency name for fundamental frequency	fundname		-	X	-
Nominal temperature	tnom		-	X	-
PAC magnitude	pacm		-	X	-
PAC phase	pacp		-	X	-
PWL period	pwlperiod		-	X	-
Edge type	edgetype	linear	-	X	-
Transition width	twidht		-	X	-
Time scaling factor	stretch		-	X	-
Temp coefficient 1	tc1		-	X	-
Temp coefficient 2	tc2		-	X	-
XF magnitude	xfm		-	X	-
Impedance	z0		X	-	-
PWL	pwl	cmdPWL Display("i pwl")	-	-	-

Note:

- When frequency/noise pairs have empty values or the number of frequency/noise pairs does not match with the FNpairs parameter, the Spectre netlister shows an error in the console.
- The default of noise_file_as_var parameter is *false*. If the noise file parameter is set as *NOISEFILE*, the netlisting behavior for noise file and noise_file_as_var parameters is as follows:

CDF Parameter	hspiceD Output	spectre Output
noise_file_as_var		
false (default)	noise file='NOISEFILE'	noise file="NOISEFILE"
true	noise file=str(NOISEFILE)	noise file=NOISEFILE

- For Hspice-format netlisting, a pair of single quotes is added to noise file parameter when noise_file_as_var is *false*. When noise_file_as_var parameter is *true*, str(NOISEFILE) is used as the value format.
- For Spectre-format netlisting, a pair of double quotes is added when noise_file_as_var parameter is *false*. No single/double quotes are added when noise_file_as_var parameter is *true*.

Example

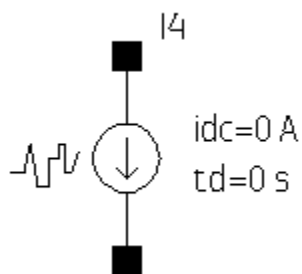
```
i4 n1 n2 dc=0 pwl ( 0 0.0 1n 0.5u 1.2n 0.5u 2n 1.2u )
```

See Also

- [PWL Source](#) in the *HSPICE® User Guide: Elements*

ipwlf

PWL source, that accepts a text file containing time and current pairs. The data in the text file can be either a table or a list.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
PWL File	fileName		X	X	
File name is variable	file_as_var	0	X	X	-
DC Current	idc	0	X	X	-
Source Type	srcType	pwl	-	X	-
Repeat	r		X	-	-
Repeat Stop Time	rstop		X	-	-
Repeat Stop Value	stopvalue		X	-	-
Repeat Stop Slope	stopslope		X	-	-
Delay Time	td	0	X	X	-
AC Magnitude	acm		X	X	-
AC Phase	acp		X	X	-
# HB Tones	hbtones	0	X	-	-
HB Magnitude 1-7	hbmag1-7		X	-	-
HB Phase1-7	hbphase1-7	0	X	-	-
HB Harmonic1-7	hbharm1-7	1	X	-	-
HB Tone1-7	hbtone1-7	1-7	X	-	-
HB ModHarm1-7	hbmodharm1-7		X	-	-
HB ModTone1-7	hbmodtone1-7		X	-	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Open DC Connection	dcopen		-	-	-
DC Resistance	rdc		X	-	-
AC Resistance	rac		X	-	-
HB Resistance	rhb		X	-	-
HBAC Resistance	rhbac		X	-	-
TRAN Resistance	rtran		X	-	-
HBAC Magnitude	hbacm		X	-	-
HBAC Phase	hbacp		X	-	-
SNAC Magnitude	snacm		X	-	-
SNAC Phase	snacp		X	-	-
Multiplier	m		X	X	-
Offset current	io		-	X	
Amplitude scale factor	scale		-	X	-
Number of noise/frequency pairs	FNpairs	0	-	X	-
Noise file name	noisefile		-	X	-
Noise file name is variable	noise_file_as_var	0	-	X	-
Frequency1-50	F1-50		-	X	-
Noise1-50	N1-50		-	X	-
Frequency name for fundamental frequency	fundname		-	X	-
Nominal temperature	tnom		-	X	-
PAC magnitude	pacm		-	X	-
PAC phase	pacp		-	X	-
PWL period	pwlperiod		-	X	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Edge type	edgetype	linear	-	X	-
Transition width	twidth		-	X	-
Desired rms value	rms		-	x	-
Time scaling factor	stretch		-	X	-
Temp coefficient 1	tc1		-	X	-
Temp coefficient 2	tc2		-	X	-
XF magnitude	xfm		-	X	-
Impedance	z0		X	-	-

Note:

- When frequency/noise pairs have empty values or the number of frequency/noise pairs does not match with the FNpairs parameter, the Spectre netlister shows an error in the console.
- The default of noise_file_as_var parameter is *false*. If the noise file parameter is set as *NOISEFILE*, the netlisting behavior for noise file and noise_file_as_var parameters is as follows:

CDF Parameter	hspiceD Output	spectre Output
noise_file_as_var		
false (default)	noisefile='NOISEFILE'	noisefile="NOISEFILE"
true	noisefile=str(NOISEFILE)	noisefile=NOISEFILE

- For Hspice-format netlisting, a pair of single quotes is added to noise file parameter when noise_file_as_var is *false*. When noise_file_as_var parameter is *true*, `str(NOISEFILE)` is used as the value format.
- For Spectre-format netlisting, a pair of double quotes is added when noise_file_as_var parameter is *false*. No single/double quotes are added when noise_file_as_var parameter is *true*.

Example

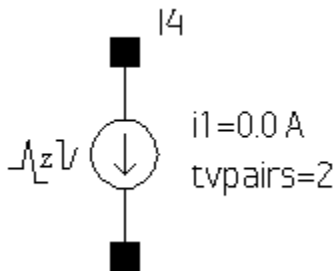
```
i4 n1 n2 dc=0 pw1 ( 0 0.0 1n 0.5u 1.2n 0.5u 2n 1.2u )
```

See Also

- [PWL Source](#) in the *HSPICE® User Guide: Elements*

ipwlz

The high Z state of the PWLZ source adds the capability to disconnect the PWL source for time periods marked with the keyword Z.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
DC Current	idc	0	X	X	-
Source Type	srcType	pwlz	-	X	-
Time Value Pairs	tvpairs		X	X	-
Time1-50	t1-50		X	X	-
Current1-50	i1-50	0	X	X	-
Z State1-50	Z1-50	0	X	X	-
Repeat	r		X	-	-
Delay Time	td	0	X	X	-
AC Magnitude	acm		X	X	-
AC Phase	acp		X	X	-
# HB Tones	hbtones	0	X	-	-
HB Magnitude 1-7	hbmag1-7		X	-	-
HB Phase1-7	hbphase1-7	0	X	-	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
HB Harmonic1-7	hbharm1-7	1	X	-	-
HB Tone1-7	hbtone1-7	1-7	X	-	-
HB ModHarm1-7	hbmodharm1-7		X	-	-
HB ModTone1-7	hbmodtone1-7		X	-	-
Open DC Connection	dcopen		-	-	-
DC Resistance	rdc		X	-	-
AC Resistance	rac		X	-	-
HB Resistance	rhb		X	-	-
HBAC Resistance	rhbac		X	-	-
TRAN Resistance	rtran		X	-	-
HBAC Magnitude	hbacm		X	-	-
HBAC Phase	hbacp		X	-	-
SNAC Magnitude	snacm		X	-	-
SNAC Phase	snacp		X	-	-
Multiplier	m		X	X	-
Delay Time	delay	0	-	X	-
Offset current	io		-	X	-
Amplitude scale factor	scale		-	X	-
Number of noise/frequency pairs	FNpairs	0	-	X	-
Noise file name	noisefile		-	X	-
Noise file name is variable	noise_file_as_var	0	-	X	-
Frequency1-50	F1-50		-	X	-
Noise1-50	N1-50		-	X	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Frequency name for fundamental frequency	fundname		-	X	-
Nominal temperature	tnom		-	X	-
PAC magnitude	pacm		-	X	-
PAC phase	pacp		-	X	-
PWL period	pwlperiod		-	X	-
Edge type	edgetype	linear	-	X	-
Transition width	twidth		-	X	-
Time scaling factor	stretch		-	X	-
Temp coefficient 1	tc1		-	X	-
Temp coefficient 2	tc2		-	X	-
XF magnitude	xfm		-	X	-
Impedance	z0		X	-	-

Note:

- When frequency/noise pairs have empty values or the number of frequency/noise pairs does not match with the FNpairs parameter, the Spectre netlister shows an error in the console.
- The default of noise_file_as_var parameter is *false*. If the noise file parameter is set as *NOISEFILE*, the netlisting behavior for noise file and noise_file_as_var parameters is as follows:

CDF Parameter	hspiceD Output	spectre Output
noise_file_as_var		
false (default)	noisefile='NOISEFILE'	noisefile="NOISEFILE"
true	noisefile=str(NOISEFILE)	noisefile=NOISEFILE

- For Hspice-format netlisting, a pair of single quotes is added to noisefile parameter when noise_file_as_var is *false*. When noise_file_as_var parameter is *true*, `str(NoiseFile)` is used as the value format.
- For Spectre-format netlisting, a pair of double quotes is added when noise_file_as_var parameter is *false*. No single/double quotes are added when noise_file_as_var parameter is *true*.

Example

```
i4 n1 n2 dc=0 pwlz ( 0 0.0 1n z 1.2n 0.5u 2n 1.2u )
```

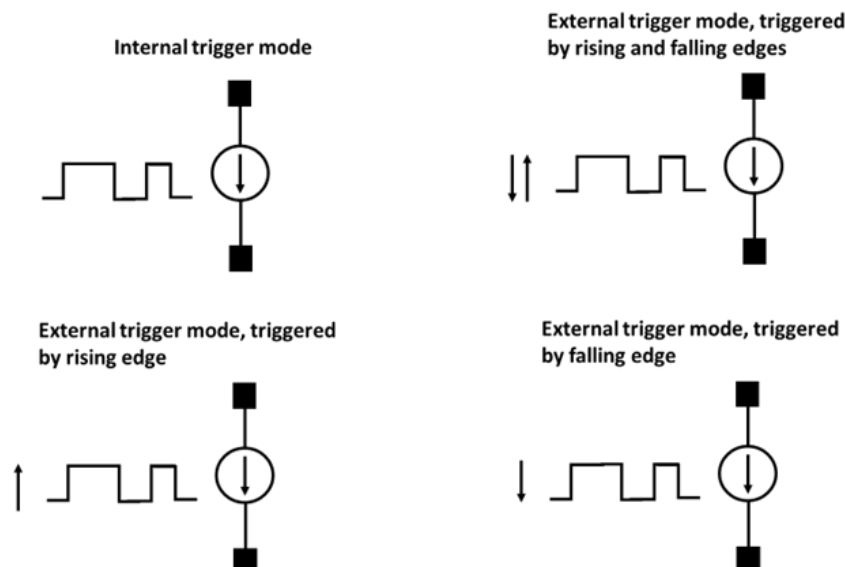
See Also

- [High Impedance State PWL \(PWLZ\)](#) in the *HSPICE® User Guide: Elements*

iprbs

Current source with pseudo random bit generator source transient function source.

The iprbs device is a PCell. The following figure shows four vprbs devices that have different trigger modes, which are Internal (on the upper-left), External by both edges (on the upper-right), External by falling edges (on the lower-left) and External by rising edges (on the lower-right). The internal trigger mode has 2 pins (PLUS and MINUS) and other three external trigger modes have 4 pins (P1, M1, P2, and M2). The netlisting procedure writes out their pins correspondingly.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
triggerValue	triggerValue	1	-	X	-
Delay time	tdPrbs		-	X	-
Zero value	val0		-	X	-
One value	val1		-	X	-
Bit period	perPrbs	1n	-	X	-
Rise time	tr		-	X	-
Fall time	tf		-	X	-
Transition reference	transitionreference		-	X	-
Lower threshold(percent)	lowerthreshold		-	X	-
Upper threshold(percent)	upperthreshold		-	X	-
Edge type	edgetype		-	X	-
Trigger	trigger	Internal	-	X	-
Threshold	triggerthreshold		-	X	-
LFSR Mode	lfsrmode	PN32	-	X	-
Browse and specify bit file	prbsFileBox	0	-	X	-
Bit data file	prbsFile		-	X	-
Seed	lfsrseed	1 3 5 7 9 11 13 15 17 19 21 23 25 27 29 31	-	X	-
Taps	lfsrtaps		-	X	-
RJ(rms)	rjrms		-	X	-
RJ(seed)	rjseed		-	X	-

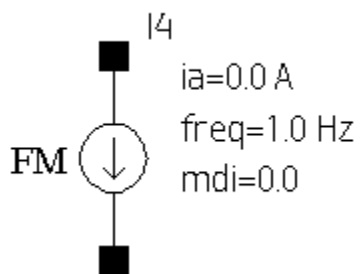
CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Number of periodic jitters	nPeriodicJitter	0	-	X	-
PJ1(amplitude)	pjump1		-	X	-
PJ1(frequency)	pjfreq1		-	X	-
PJ1(type)	pjtype1	sine	-	X	-
PJ2(amplitude)	pjump2		-	X	-
PJ2(frequency)	pjfreq2		-	X	-
PJ2(type)	pjtype2	sine	-	X	-
PJ3(amplitude)	pjump3		-	X	-
PJ3(frequency)	pjfreq3		-	X	-
PJ3(type)	pjtype3	sine	-	X	-
Pattern Parameter data	data		-	X	-
Reference Value	ref		-	X	-
Random delay time	jitter		-	X	-
Seed	seed		-	X	-
Feedback Shift Register	taps		-	X	-
Source type	srcType	prbs	-	X	-
Version	int	1	-	-	-

Controlling Version Number for Legacy Designs

To control the version number for legacy designs, create and use the [dbAnalogLibIprbsVersion](#) Tcl preference before the analogLib library is loaded.

isffm

Single-frequency FM current source.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
DC Current	idc	0	X	-	-
Source Type	srcType	sffm	-	-	-
Output Amplitude	ia	0	X	-	-
Offset Current	io	0	X	-	-
Carrier Frequency	freq	1	X	-	-
Modulation Index	mdi	0	X	-	-
Signal Frequency	fs		X	-	-
AC Magnitude	acm		X	-	-
AC Phase	acp		X	-	-
# HB Tones	hbtones	0	X	-	-
HB Magnitude 1-7	hbmag1-7		X	-	-
HB Phase1-7	hbphase1-7	0	X	-	-
HB Harmonic1-7	hbharm1-7	1	X	-	-
HB Tone1-7	hbtone1-7	1-7	X	-	-
HB ModHarm1-7	hbmodharm1-7		X	-	-
HB ModTone1-7	hbmodtone1-7		X	-	-
Open DC Connection	dcopen		-	-	-
DC Resistance	rdc		X	-	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
AC Resistance	rac		X	-	-
HB Resistance	rhb		X	-	-
HBAC Resistance	rhbac		X	-	-
TRAN Resistance	rtran		X	-	-
HBAC Magnitude	hbacm		X	-	-
HBAC Phase	hbacp		X	-	-
SNAC Magnitude	snacm		X	-	-
SNAC Phase	snacp		X	-	-
Multiplier	m		X	-	-
Impedance	z0		X	-	-

Example

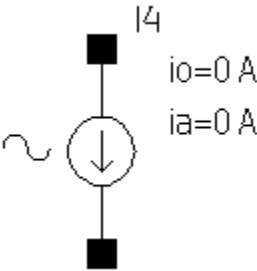
```
i4 n1 gnd! dc=25u sffm ( 0.0 30u 20k 10 5k )
```

See Also

- [Single-frequency FM Source](#) in the *HSPICE® User Guide: Elements*

isin

Sinusoidal current source that can be either a constant sinusoid or a damped, that produces a dying exponential with a sine wave.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
DC Current	idc	0	X	X	-
Source Type	srcType	pwlz	-	X	-
Current Offset	io		X	X	-
Current Amplitude	ia		X	X	-
Frequency	freq		X	X	-
Delay Time	td		X	X	-
Damping Factor	theta		X	X	-
Phase for sinusoid	sinephase		X	X	-
AC Magnitude	acm		X	X	-
AC Phase	acp		X	X	-
# HB Tones	hbtones	0	X	-	-
HB Magnitude 1-7	hbmag1-7		X	-	-
HB Phase1-7	hbphase1-7	0	X	-	-
HB Harmonic1-7	hbharm1-7	1	X	-	-
HB Tone1-7	hbtone1-7	1-7	X	-	-
HB ModHarm1-7	hbmodharm1-7		X	-	-
HB ModTone1-7	hbmodtone1-7		X	-	-
Open DC Connection	dcopen		-	-	-
DC Resistance	rdc		X	-	-
AC Resistance	rac		X	-	-
HB Resistance	rhb		X	-	-
HBAC Resistance	rhbac		X	-	-
TRAN Resistance	rtran		X	-	-
HBAC Magnitude	hbacm		X	-	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
HBAC Phase	hbacp		X	-	-
SNAC Magnitude	snacm		X	-	-
SNAC Phase	snacp		X	-	-
Multiplier	m		X	X	-
Phase shift	phase		-	X	
Amplitude 2	ia2		-	X	-
Amp modulation frequency	ammodfreq		-	X	-
Amp modulation index	ammodindex		-	X	-
Amp modulation phase	ammodphase		-	X	-
Number of FM Files	filenums		-	X	-
Name of FM File1	fmmodfile1		-	X	-
FM file1 name is variable	fmmodfile1_as_0 _var		-	X	-
Name of FM file 2	fmmodfile2		-	X	-
FM file2 name is variable	fmmodfile2_as_0 _var		-	X	-
FM frequency	fmmodfreq		-	X	-
FM index	fmmodindex		-	X	-
Frequency 2	freq2		-	X	-
Number of noise/frequency pairs	FNpairs	0	-	X	-
Noise file name	noisefile		-	X	-
Noise file name is variable	noise_file_as_0 var		-	X	-
Frequency1-50	F1-50		-	X	-
Noise1-50	N1-50		-	X	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Frequency name for fundamental frequency	fundname	-	-	X	-
Second frequency name	fundname2	-	-	X	-
Noise temperature	noisetemp	-	-	X	-
Nominal temperature	tnom	-	-	X	-
PAC magnitude	pacm	-	-	X	-
PAC phase	pacp	-	-	X	-
Phase for sinusoid 2	sinephase2	-	-	X	-
Temp coefficient 1	tc1	-	-	X	-
Temp coefficient 2	tc2	-	-	X	-
XF magnitude	xfm	-	-	X	-
Impedance	z0	X	-	-	-

Note:

- When frequency/noise pairs have empty values or the number of frequency/noise pairs does not match with the FNpairs parameter, the Spectre netlister shows an error in the console.
- The default of noise_file_as_var parameter is *false*. If the noise file parameter is set as *NOISEFILE*, the netlisting behavior for noise file and noise_file_as_var parameters is as follows:

CDF Parameter	hspiceD Output	spectre Output
noise_file_as_var		
false (default)	noise file='NOISEFILE'	noise file="NOISEFILE"
true	noise file=str(NOISEFILE)	noise file=NOISEFILE

- For Hspice-format netlisting, a pair of single quotes is added to noisefile parameter when noise_file_as_var is *false*. When noise_file_as_var parameter is *true*, `str(NoiseFile)` is used as the value format.
- For Spectre-format netlisting, a pair of double quotes is added when noise_file_as_var parameter is *false*. No single/double quotes are added when noise_file_as_var parameter is *true*.

Example

```
i4 n1 0 dc=0 sin ( 0 100u 100Meg 1n 1e+6 0 )
```

This damped sinusoidal source connects between nodes n1 and 0. In this waveform:

- Peak value is 100 uA
- Offset is 0 A
- Frequency is 100 MHz
- Time delay is 1 ns
- Damping factor is 1e+6
- Phase delay is 0 deg

See Also

- [Sinusoidal Source Function](#) in the *HSPICE® User Guide: Elements*

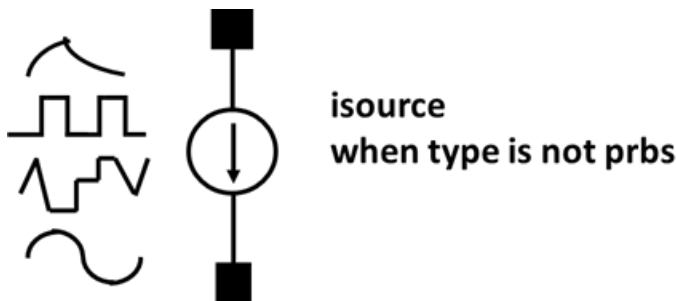
Controlling Version Number for Legacy Designs

To control the version number for legacy designs, create and use the `dbAnalogLibIsinVersion` Tcl preference before the analogLib library is loaded.

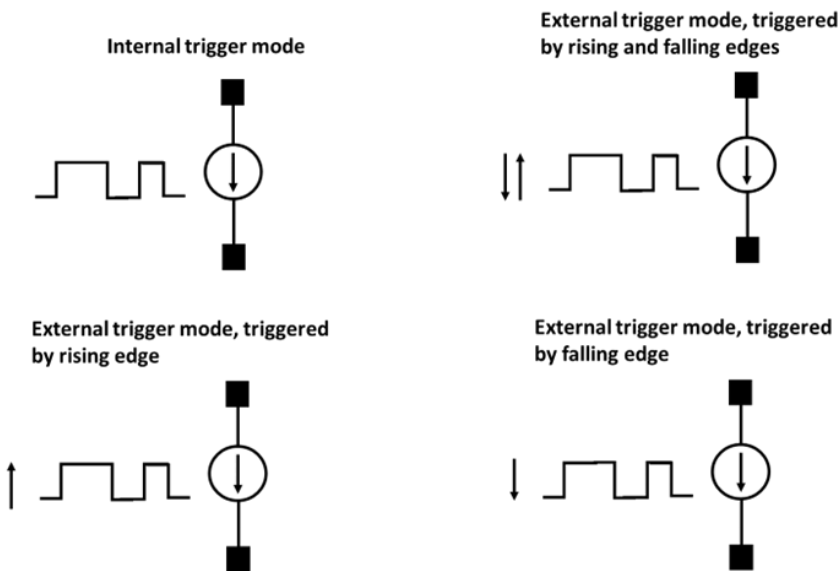
isource

Current source with various parameters settings.

The isource device is a PCell. Its PCell behavior is controlled by its source type. For source type other than prbs, it is a 2-pin device and its symbol looks as follows:



For prbs source type, its PCell behavior is affected by the trigger parameter. The following figure shows four isource devices that have different trigger modes, which are Internal (on the upper-left), External by rising edges (on the upper-right), External by falling edges (on the lower-left) and External by both edges (on the lower-right). The internal trigger mode has 2 pins (PLUS and MINUS) and other three external trigger modes have 4 pins (P1, M1, P2, and M2). The netlisting procedure writes out their pins correspondingly.



The bit source type is mainly supported for Spectre netlisting.

CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
DC Current	idc	0	X	X	-
Source Type	srcType	dc	X	X	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Frequency name 1	fundname		-	X	-
Waveform entry method	pwlEntryMethod	File	X	X	-
File name	fileName		X	X	-
File name is variable	file_as_var	0	X	X	-
Time Value Pairs	tvpairs	0	X	X	-
Time1	t1		X	X	-
Current1	i1		X	X	-
Time2	t2		X	X	-
Current2	i2		X	X	-
Zero value	val0		X	X	-
One value	val1		X	X	-
Rise Delay Time	td1	0	X	X	-
Fall Delay Time	td2	0	X	X	-
Rise Time Constant	tau1	0	X	X	-
Fall Time Constant	tau2	0	X	X	-
Exp period	expperiod		-	x	-
Rise Time	tr		X	X	-
Fall Time	tf		X	X	-
Pulse Width	pw		X	X	-
Period	per		X	X	-
Time3-50	t3-50		-	X	-
Current3-50	i3-50		-	X	-
Repeat	r		X	-	-
Repeat Stop Time	rstop		X	-	-
Repeat Stop Value	stopvalue		X	-	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Repeat Stop Slope	stopslope		X	-	-
Current Offset	io		X	-	-
Frequency	freq		X	X	-
Current Amplitude	ia		X	X	-
Phase for sinusoid	sinephase	0	X	X	-
Sine DC level	sinedc		-	X	-
Delay Time	td	0	X	X	-
Damping Factor	theta	0	X	X	-
# HB Tones	hbtones	0	X	-	-
HB Magnitude 1-7	hbmag1-7		X	-	-
HB Phase1-7	hbphase1-7	0	X	-	-
HB Harmonic1-7	hbharm1-7	1	X	-	-
HB Tone1-7	hbtone1-7	1-7	X	-	-
HB ModHarm1-7	hbmodharm1-7		X	-	-
HB ModTone1-7	hbmodtone1-7		X	-	-
Power Switch	power	V	X	-	-
Impedance	z0		X	-	-
Open DC Connection	dcopen		-	-	-
Display second sinusoid	numofsines	0	-	X	-
Frequency name 2	fundname2		-	X	-
Frequency 2	freq2		-	X	-
Amplitude 2 (lpk)	ia2		-	X	-
Phase for sinusoid 2	sinephase2		-	X	-
Display multi sinusoid	vecofsines	0	-	X	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Number of frequencies	freqvec	0	X	X	-
Sinusoid frequency 1-9	freqv1-9		-	X	
Sinusoid amplitude 1-9	vav1-9		-	-	-
Sinusoid phase 1-9	phasev1-9		-	X	-
Sinusoid max harmonic 1-9	harmv1-9		-	X	-
Display modulation params	modulation	0	-	X	-
Amp modulation frequency 1	ammodfreq		-	X	-
Amp modulation index 1	ammodindex		-	X	-
Amp modulation phase 1	ammodphase		-	X	-
Number of FM Files	filenums	none	-	X	-
Name of FM File1	fmmodfile1		-	X	-
FM file1 name is variable	fmmodfile1_as_var	0	-	X	-
Name of FM file 2	fmmodfile2		-	X	-
FM file2 name is variable	fmmodfile2_as_var	0	-	X	-
FM index 1	fmmodindex		-	X	-
FM frequency 1	fmmodfreq		-	X	-
IQ modulation I File	imodfile		-	X	-
I File name is variable	imodfile_as_var	0	-	X	-
IQ modulation Q File	qmodfile		-	X	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Q File name is variable	qmodfile_as_var	0	-	X	-
DC offset	offset		-	X	-
Pattern Parameter data	data		-	X	-
Pattern Parameter data is variable	data_as_var		-	X	-
Pattern Parameter rptstart	rptstart	1	-	X	-
Pattern Parameter rpttimes	rpttimes	0	-	X	-
Amplitude scale factor	scale		-	X	-
Desired rms value	rms		-	x	-
Time scaling factor	stretch		-	X	-
Filter for pwl source	pwlfilter		-	X	-
Rolloff factor	rolloff		-	X	-
Bandwidth of the pwl signal	pwlbandwidth		-	X	-
PWL period	pwlperiod		-	X	-
Edge type	edgetype		X	X	-
RMS Jitter	perjitter		X	-	-
Period Jitter	perjitter		X	-	-
Width of transition	twidh		-	X	-
Breakpoints	allbrkpts	no	-	X	-
Period start time	pwlperiodstart		-	X	-
Delay time	tdPrbs		-	X	-
Bit period	perPrbs		-	X	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Transition reference	transitionreference		-	X	-
Lower threshold(percent)	lowerthreshold		-	X	-
Upper threshold(percent)	upperthreshold		-	X	-
Edge type	edgetypePrbs		-	X	-
Rise delay	td01		-	X	-
Fall delay	td10		-	X	-
PAM4 modulation	pam4_modulation	none	-	X	-
PAM4 mapping	pam4_mapping		-	X	-
triggerValue	triggerValue	1	-	X	-
Trigger	trigger	Internal	-	X	-
Threshold	triggerthreshold		-	X	-
LFSR Mode	lfsrmode		-	X	-
Browse and specify bit file	prbsFileBox	0	-	-	-
Bit data file	prbsFile		-	X	-
Bit data file name is variable	prbsFile_as_var	0	-	X	-
Seed	lfsrseed	1 3 5 7 9 11 13 15 17 19 21 23 25 27 29 31	-	X	-
Taps	lfsrtaps		-	X	-
RJ(rms)	rjrms		-	X	-
RJ(seed)	rjseed		-	X	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Number of periodic jitters	nPeriodicJitter	0	-	X	-
PJ1(amplitude)	pjump1		-	X	-
PJ1(frequency)	pjfreq1		-	X	-
PJ1(type)	pjtype1	sine	-	X	-
PJ2(amplitude)	pjump2		-	X	-
PJ2(frequency)	pjfreq2		-	X	-
PJ2(type)	pjtype2	sine	-	X	-
PJ3(amplitude)	pjump3		-	X	-
PJ3(frequency)	pjfreq3		-	X	-
PJ3(type)	pjtype3	sine	-	X	-
Display small signal params	smallSig	0	X	X	-
AC Magnitude	acm		X	X	-
AC Phase	acp		X	X	-
PAC magnitude	pacm		-	X	-
PAC phase	pacp		-	X	-
XF magnitude	xfm		-	X	-
HBAC Magnitude	hbacm		X	-	-
HBAC Phase	hbacp		X	-	-
SNAC Magnitude	snacm		X	-	-
SNAC Phase	snacp		X	-	-
Display resistance parameters	resistanceParam	0	X	-	-
DC Resistance	rdc		X	-	-
TRAN Resistance	rtran		X	-	-
AC Resistance	rac		X	-	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
HB Resistance	rhb		X	-	-
HBAC Resistance	rhbac		X	-	-
Display temperature params	tempParam	0	-	X	-
Nominal temperature	tnom		-	X	-
Linear temperature coefficient	tc1		-	X	-
Quadratic temperature coefficient	tc2		-	X	-
Display noise parameters	noiseParam	0	-	X	-
Generate noise?	isnoisyNoiseModel		-	X	-
Noise entry method	noiseEntryMethod	File	-	X	-
Noise file name	noisefile		-	X	-
Noise file name is variable	noise_file_as_var	0	-	X	-
Noise type	noiseModelType	elsource	-	X	-
Interpolation method	noiseModelInterMethod	source	-	X	-
Number of noise/frequency pairs	FNpairs	0	-	X	-
Frequency1-50	F1-50		-	X	-
Noise1-50	N1-50		-	X	-
Noise temperature	noisetemp		-	X	-
This dummy parameter exists only for using its default value	paramForDefault	evalDefaultZeroOrEmpty("")	-	-	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Multiplier	m		-	X	-
Version	version	3	-	-	-

Note:

- When frequency/noise pairs have empty values or the number of frequency/noise pairs does not match with the FNpairs parameter, the Spectre netlister shows an error in the console.
- The default of noise_file_as_var parameter is *false*. If the noiseFile parameter is set as *NOISEFILE*, the netlisting behavior for noiseFile and noise_file_as_var parameters is as follows:

CDF Parameter	hspiceD Output	spectre Output
noise_file_as_var		
false (default)	noiseFile='NOISEFILE'	noiseFile="NOISEFILE"
true	noiseFile=str(NOISEFILE)	noiseFile=NOISEFILE

- For Hspice-format netlisting, a pair of single quotes is added to noiseFile parameter when noise_file_as_var is *false*. When noise_file_as_var parameter is *true*, str(NOISEFILE) is used as the value format.
- For Spectre-format netlisting, a pair of double quotes is added when noise_file_as_var parameter is *false*. No single/double quotes are added when noise_file_as_var parameter is *true*.
- In Spectre-format netlisting, you can write out frequency/noise pairs (FNpairs) or noiseFile parameters even they are hidden by noiseParam. To skip frequency/noise pairs (FNpairs) or noiseFile parameters for legacy designs, create and use the dbAnalogLibIsSourceVersion Tcl preference.

Example

```
i4 n1 n2 dc=10u power=0
i4 n1 n2 dc=0 exp ( 0 100u 0 20n 0 30n )
i4 n1 n2 dc=10u pulse ( 0 100u 1n 1n 1n 10n 20n )
```

This damped sinusoidal source connects between nodes n1 and 0. In this waveform:

- Peak value is 100 uA
- Offset is 0 A
- Frequency is 100 MHz
- Time delay is 1 ns
- Damping factor is 1e+6
- Phase delay is 0 deg

See Also

- [Independent Sources and Stimuli](#) in the *PrimeSim Continuum User Guide: Elements*

Controlling Version Number for Legacy Designs

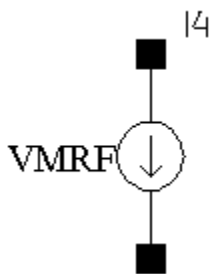
To control the version number for legacy designs, create and use the [dbAnalogLibIsourceVersion](#) Tcl preference before the analogLib library is loaded.

Using isource to Set up Harmonic Balance or Shooting PSS Simulations

isource can be used as stimulus for one-, two-, and N-tone Harmonic Balance and Shooting Newton simulations. Its use is analogous to vsource and similar to port. Refer to [vsource](#) and [port](#) documentation for details of isource setup.

ivmrf

Vector Modulated RF current source. It allows digital modulation of an RF carrier using in-phase and quadrature components created from a binary data stream.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
DC Current	idc	0	X	-	-
Source Type	srcType	dc	-	-	-
Current Amplitude	amp		X	-	-
Carrier Frequency	freq		X	-	-
Carrier Phase	phase		X	-	-
Modulation Method	mod		X	-	-
Filter Method	filter		X	-	-
Filter Parameter	filcoef		X	-	-
Bit Rate	rate		X	-	-
Bitstream	bitstream		X	-	-
AC Magnitude	acm		X	-	-
AC Phase	acp		X	-	-
# HB Tones	hbtones	0	X	-	-
HB Magnitude 1-7	hbmag1-7		X	-	-
HB Phase1-7	hbphase1-7	0	X	-	-
HB Harmonic1-7	hbharm1-7	1	X	-	-
HB Tone1-7	hbtone1-7	1-7	X	-	-
HB ModHarm1-7	hbmodharm1-7		X	-	-
HB ModTone1-7	hbmodtone1-7		X	-	-
Open DC Connection	dcopen		-	-	-
DC Resistance	rdc		X	-	-
AC Resistance	rac		X	-	-
HB Resistance	rhb		X	-	-
HBAC Resistance	rhbac		X	-	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
TRAN Resistance	rtran		X	-	-
HBAC Magnitude	hbacm		X	-	-
HBAC Phase	hbacp		X	-	-
SNAC Magnitude	snacm		X	-	-
SNAC Phase	snacp		X	-	-
Multiplier	m		X	-	-
Impedance	z0		X	-	-

Example

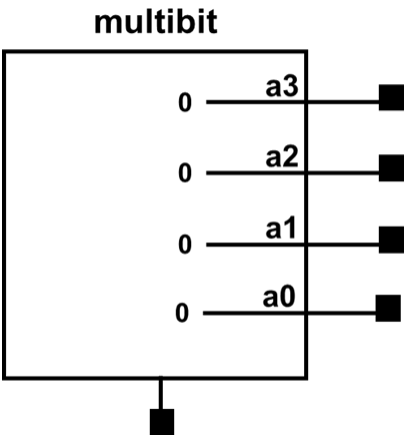
```
i55 n1 n2 dc=0 vmrf amp=100u freq=10k phase=0 rate=100Meg
+ mod=bpsk filter=rect bitstream=01010010011100b
```

See Also

- [Vector-modulated RF \(VMRF\) Source](#) in the *HSPICE® User Guide: Elements*

multibit

Generate a subcircuit of vdc sources, set the dc value according to the pattern specified.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Input Mode	mode1	Binary	X	X	-
Number of Bits	numbits	4	X	X	-
Expand Bus	expand	Yes	X	X	-
Bit Pattern(MSB...LSB)	pattern	0000	X	X	-
Decimal	dec	0	X	X	-
Bit 1 voltage level	vbit1	1	X	X	-
Bit 0 voltage level	vbit0	0	X	X	-

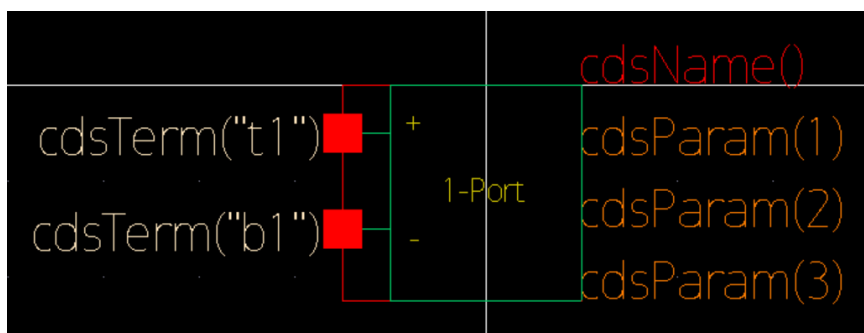
Examples

```
subckt multibit_1 ( a0 a1 a2 a3 ref)
parameters a0=fmod(11, 2) a1=fmod(5, 2) a2=fmod(2, 2) a3=fmod(1, 2)
vbit0=0 \
vbit1=1
V3 (a3 ref) vsource type=dc dc=( (a3>0) ? vbit1 : vbit0 )
V2 (a2 ref) vsource type=dc dc=( (a2>0) ? vbit1 : vbit0 )
V1 (a1 ref) vsource type=dc dc=( (a1>0) ? vbit1 : vbit0 )
V0 (a0 ref) vsource type=dc dc=( (a0>0) ? vbit1 : vbit0 )
ends multibit_1

I125 (net296 net295 net294 net293 net297) multibit_1 vbit1=1 vbit0=0
a0=fmod(int((11/1)), 2) \
a1=fmod(int((11/2)), 2) a2=fmod(int((11/4)), 2) a3=fmod(int((11/8)), 2)
```

n1port

The n1port element is a special case of the generic nport element with specific port number.



Note:

The **nport** element has certain behavior which depends on S, Y or Z-parameter files.

CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Browse and select s-data file	nportFileB	0	X	X	-
Model Type	modelFileType	TSTONE FILE	X	-	-
S-parameter data file	dataFile		X	X	-
Interpolation method	interp	linear	-	X	-
Tran convolution parameters	tranParaLabel	1	-	X	-
Accuracy	accuracyMode	default	-	X	-
Advanced transient parameters	tranAdvanPara Label	0	X	X	-
Max sampling points	maxn		-	X	-
Max frequency of interest	fmax		X	X	-
Impulse response truncation	imptrunc		-	X	-
ROM data file	romdatfile		-	X	-
ROM data file name is variable	romdatfile_as_var	0	-	X	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Rational order	ratorder		-	X	-
Relative error	relerr		-	X	-
Absolute error	abserr		-	X	-
Noise parameters	noiseParaLabel	0	X	X	-
Thermal noise	thermalnoise	Yes	X	X	-
Thermal noise model	noisemodel			X	-
Rarely used parameters	otherParaLabel	0	X	X	-
S-parameter data format	datafmt		-	-	-
Frequency sampling interval	fdelta		X	X	-
Use smooth data windowing	usewindow		-	X	-
Multiplier	m		X	X	-
Scale factor	scale		-	X	-
Model name	hmname	MDN	X	-	-
Enable mixed mode	mixedmode	0	X	-	-
The order of indices	datatype		X	-	-
Characteristic impedance	z0	50	X	-	-
Hspice S-parameter data format	datafmtHspice	touchstone	X	X	-
Enable passive checker	passive	0	X	-	-
Hspice Interpolation method	interpolation		X	-	-
Delay frequency	delayfreq		X	-	-

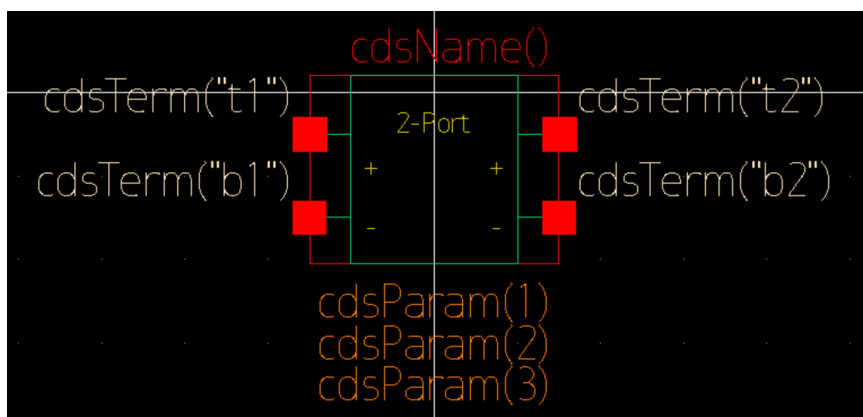
CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Extracts a system delay	delayhandle	0	X	-	-
Temperature difference	dtemp		X	-	-
High freq extrapolate method	highpass	3	X	-	-
Linear interpolation data type	intdattyp	MA	X	-	-
Low freq extrapolate method	lowpass	1	X	-	-
Enable noise passive checker	noipassivechk	1	X	-	-
Precondition factor keyword	prefac	0.75	X	-	-
Enable rational function	rational_func	0	X	-	-
Reuse rational function data	rational_func_reuse	1	X	-	-
Method of smooth	smooth	0	X	-	-
Width of the smoothing window	smoothpts		X	-	-
Stamping method	stamp		X	-	-

Controlling Version Number for Legacy Designs

To control the version number for legacy designs, create and use the [dbAnalogLibNxPortVersion](#) Tcl preference before the analogLib library is loaded.

n2port

The n2port element is a special case of the generic nport element with specific port number.



Note:

The [nport](#) element has certain behavior which depends on S, Y or Z-parameter files.

CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Browse and select s-data file	nportFileB	0	X	X	-
Model Type	modelFileType	TSTONE FILE	X	-	-
S-parameter data file	dataFile		X	X	-
Interpolation method	interp	linear	-	X	-
Tran convolution parameters	tranParaLabel	1	-	X	-
Accuracy	accuracyMode	default	-	X	-
Advanced transient parameters	tranAdvanPara Label	0	X	X	-
Max sampling points	maxn		-	X	-
Max frequency of interest	fmax		X	X	-
Impulse response truncation	imptrunc		-	X	-
ROM data file	romdatfile		-	X	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
ROM data file name is variable	romdatfile_as_var	0	-	X	-
Rational order	ratorder		-	X	-
Relative error	relerr		-	X	-
Absolute error	abserr		-	X	-
Noise parameters	noiseParaLabel	0	X	X	-
Thermal noise	thermalnoise	Yes	X	X	-
Thermal noise model	noisemodel		-	X	-
Rarely used parameters	otherParaLabel	0	X	X	-
S-parameter data format	datafmt		-	-	-
Frequency sampling interval	fdelta		X	X	-
Use smooth data windowing	usewindow		-	X	-
Multiplier	m		X	X	-
Scale factor	scale			X	-
Model name	hmname	MDN	X	-	-
Enable mixed mode	mixedmode	0	X	-	-
The order of indices	datatype		X	-	-
Characteristic impedance	z0	50	X	-	-
Hspice S-parameter data format	datafmtHspice	touchstone	X	X	-
Enable passive checker	passive	0	X	-	-
Hspice Interpolation method	interpolation		X	-	-

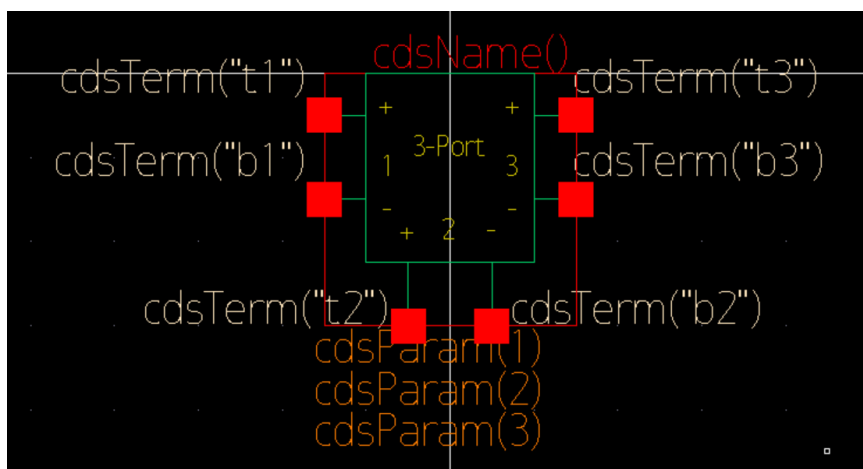
CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Delay frequency	delayfreq		X	-	-
Extracts a system delay	delayhandle	0	X	-	-
Temperature difference	dtemp		X	-	-
High freq extrapolate method	highpass	3	X	-	-
Linear interpolation data type	intdattyp	MA	X	-	-
Low freq extrapolate method	lowpass	1	X	-	-
Enable noise passive checker	noipassivechk	1	X		-
Precondition factor keyword	precfac	0.75	X	-	-
Enable rational function	rational_func	0	X	-	-
Reuse rational function data	rational_func_reuse	1	X	-	-
Method of smooth	smooth	0	X	-	-
Width of the smoothing window	smoothpts		X	-	-
Stamping method	stamp		X	-	-

Controlling Version Number for Legacy Designs

To control the version number for legacy designs, create and use the [dbAnalogLibNxPortVersion](#) Tcl preference before the analogLib library is loaded.

n3port

The n3port element is a special case of the generic nport element with specific port number.



Note:

The **nport** element has certain behavior which depends on S, Y, or Z-parameter files.

CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Browse and select s-data file	nportFileB	0	X	X	-
S-parameter data file	dataFile		X	X	-
Interpolation method	interp	linear	-	X	-
Tran convolution parameters	tranParaLabel	1	-	X	-
Accuracy	accuracyMode	default	-	X	-
Advanced transient parameters	tranAdvanPara Label	0	X	X	-
Max sampling points	maxn		-	X	-
Max frequency of interest	fmax		X	X	-
Impulse response truncation	imptrunc		-	X	-
ROM data file	romdatfile		-	X	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
ROM data file name is variable	romdatfile_as_var	0	-	X	-
Rational order	ratorder		-	X	-
Relative error	relerr		-	X	-
Absolute error	abserr		-	X	-
Noise parameters	noiseParaLabel	0	X	X	-
Thermal noise	thermalnoise	Yes	X	X	-
Thermal noise model	noisemodel		-	X	-
Rarely used parameters	otherParaLabel	0	X	X	-
S-parameter data format	datafmt		-	-	-
Frequency sampling interval	fdelta		X	X	-
Use smooth data windowing	usewindow		-	X	-
Multiplier	m		X	X	-
Scale factor	scale		-	X	-
Model name	hmname	MDN	X	-	-
Enable mixed mode	mixedmode	0	X	-	-
The order of indices	datatype		X	-	-
Characteristic impedance	z0	50	X	-	-
Hspice S-parameter data format	datafmtHspice	touchstone	X	X	-
Enable passive checker	passive	0	X	-	-
Hspice Interpolation method	interpolation		X	-	-

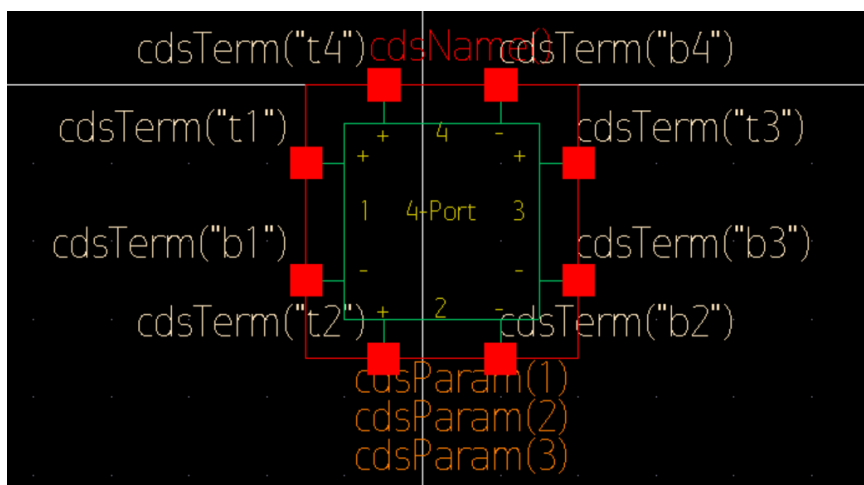
CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Delay frequency	delayfreq		X	-	-
Extracts a system delay	delayhandle	0	X	-	-
Temperature difference	dtemp		X	-	-
High freq extrapolate method	highpass	3	X	-	-
Linear interpolation data type	intdattyp	MA	X	-	-
Low freq extrapolate method	lowpass	1	X	-	-
Enable noise passive checker	noipassivechk	1	X	-	-
Precondition factor keyword	precfac	0.75	X	-	-
Enable rational function	rational_func	0	X	-	-
Reuse rational function data	rational_func_reuse	1	X	-	-
Method of smooth	smooth	0	X	-	-
Width of the smoothing window	smoothpts		X	-	-
Stamping method	stamp		X	-	-

Controlling Version Number for Legacy Designs

To control the version number for legacy designs, create and use the [dbAnalogLibNxPortVersion](#) Tcl preference before the analogLib library is loaded.

n4port

The n4port element is a special case of the generic nport element with specific port number.



Note:

The [nport](#) element has certain behavior which depends on S, Y or Z-parameter files.

CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Browse and select s-data file	nportFileB	0	X	X	-
S-parameter data file	dataFile		X	X	-
Interpolation method	interp	linear	-	X	-
Tran convolution parameters	tranParaLabel	1	-	X	-
Accuracy	accuracyMode	default	-	X	-
Advanced transient parameters	tranAdvanPara Label	0	X	X	-
Max sampling points	maxn		-	X	-
Max frequency of interest	fmax		X	X	-
Impulse response truncation	imptrunc		-	X	-
ROM data file	romdatfile		-	X	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
ROM data file name is variable	romdatfile_as_var	0	-	X	-
Rational order	ratorder		-	X	-
Relative error	relerr		-	X	-
Absolute error	abserr		-	X	-
Noise parameters	noiseParaLabel	0	X	X	-
Thermal noise	thermalnoise	Yes	X	X	-
Thermal noise model	noisemodel		-	X	-
Rarely used parameters	otherParaLabel	0	X	X	-
S-parameter data format	datafmt		-	-	-
Frequency sampling interval	fdelta		X	X	-
Use smooth data windowing	usewindow		-	X	-
Multiplier	m		X	X	-
Scale factor	scale			X	-
Model name	hmname	MDN	X	-	-
Enable mixed mode	mixedmode	0	X	-	-
The order of indices	datatype		X	-	-
Characteristic impedance	z0	50	X	-	-
Hspice S-parameter data format	datafmtHspice	touchstone	X	X	-
Enable passive checker	passive	0	X	-	-
Hspice Interpolation method	interpolation		X	-	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Delay frequency	delayfreq		X	-	-
Extracts a system delay	delayhandle	0	X	-	-
Temperature difference	dtemp		X	-	-
High freq extrapolate method	highpass	3	X	-	-
Linear interpolation data type	intdattyp	MA	X	-	-
Low freq extrapolate method	lowpass	1	X	-	-
Enable noise passive checker	noipassivechk	1	X	-	-
Precondition factor keyword	precfac	0.75	X	-	-
Enable rational function	rational_func	0	X	-	-
Reuse rational function data	rational_func_reuse	1	X	-	-
Method of smooth	smooth	0	X	-	-
Width of the smoothing window	smoothpts		X	-	-
Stamping method	stamp		X	-	-

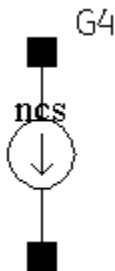
Controlling Version Number for Legacy Designs

To control the version number for legacy designs, create and use the [dbAnalogLibNxPortVersion](#) Tcl preference before the analogLib library is loaded.

ncs

Behavioral noise source.

In HSPICE advanced analog analyses, you can use the G-element to specify noise sources. Frequency domain noise analyses (.NOISE, .HBNOISE, and .PHASENOISE) take these noise sources into account.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Noise Expression	noise		X	-	-
Noise File	noisefile		X	-	-
Noise file name is variable	noise_file_as_var	0	X	-	-
Noise entry method	noiseEntryMethod	file	X	-	-

Note:

- The default of noise_file_as_var parameter is *false*. If the noisefile parameter is set as *NOISEFILE*, the netlisting behavior for noisefile and noise_file_as_var parameters is as follows:

CDF Parameter	hspiceD Output	spectre Output
noise_file_as_var		
false (default)	noisefile='NOISEFILE'	noisefile="NOISEFILE"
true	noisefile=str(NOISEFILE)	noisefile=NOISEFILE

- For Hspice-format netlisting, a pair of single quotes is added to noisefile parameter when noise_file_as_var is *false*. When noise_file_as_var parameter is *true*, str(NOISEFILE) is used as the value format.

- The `noiseEntryMethod` parameter controls both Hspice-format and Spectre-format netlisting behavior. This `noiseEntryMethod` parameter has two choices: `file` and `expression`. Based on its value, the Property Editor shows corresponding parameters such as `noise`, `noisefile`, and `noise_file_as_var`. With the `noiseEntryMethod` parameter, either `noise` or `noisefile` is netlisted in the Hspice-format and Spectre-format netlisting. The netlisting results are mutually exclusive to choose either `noise` or `noisefile`.

To hide parameters such as `noise`, `noisefile`, and `noise_file_as_var` for legacy designs, create and use the [dbAnalogLibNcsVersion](#) Tcl preference.

Example

Noise expression form

```
g4 n1 n2 noise='4*1.3806266e-23*(TEMPER+273.15)*0.001'
```

Noise file form

```
g4 n1 n2 noisefile='out_noise.txt'
```

See Also

- [Behavioral Current Noise Source](#) in the *HSPICE® User Guide: Elements*
- [Behavioral Noise Sources](#) in the *HSPICE® User Guide: Advanced Analog Simulation and Analysis*

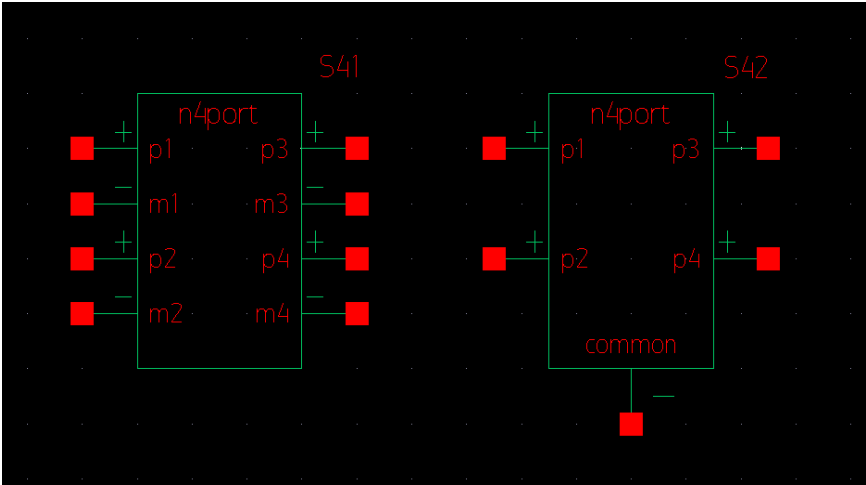
Controlling Version Number for Legacy Designs

To control the version number for legacy designs, create and use the [dbAnalogLibNcsVersion](#) Tcl preference before the `analogLib` library is loaded.

nport

The `nport` cell is a PCell device that is controlled by version parameter.

The following figure shows two `nport` devices that have `numports == 4` (`p == 4`). The one on the left is with common mode off and another one on the right is with common mode on.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Common reference	nmode	0	-	X	-
numports	numports	1	X	-	-
Model Name	mname		X	-	-
Model File	modelFile		X	-	-
Data Source	fileOrCV	file	-	X	-
S-parameter data file	dataFile		-	X	-
S-parameter file as Design Var?	sparam_file_as _var	0	-	X	-
Model Type	modelFileType	TSTONE FILE	X	-	-
S-parameter Lib	sparamLib		-	X	-
S-parameter Cell	sparamCell		-	X	-
S-parameter View	sparamView		-	X	-
Browse and specify S-parameter LCV	sparamlcvB	0	-	-	-
Resistance	r	50	-	-	-
Rational Func Approx	rational_func		X	-	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
checkFlag	checkFlag	1	-	X	-
Number of ports	p	1	-	X	X
Browse and select s-data file	nportFileB	0	-	X	-
Interpolation method	interp	linear	-	X	-
Passivity	passivity	check	-	X	-
Passivity	passivity_bbspace	fit_enforce	-	X	-
Tran convolution parameters	tranParaLabel	1	-	X	-
Accuracy	accuracyMode	default	-	X	-
Advanced transient parameters	tranAdvanParaLabel	0	-	X	-
Max sampling points	maxn		-	X	-
Base Frequency	fbase		X	-	-
Max Frequency	fmax		X	-	-
Impulse response truncation	imptrunc		-	X	-
Causality correction	causality	fmax	-	X	-
Enforce causality	s_causality		X	-	-
Max frequency multiplier for causality enforcement	causality_fmax_scale		X	-	-
DC extrapolation	dcextrap	constant	-	X	-
ROM data file	romdatfile		-	X	-
ROM data file name is variable	romdatfile_as_var	0	-	X	-
Rational order	ratorder		-	X	-
Relative error	relerr		-	X	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Absolute error	abserr		-	X	-
Noise parameters	noiseParaLabel	0	-	X	-
Generate Noise?	noise	1	X	-	-
Noise Passive Check	noipassivechk		X	-	-
Temp Difference	dtemp		X	-	-
Thermal noise	thermalnoise	yes	-	X	-
Thermal noise model	noisemodel		-	X	-
Noise correlation	noisecorr		-	X	-
Rarely used parameters	otherParaLabel	0	-	X	-
Interpolation Method	interpolation		X	-	-
Model Name	mname		X		-
Modeling method (rational_func)	rational_func_for_spectre		-	X	-
Parameter Type	type	S	X	-	-
Frequency Behavior	fqmodel		X	-	-
Characteristic Impedance	z0	50	X	-	-
Precondition Factor	precfac	0.75	X	-	-
Delay Handler	delayhandle		X	-	-
Delay Frequency	delayfreq	iPar("fmax")	X	-	-
Interpolation Data Type	intdattyp	MA	X	-	-
High Extrapolation Method	highpass	3	X	-	-
Low Extrapolation Method	lowpass	2	X	-	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Stamp	stamp		X	-	-
Mixed Mode	mixedmode	0	X	-	-
Data Type String	datatype		X	-	-
Write causal S-param?	write_causality_enforced_file		X	-	-
Causal s-param output file	outFile		-	X	-
Rational Func Reuse	rational_func_reuse		X	-	-
Use smooth data windowing	usewindow		-	X	-
S-parameter data format	datafmt		-	X	-
Passivity tolerance	pabstol		-	X	-
Data truncation threshold	datatrunc		-	X	-
Frequency sampling interval	fdelta		-	X	-
Multiplier	m		X	X	-
Scale factor	scale		-	X	-
High freq extrapolation	hfextrap		-	X	-
Flag for matrix form input	matrixform		-	X	-
Matrix entry data file	matrixfile		-	X	-
Matrix entry data file name is variable	matrixfile_as_var	0	-	X	-
Type of port 1-20	porttype1-20		-	X	-
Quantity of Port 1-20	portquantity1-20		-	X	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Prioritize Accuracy Range	prioritizeAccuracyRange	0	-	X	-
Start Frequency	startFrequency		-	X	-
End Frequency	endFrequency		-	X	-
Additional parameter list	additionalParam		-	X	-
Passive Mode	passive	0	X	-	-
Enforce Passivity	enforce_passive		X	-	-
Additional parameter list	additionalSpiceParam		X	-	-
AC Model	acModel	freqdomain	-	X	-
RLGC data file	file		-	-	-
check box status	checkBoxValue	1	-	X	-
Version	version	2	-	-	-

Example

```
s7 p1 m1 p2 m2 mname=mymodel type='S' z0=50 fbase=5G
+ fmax=15G precfac=0.75 delayfreq=15G
+ interpolation='LINEAR' intdattyp='MA' highpass=3
+ lowpass=2 mixedmode=0 noise=1 passive=0
.model mymodel s TSTONEFILE=<path_to_file>
```

See Also

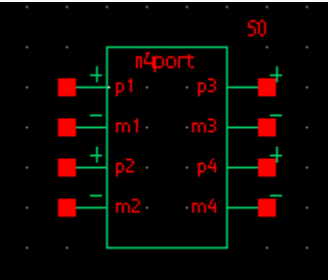
- https://solvnet.synopsys.com/dow_retrieve/latest/hspice/hspice_olh/index.htm#context/cidSynopsysCustom/s-parameter_modeling

Using nport to Run PrimeSim Analyses

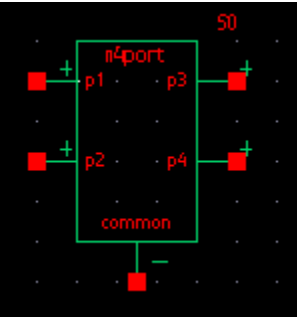
Use the nport element to specify S-parameters. In many cases, the number of ports and the file name containing S-parameters (usually in Touchstone format) is the only required setup in the Property Editor.

Prompt	Value
Common reference	☐
numports	4
Model File	
Model Type	TSTONEFILE
Modeling method (rational_func)	
Advanced transient parameters	☐
Noise parameters	☐
Other Parameters	☐

In its default form, the nport symbol has a return terminal per port, so that the total number of terminals is 2N, as shown in the following figure.



Check the **Common reference** option in the Property Editor to create a symbol with a shared return terminal, as a way of simplifying wiring of nports with many terminals, as shown in the following figure.



In earlier versions of Custom Compiler, nport required you to specify the name of a model netlisted as an HSPICE `.model` statement. This specification is no longer required, and the netlister automatically handles requirements such as the insertion of model statements.

Prompt	Value
Common reference	☒ 
numports	4
Model File	test.s4p
Model Type	TSTONEFILE
Modeling method (rational_func)	

The **Modeling method** option controls the behavior of S-parameters in transient, frequency-domain, and Shooting Newton simulations:

	rational_func=0	rational_func=1 or unspecified	rational_func=2
Transient (including HB initial transient)	IFFT	Rational	Rational
Any frequency dependent analyses	Raw data	Raw data	Rational
Shooting Newton	Rational	Rational	Rational

Set **Modeling method** to 0 to use IFFT (convolution) method in transient. Raw data from the S-parameter file is used for frequency domain methods, except when overridden by setting **Modeling method=2** to use rational fitting. Shooting Newton always uses rational modeling.

In recent versions of Custom Compiler and PrimeSim, the nport interface and S-element technology were simplified and improved to require, in most cases, specifying only **Model File**, **numports** and **Modeling method** options. The remaining S-element parameters are exposed by checking the **Advanced transient parameters**, **Noise parameters**, and **Other Parameters** options.

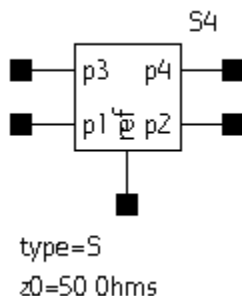
Prompt	
Common reference	☒
numports	4
Model File	
Model Type	TSTONEFILE
Modeling method (rational_func)	
Advanced transient parameters	☐
Noise parameters	☐
Other Parameters	☐

Refer to S-element (Scattering Parameter Data) in the *PrimeSim Continuum User Guide: Elements* for details of S-element options. In most cases, you should leave them at default values and allow simulation engines to determine optimal settings automatically. The following list describes a few noteworthy parameters in **Advanced transient parameters** and **Other Parameters** categories:

- **Enforce causality:** Newer versions of PrimeSim SPICE support IFFT causality enforcement. Causality enforcement is on by default and recommended to leave unspecified (preferred) or set to 1.
- **Max frequency multiplier for causality enforcement:** When causality is enforced, PrimeSim extrapolates raw data to an automatically selected frequency larger than the maximum data file frequency. If you observe log file messages warning about differences between causal and raw data, increase this parameter to between 2 and 10. The same suggestion applies if you run into convergence problems or suspected accuracy issues in IFFT.
- **Max frequency of interest:** It is recommended to leave this setting unspecified.
- **Interpolation Method:** In recent versions of PrimeSim SPICE, **Interpolation Method** controls raw data interpolation in frequency domain analyses. **Modeling method (rational_func)** is used to control rational fitting vs. convolution (IFFT) methods. Spline and Linear methods work best. The behavior of **Interpolation Method=rational** is not defined and should not be used.
- **Additional parameter list:** Specify a list of parameter-value pairs in the form p1=val1 p2=val2... to send to the simulator any simulation parameters not exposed in the Property Editor. An additional parameter list is appended to the end of the S-element instance line.

nport_ref

You can use the nport_ref element to describe a multi-terminal network circuit analyses within most HSPICE and RF analyses.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Model Name	mname		X	-	-
Model File	modelFile		X	-	-
Model Type	modelFileType	TSTONE FILE	X	-	-
Frequency Behavior	mqmodel		X	-	-
Parameter Type	type	S	X	-	-
Characteristic Impedance	z0	50	X	-	-
Resistance	r	50	-	-	-
Base Frequency	fbase		X	-	-
Max Frequency	fmax		X	-	-
Precondition Factor	precfac	0.75	X	-	-
Delay Handler	delayhandle		X	-	-
Delay Frequency	delayfreq	iPar("fmax")	X	-	-
Interpolation Method	interpolation	LINEAR	X	-	-
Interpolation Data Type	intdattyp	MA	X	-	-
High Extrapolation Method	highpass	3	X	-	-
Low Extrapolation Method	lowpass	2	X	-	-
Mixed Mode	mixedmode	0	X	-	-
Data Type String	datatype		X	-	-
Generate Noise?	noise	1	X	-	-
Noise Passive Check	noipassivechk		X	-	-
Temp Difference	dtemp		X	-	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Passive Mode	passive	0	X	-	-
Enforce Passive Mode	enforce_passive		X	-	-
Rational Func Approx	rational_func		X	-	-
checkFlag	checkFlag	1	-	X	-
S-parameter file as Design Var?	sparam_file_as_var	0	-	X	-
Rational Func Reuse	rational_func_reuse		X	-	-
Stamp	stamp		X	-	-
Multiplier	m		X	-	-

Example

```
s7 m1 m2 p1 p2 ref mname=mymodel type='S' z0=50 fbase=5G
+ fmax=15G precfac=0.75 delayfreq=15G interpolation='LINEAR'
+ intdattyp='MA' highpass=3 lowpass=2 mixedmode=1 noise=1
+ passive=0
.model mymodel s TSTONEFILE=<path_to_file>
```

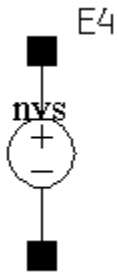
See Also

- [Mixed-Mode S-parameters](#) in the *HSPICE® User Guide: Signal Integrity Modeling and Analysis*

nvs

Behavioral noise voltage source.

In HSPICE advanced analog analyses, you can use the E-element to specify noise voltage sources.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Noise Expression	noise		X	-	-
Noise File	noisefile		X	-	-
Noise file name is variable	noise_file_as_var	0	X	-	-
Noise entry method	noiseEntryMethod	file	X	-	-

Note:

- The default of noise_file_as_var parameter is *false*. If the noisefile parameter is set as *NOISEFILE*, the netlisting behavior for noisefile and noise_file_as_var parameters is as follows:

CDF Parameter	hspiceD Output	spectre Output
noise_file_as_var		
false (default)	noisefile='NOISEFILE'	noisefile="NOISEFILE"
true	noisefile=str(NOISEFILE)	noisefile=NOISEFILE

- For Hspice-format netlisting, a pair of single quotes is added to noisefile parameter when noise_file_as_var is *false*. When noise_file_as_var parameter is *true*, str(NOISEFILE) is used as the value format.
- The noiseEntryMethod parameter controls both Hspice-format and Spectre-format netlisting behavior. This noiseEntryMethod parameter has two choices: *file* and *expression*. Based on its value, the Property Editor shows corresponding parameters such as noise, noisefile, and

noise_file_as_var. With the noiseEntryMethod parameter, either noise or noisefile is netlisted in the Hspice-format and Spectre-format netlisting. The netlisting results are mutually exclusive to choose either noise or noisefile.

To hide parameters such as noise, noisefile, and noise_file_as_var for legacy designs, create and use the dbAnalogLibNvsVersion Tcl preference.

Example

```
e4 n1 n2 noisefile='noise_v.txt'  
e4 n1 n2 noise='4*1.3806266e-23*(TEMPER+273.15)*0.001'
```

See Also

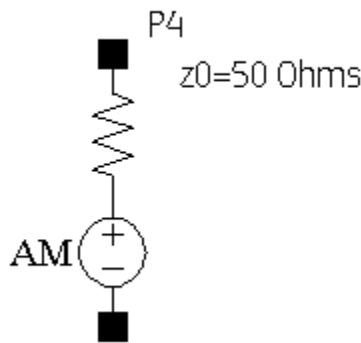
- Behavioral Voltage Noise Source in the HSPICE® User Guide: Elements

Controlling Version Number for Legacy Designs

To control the version number for legacy designs, create and use the dbAnalogLibNvsVersion Tcl preference before the analogLib library is loaded.

pam

Amplitude modulated port element.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Source Type	sourceType	port	-	-	-
Source Type	srcType	am	-	-	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Port Number	num		X	-	-
Impedance	z0	50	X	-	-
Resistance	r	50	-	-	-
Power Switch	power		X	-	-
DC Value	dc		X	-	-
DC Voltage	vdc		-	-	-
Signal Amplitude	sa	0	X	-	-
Carrier Frequency	fc	1	X	-	-
Modulation Frequency	fm	0	X	-	-
Offset Constant	oc	0	X	-	-
Delay Time	td	0	X	-	-
AC Magnitude	acm		X	-	-
AC Phase	acp		X	-	-
# HB Tones	hbtones	0	X	-	-
HB Magnitude 1-7	hbmag1-7		X	-	-
HB Phase1-7	hbphase1-7	0	X	-	-
HB Harmonic1-7	hbharm1-7	1	X	-	-
HB Tone1-7	hbtone1-7	1-7	X	-	-
HB ModHarm1-7	hbmodharm1-7		X	-	-
HB ModTone1-7	hbmodtone1-7		X	-	-
Open DC Connection	dcopen		X	-	-
DC Resistance	rdc		X	-	-
AC Resistance	rac		X	-	-
HB Resistance	rhb		-	-	-
HBAC Resistance	rhbac		-	-	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
TRAN Resistance	rtran		X	-	-
HBAC Magnitude	hbacm		X	-	-
HBAC Phase	hbacp		X	-	-
SNAC Magnitude	snacm		X	-	-
SNAC Phase	snacp		X	-	-

Example

```
p4 n1 gnd! port=1 am ( 0.0 0.0 10k 1k 0 ) z0=50
```

See Also

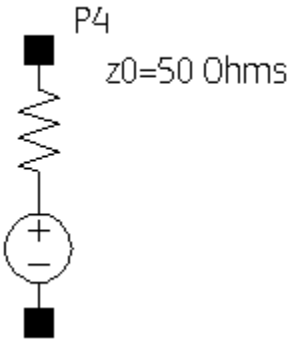
- [Port Element](#) in the *HSPICE® User Guide: Elements*

Updating Port Number Automatically

You can create and use the [dbAnalogLibAutoUpdatePortNum](#) Tcl preference to control whether to automatically update the port number of port family devices.

pdc

DC port element.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Source Type	sourceType	port	-	-	-
Source Type	srcType	dc	-	X	-
Port Number	num		X	X	-
Impedance	z0	50	X	-	-
Resistance	r	50	-	X	-
Power Switch	power		X	-	-
DC Value	dc		X	-	-
DC Voltage	vdc		-	X	-
AC Magnitude	acm		X	X	-
AC Phase	acp		X	X	-
# HB Tones	hbtones	0	X	-	-
HB Magnitude 1-7	hbmag1-7		X	-	-
HB Phase1-7	hbphase1-7	0	X	-	-
HB Harmonic1-7	hbharm1-7	1	X	-	-
HB Tone1-7	hbtone1-7	1-7	X	-	-
HB ModHarm1-7	hbmodharm1-7		X	-	-
HB ModTone1-7	hbmodtone1-7		X	-	-
Open DC Connection	dcopen		X	-	-
DC Resistance	rdc		X	-	-
AC Resistance	rac		X	-	-
HB Resistance	rhb		X	-	-
HBAC Resistance	rhbac		X	-	-
TRAN Resistance	rtran		X	-	-
HBAC Magnitude	hbacm		X	-	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
HBAC Phase	hbacp		X	-	-
SNAC Magnitude	snacm		X	-	-
SNAC Phase	snacp		X	-	-
Number of noise/frequency pairs	FNpairs	0	-	X	-
Noise file name	noisefile		-	X	-
Noise file name is variable	noise_file_as_var	0	-	X	-
Frequency1-50	F1-50		-	X	-
Noise1-50	N1-50		-	X	-
Multiplier	m			X	
Nominal temperature	tnom		-	X	-
PAC magnitude	pacm		-	X	-
PAC phase	pacp		-	X	-
Temp coefficient 1	tc1		-	X	-
Temp coefficient 2	tc2		-	X	-
XF magnitude	xfm		-	X	-

Note:

- When frequency/noise pairs have empty values or the number of frequency/noise pairs does not match with the FNpairs parameter, the Spectre netlister shows an error in the console.
- The default of noise_file_as_var parameter is *false*. If the noisefile parameter is set as *NOISEFILE*, the netlisting behavior for noisefile and noise_file_as_var parameters is as follows:

CDF Parameter	hspiceD Output	spectre Output
noise_file_as_var		
false (default)	noisefile='NOISEFILE'	noisefile="NOISEFILE"

CDF Parameter	hspiceD Output	spectre Output
true	noisefile=str(NOISEFILE)	noisefile=NOISEFILE

- For Hspice-format netlisting, a pair of single quotes is added to noisefile parameter when noise_file_as_var is *false*. When noise_file_as_var parameter is *true*, str(NOISEFILE) is used as the value format.
- For Spectre-format netlisting, a pair of double quotes is added when noise_file_as_var parameter is *false*. No single/double quotes are added when noise_file_as_var parameter is *true*.

Example

```
p4 n1 gnd! port=1 dc=3.3 z0=50
```

See Also

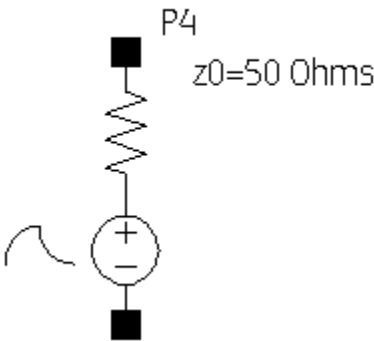
- [Port Element](#) in the *PrimeSim Continuum User Guide: Elements*

Updating Port Number Automatically

You can create and use the [dbAnalogLibAutoUpdatePortNum](#) Tcl preference to control whether to automatically update the port number of port family devices.

pexp

Exponential port element.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Source Type	sourceType	port	-	-	-
Source Type	srcType	exp	-	X	-
Port Number	num		X	X	-
Impedance	z0	50	X	-	-
Resistance	r	50	-	X	-
Power Switch	power		X	-	-
DC Value	dc		X	-	-
DC Voltage	vdc		-	X	-
Value 1	v1	0	X	X	-
Value 2	v2	0	X	X	-
Rise Delay Time	td1	0	X	X	-
Fall Delay Time	td2		X	X	-
Rise Time Constant	tau1		X	X	-
Fall Time Constant	tau2	0	X	X	-
AC Magnitude	acm		X	X	-
AC Phase	acp		X	X	-
# HB Tones	hbtones	0	X	-	-
HB Magnitude 1-7	hbmag1-7		X	-	-
HB Phase1-7	hbphase1-7	0	X	-	-
HB Harmonic1-7	hbharm1-7	1	X	-	-
HB Tone1-7	hbtone1-7	1-7	X	-	-
HB ModHarm1-7	hbmodharm1-7		X	-	-
HB ModTone1-7	hbmodtone1-7		X	-	-
Open DC Connection	dcopen		X	-	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
DC Resistance	rdc		X	-	-
AC Resistance	rac		X	-	-
HB Resistance	rhb		X	-	-
HBAC Resistance	rhbac		X	-	-
TRAN Resistance	rtran		X	-	-
HBAC Magnitude	hbacm		X	-	-
HBAC Phase	hbacp		X	-	-
SNAC Magnitude	snacm		X	-	-
SNAC Phase	snacp		X	-	-
Number of noise/frequency pairs	FNpairs	0	-	X	-
Noise file name	noisefile		-	X	-
Noise file name is variable	noise_file_as_var	0	-	X	-
Frequency1-50	F1-50		-	X	-
Noise1-50	N1-50		-	X	-
Multiplier	m			X	
Nominal temperature	tnom		-	X	-
PAC magnitude	pacm		-	X	-
PAC phase	pacp		-	X	-
Temp coefficient 1	tc1		-	X	-
Temp coefficient 2	tc2		-	X	-
XF magnitude	xfm		-	X	-
Delay time	td		-	X	

Note:

- When frequency/noise pairs have empty values or the number of frequency/noise pairs does not match with the FNpairs parameter, the Spectre netlister shows an error in the console.
- The default of noise_file_as_var parameter is *false*. If the noise file parameter is set as *NOISEFILE*, the netlisting behavior for noise file and noise_file_as_var parameters is as follows:

CDF Parameter	hspiceD Output	spectre Output
noise_file_as_var		
false (default)	noisefile='NOISEFILE'	noisefile="NOISEFILE"
true	noisefile=str(NOISEFILE)	noisefile=NOISEFILE

- For Hspice-format netlisting, a pair of single quotes is added to noise file parameter when noise_file_as_var is *false*. When noise_file_as_var parameter is *true*, str(NOISEFILE) is used as the value format.
- For Spectre-format netlisting, a pair of double quotes is added when noise_file_as_var parameter is *false*. No single/double quotes are added when noise_file_as_var parameter is *true*.

Example

```
p4 n1 gnd! port=1 dc=3.3 exp ( 0 2.5 5n 1n 15n 2n ) z0=50
```

See Also

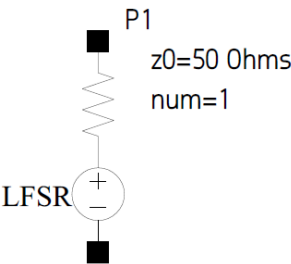
- [Port Element](#) in the *HSPICE® User Guide: Elements*

Updating Port Number Automatically

You can create and use the [dbAnalogLibAutoUpdatePortNum](#) Tcl preference to control whether to automatically update the port number of port family devices.

plfsr

The plfsr element specifies lfsr waveform type and analysis settings for P-element.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Source Type	sourceType	port	-	-	-
Source Type	srcType	lfsr	-	-	-
Port Number	num				
Resistance	r	50	X	-	-
Power Switch	power		X	-	-
DC Value	dc		X	-	-
DCVoltage	vdc		X	-	-
Low Value	vl	0	X	-	-
High Value	vh	1	X	-	-
Delay Time	td	0	X	-	-
Rise Time	tr	0	X	-	-
Fall Time	tf	0	X	-	-
Bit Rate	rate	1M	X	-	-
Seed	seed	1	X	-	-
Taps	taps	[1,2]	X	-	-
Output Resistance	rout		X	-	-
AC Magnitude	acm		X	-	-
AC Phase	acp		X	-	-
# HB Tones	hbtones	0	X	-	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
HB Magnitude 1-7	hbmag1-7		X	-	-
HB Phase1-7	hbphase1-7	0	X	-	-
HB Harmonic1-7	hbharm1-7	1	X	-	-
HB Tone1-7	hbtone1-7	1-7	X	-	-
HB ModHarm1-7	hbmodharm1-7		X	-	-
HB ModTone1-7	hbmodtone1-7		X	-	-
Open DC Connection	dcopen		X	-	-
DC Resistance	rdc		X	-	-
AC Resistance	rac		X	-	-
HB Resistance	rhb		X	-	-
HBAC Resistance	rhbac		X	-	-
TRAN Resistance	rtran		X	-	-
HBAC Magnitude	hbacm		X	-	-
HBAC Phase	hbacp		X	-	-
SNAC Magnitude	snacm		X	-	-
SNAC Phase	snacp		X	-	-
PAM4 modulation	pam4_modulation	none	X	-	-
PAM4 mapping	pam4_mapping		X	-	-
Impedance	z0		X	-	-

Example

```
p1 p gnd! port=1 dc=0 lfsr ( 0 1 0 0 0 1meg 1 [1,2] ) z0=50
```

See Also

- [Port Element](#) in the *HSPICE® User Guide: Elements*

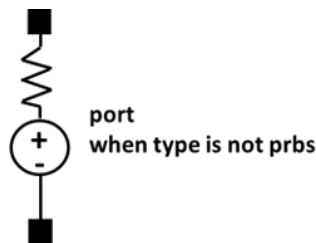
Updating Port Number Automatically

You can create and use the `dbAnalogLibAutoUpdatePortNum` Tcl preference to control whether to automatically update the port number of port family devices.

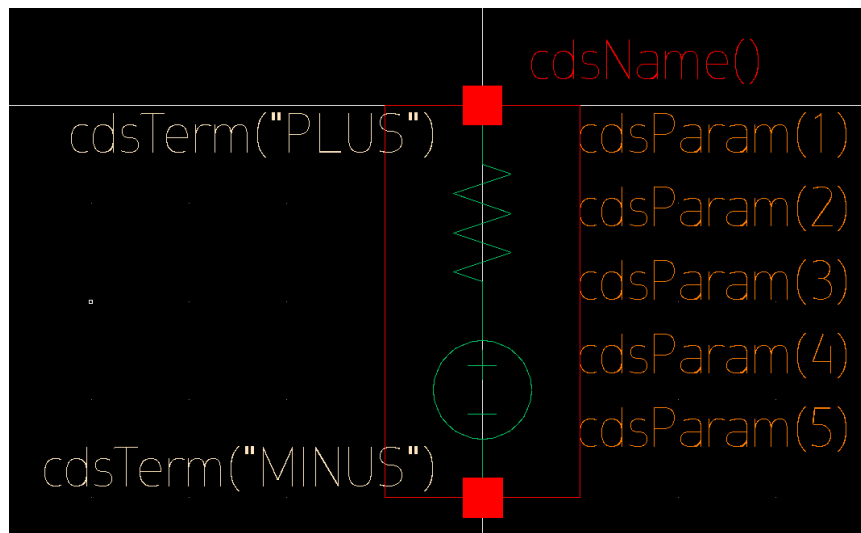
port

Port element with selectable transient waveform type. Source type can be dc, exp, pulse, pwl, sine, and prbs.

This port element can also be used for Algorithmic Modeling Interface (AMI) to enable StatEye analysis of systems with Serializer-Deserializer (SERDES) devices for equalizers in both transmitter and receiver sides.

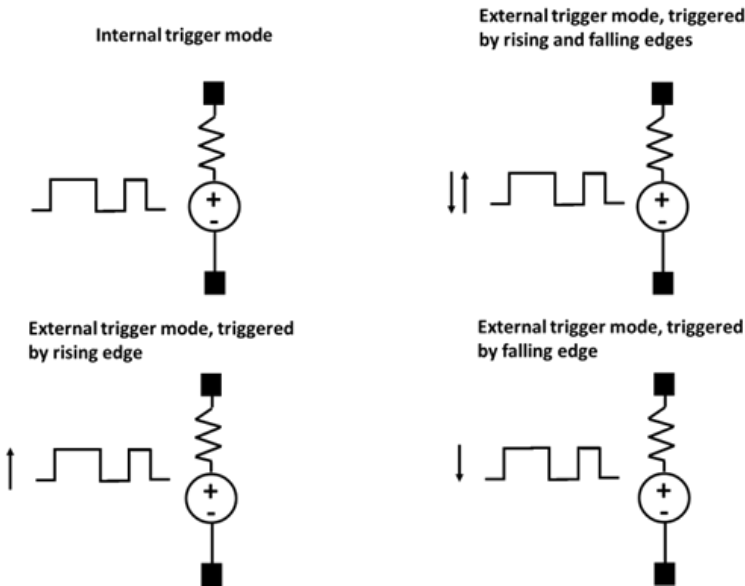


The port device is a PCell. Its PCell behavior is affected by its source type. For source type other than prbs, it is a 2-pin device and its symbol looks as follows:



For prbs source type, its PCell behavior is affected by its triggerValue parameter. The following figure shows four vprbs devices that have different trigger modes, which are Internal (on the upper-left), External by both edges (on the upper-right), External by falling edges (on the lower-left), and External by rising edges (on the lower-right) modes. The internal trigger mode has 2 pins (PLUS and MINUS) and other three external trigger

modes have 4 pins (P1, M1, P2, and M2). The netlisting procedure writes out their pins accordingly.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Source Type	srcType	dc	X	X	-
Source Type	sourceType	port	-	X	-
Port Number	num		X	X	-
Impedance	z0		X	-	-
Resistance	r	50	-	X	-
Reactance	x		-	X	-
Power Switch	power		X	-	-
DC Value	dc		X	-	-
DC Voltage	vdc		-	X	-
Waveform entry method	pwlEntryMethod	File	-	X	-
File name	fileName		-	X	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
File name is variable	file_as_var	0	-	X	-
Time Value Pairs	tvpairs	0	X	X	-
Time1-50	t1-50		X	X	-
Z State1-50	Z1-50	0	-	-	-
Value1-50	v1-50		X	X	-
Zero value	val0		-	X	-
One value	val1		-	X	-
Rise Delay Time	td1	0	X	X	-
Fall Delay Time	td2	0	X	X	-
Rise Time Constant	tau1	0	X	X	-
Fall Time Constant	tau2	0	X	X	-
Exp period	exppperiod		-	x	-
Rise Time	tr		X	X	-
Fall Time	tf		X	X	-
Pulse width	pw		X	X	-
Period	per		X	X	-
Offset Value	vo	0	X	X	-
DC offest	offset		-	X	-
Desired rms value	rms		-	x	-
Time scaling factor	stretch		-	X	-
Repeat	rpt		X	-	-
Repeat Stop Time	rstop		X	-	-
Repeat Stop Value	stopvalue		X	-	-
Repeat Stop Slope	stopslope		X	-	-
PWL period	pwlperiod		-	X	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Edge type	edgetype		-	X	-
Breakpoints	allbrkpts	no	-	X	-
Width of transition	twidht		-	X	-
Frequency name 1	fundname		-	X	-
Frequency	freq		X	X	-
Amplitude Value	va	0	X	X	-
Amplitude 1 (dBm)	vaDBm		-	X	-
Phase for sinusoid	sinephase	0	X	X	-
Voltage level	sinedc		-	X	-
Signal Amplitude	sa	0	X	-	-
Carrier Frequency	fc	1	X	-	-
Modulation Frequency	fm	0	X	-	-
Offset Constant	oc	0	X	-	-
Voltage High	vh	1	X	-	-
Voltage Low	vl	0	X	-	-
Sample Time	tsample	1n	X	-	-
Data	data	b1010 110	X	-	-
Pattern Parameter data is variable	data_as_var		-	X	-
Pattern Parameter rptstart	rptstart	1	X	X	-
Pattern Parameter rpttimes	rpttimes	0	X	X	-
Repeat Bit	rb		X	-	-
Modulation Index	mdi	0	X	-	-
Signal Frequency	fs		X	-	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Voltage Low	vlow	0	X	-	-
Voltage High	vhigh	1	X	-	-
Delay Time	td	0	X	X	-
Bit Rate	rate	1M	X	-	-
Seed	seed	1	X	-	-
Taps	taps	[1,2]	X	-	-
Output Resistance	rout		X	-	-
Encode (DW8b10b)	enc	0	X	-	-
Initial Running Disparity	ird		X	-	-
Damping Factor	theta	0	X	X	-
# HB Tones	hbtones	0	X	-	-
HB Magnitude 1-7	hbmag1-7		X	-	-
HB Phase1-7	hbphase1-7	0	X	-	-
HB Harmonic1-7	hbharm1-7	1	X	-	-
HB Tone1-7	hbtone1-7	1-7	X	-	-
HB ModHarm1-7	hbmodharm1-7		X	-	-
HB ModTone1-7	hbmodtone1-7		X	-	-
SI Parameters	siParameters	0	X	-	-
Emphasis Level	emphasisLevel		X	-	-
Emphasis Time	emphasisTime		X	-	-
Tap Coef. Pairs	dfepairs	0	X	-	-
DFE Tap 1-50	tap1-50		X	-	-
Coefficient 1-50	coef1-50		X	-	-
DFE Vth	dfe_vth	0	X	-	-
DFE Clk	dfe_clk	0	X	-	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Duty Cycle Distortion	dcd		X	-	-
Duty Cycle Distortion Variation Type	dcdType		X	-	-
Periodic Jitter Voltage	pj		X	-	-
Periodic Jitter Type	pjType		X	-	-
Random Jitter	rj		X	-	-
Voltage Noise	vn		X	-	-
Algorithmic Modeling Options	algorithmicModeling	n/a	X	-	-
Algorithmic Modeling Interface Object	amiObj		X	-	-
Algorithmic Modeling Interface Parameter	amiParam		X	-	-
AMI Param XX Name	amiParamXxName		X	-	-
AMI Param XX Value	amiParamXxValue		X	-	-
Index number of AMI corner	amiCorner		X	-	-
AMI Clock Data Recovery	amiCdr		X	-	-
Rise delay	td01		-	X	-
Fall delay	td10		-	X	-
PAM4 modulation	pam4_modulation	none	X	X	-
PAM4 mapping	pam4_mapping		X	X	-
IBIS AMI file	ibisAmiFile		X	-	-
IBIS AMI model	ibisAmiModel		X	-	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Use VRANGE value info in *.ibs file in IBIS-AMI simulation	ibisAmiVrange Mode		X	-	-
Use RAMP info in *.ibs file	ibisAmiUseRamp		X	-	-
Use VRANGE info in *.ibs file	ibisAmiUseVrange		X	-	-
Use linearized pull-up or pull-down info in *.ibs file	ibisAmiUseZsrc		X	-	-
Display second sinusoid	numofsines	0	-	X	-
Frequency name 2	fundname2		-	X	-
Frequency 2	freq2		-	X	-
Amplitude 2 (Vpk)	va2		-	X	-
Amplitude 2 (dBm)	vaDBm2		-	X	-
Phase for sinusoid 2	sinephase2		-	X	-
Display multi sinusoid	vecofsines	0	-	X	-
Number of frequencies	freqvec	0	-	X	-
Sinusoid frequency 1-9	freqv1-9		-	X	-
Sinusoid amplitude 0-9	vav0-9		-	X	-
Sinusoid amplitude 1-9 (DBm)	vaDBmv1-9		-	X	-
Sinusoid phase 1-9	phasev1-9		-	X	-
Sinusoid max harmonic 1-9	harmv1-9		-	X	-
Display modulation params	modulation	0	-	X	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Amp modulation index 1	ammodindex		-	X	-
Amp modulation frequency 1	ammodfreq		-	X	-
Amp modulation phase 1	ammodphase		-	X	-
FM index 1	fmmodindex		-	X	-
FM frequency 1	fmmodfreq		-	X	-
IQ modulation I File	imodfile		-	X	-
I File name is variable	imodfile_as_var	0	-	X	-
IQ modulation Q File	qmodfile		-	X	-
Q File name is variable	qmodfile_as_var	0	-	X	-
Number of FM files	filenums	none	-	X	-
Name of FM file 1	fmmodfile1		-	X	-
FM file1 name is variable	fmmodfile1_as_var	0	-	X	-
Name of FM file 2	fmmodfile2		-	X	-
FM file2 name is variable	fmmodfile2_as_var	0	-	X	-
Amplitude scale factor	scale		-	X	-
Power of PWL waveform	pwldbm		-	X	-
Filter for pwl source	pwlfilter		-	X	-
Rolloff factor	rolloff		-	X	-
Bandwidth of the pwl signal	pwlbandwidth		-	X	-
Delay time	tdPrbs		-	X	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Bit period	perPrbs		-	X	-
Transition reference	transitionreference		-	X	-
Lower threshold(percent)	lowerthreshold		-	X	-
Upper threshold(percent)	upperthreshold		-	X	-
Edge type	edgetypePrbs		-	X	-
triggerValue	triggerValue	1	-	X	-
Trigger	trigger	Internal	-	X	-
Threshold	triggerthreshold		-	X	-
LFSR Mode	lfsrmode		-	X	-
Browse and specify bit file	prbsFileBox	0	-	-	-
Bit data file	prbsFile		-	X	-
Bit data file name is variable	prbsFile_as_var	0	-	X	-
Seed	lfsrseed	1 3 5 7 9 11 13 15 17 19 21 23 25 27 29 31	-	X	-
Taps	lfsrtaps		-	X	-
RJ(rms)	rjrms		-	X	-
RJ(seed)	rjseed		-	X	-
Number of periodic jitters	nPeriodicJitter	0	-	X	-
PJ1(amplitude)	pjump1		-	X	-
PJ1(frequency)	pjfreq1		-	X	-
PJ1(type)	pjtype1	sine	-	X	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
PJ2(amplitude)	pjamp2		-	X	-
PJ2(frequency)	pjfreq2		-	X	-
PJ2(type)	pjtype2	sine	-	X	-
PJ3(amplitude)	pjamp3		-	X	-
PJ3(frequency)	pjfreq3		-	X	-
PJ3(type)	pjtype3	sine	-	X	-
Display small signal params	smallSig	0	X	X	-
HBAC Magnitude	hbacm		X	-	-
HBAC Phase	hbacp		X	-	-
SNAC Magnitude	snacm		X	-	-
SNAC Phase	snacp		X	-	-
PAC magnitude	pacm		-	X	-
PAC magnitude (dBm)	pacDBm		-	X	-
PAC phase	pacp		-	X	-
AC Magnitude	acm		X	X	-
AC Phase	acp		X	X	-
XF magnitude	xfm		-	X	-
Display resistance parameters	resistanceParam	0	X	-	-
Open DC Connection	dcopen	""	X	-	-
DC Resistance	rdc		X	-	-
AC Resistance	rac		X	-	-
HB Resistance	rhb		X	-	-
HBAC Resistance	rhbac		X	-	-
TRAN Resistance	rtran		X	-	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Display temperature params	tempParam	0	-	X	-
Linear temperature coefficient	tc1		-	-	-
Quadratic temperature coefficient	tc2		-	X	-
Nominal temperature	tnom		-	X	-
Display noise parameters	noiseParam	0	-	X	-
Generate noise?	isnoisyNoiseModel		-	X	-
Noise temperature	noisetemp		-	-	-
Noise entry method	noiseEntryMethod	File	-	X	-
Number of noise/frequency pairs	FNpairs	0	-	X	-
Frequency1-50	F1-50		-	X	-
Noise1-50	N1-50		-	X	-
Noise file name	noisefile		-	X	-
Noise file name is variable	noise_file_as_var	0	X	X	-
Noise type	noiseModelType	Noise Voltage(V ² /Hz)	-	X	-
Interpolation method	noiseModelInterMethod		-	X	-
High-Z impedance	highz		-	X	-
Min high-Z transition width	min_z_transition_width		-	X	-
Multiplier	m		-	X	-
Version	version	2	-	-	-

Note:

- When frequency/noise pairs have empty values or the number of frequency/noise pairs does not match with the FNpairs parameter, the Spectre netlister shows an error in the console.
- The default of noise_file_as_var parameter is *false*. If the noise_file parameter is set as *NOISEFILE*, the netlisting behavior for noise_file and noise_file_as_var parameters is as follows:

CDF Parameter	hspiceD Output	spectre Output
noise_file_as_var		
false (default)	noisefile='NOISEFILE'	noisefile="NOISEFILE"
true	noisefile=str(NOISEFILE)	noisefile=NOISEFILE

- For Hspice-format netlisting, a pair of single quotes is added to noise_file parameter when noise_file_as_var is *false*. When noise_file_as_var parameter is *true*, `str(NOISEFILE)` is used as the value format.
- For Spectre-format netlisting, a pair of double quotes is added when noise_file_as_var parameter is *false*. No single/double quotes are added when noise_file_as_var parameter is *true*.

Controlling Version Number for Legacy Designs

To control the version number for legacy designs, create and use the `dbAnalogLibPortVersion` Tcl preference before the analogLib library is loaded.

Updating Port Number Automatically

You can create and use the `dbAnalogLibAutoUpdatePortNum` Tcl preference to control whether to automatically update the port number of port family devices.

Using port to Set up Harmonic Balance or Shooting PSS simulations

The sine type is commonly used to set up single- and multi-tone Harmonic Balance (HB) and Shooting Newton simulations. The parameter **Impedance** sets the value of the port impedance (default 50 Ohms). DC Value is used to set the value to use in DC simulations, which defaults to the waveform's value at t=0.

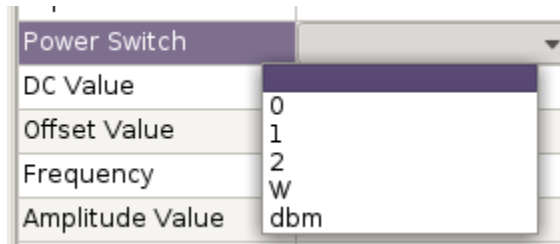
Note:

HB Tones parameters are supported only for legacy reasons in the PrimeSim HSPICE tool, but their ongoing use is not recommended.

Parameters	
Prompt	Value
Source Type	sine
Port Number	1
Impedance	
Power Switch	
DC Value	

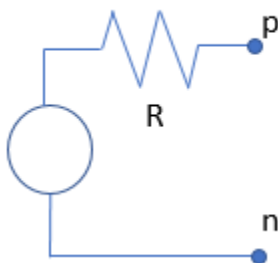
Use the **Power Switch** parameter to control how the port's amplitude is interpreted:

- 0 or unspecified if it specifies the open-circuit voltage
- 1 or W if it specifies available power in Watts
- 2 or dbm for available power in dbm



A note about port's amplitude definition:

The port's equivalent model is shown in the following figure.



When **Power Switch=0**, the port's amplitude is defined as the open-circuit voltage across (p, n). Starting with FineSim and PrimeSim SPICE, the option `.option port_element_voltage_matchload` can be used to tell the engines to treat the amplitude as the voltage across (p, n) when a matched resistor is connected across (p, n).

Single-sinusoid Specification

The single-sine source implements the function:

$$v(t) = v_0 + v_a e^{-q(t-t_d)} \sin(2\pi f(t - t_d) + \varphi)$$

Offset Value	0
Frequency	
Amplitude Value	
Phase for sinusoid	
Delay Time	
Damping Factor	

Simulation defaults are zero for each of the parameters. The mapping between the parameters and the functional expression is as shown in the following table.

Parameter	Maps to ...
Offset Value	v_0
Frequency	f
Amplitude Value	v_a
Phase for sinusoid	φ
Delay Time	t_d
Damping Factor	q

Damping factor is used in transient simulations, but ignored in HB and Shooting Newton simulations.

Two-sinusoid specification

To set up a two-tone source, click **Display second sinusoid**.

Display second ...	☒	
Frequency 2		
Amplitude 2 (Vpk)		
Phase for sinus...		

The meaning of the second sinusoid parameters is analogous to the first. It implements the function:

$$v(t) = v_0 + e^{-q(t-t_d)} [v_{a1} \sin(2\pi f_1(t - t_d) + \varphi_1) + v_{a2} \sin(2\pi f_2(t - t_d) + \varphi_2)]$$

As before, **Power Switch** determines the meaning of v_{a2} , whether it is measured as open-circuit voltage or as available power in Watts or dbm.

Multi-sinusoid specification

You can add additional sinusoids using **Display multi sinusoid** and specifying the number of frequencies to add.

Number of frequencies	3
Sinusoid Frequency 1	
Sinusoid Ampl 1 (Vpk)	
Sinusoid Ampl 1 (dbm)	
Sinusoid Phase 1	
Sinusoid Maxharm 1	
Sinusoid Frequency 2	
Sinusoid Ampl 2 (Vpk)	
Sinusoid Ampl 2 (dbm)	
Sinusoid Phase 2	
Sinusoid Maxharm 2	
Sinusoid Frequency 3	
Sinusoid Ampl 3 (Vpk)	
Sinusoid Ampl 3 (dbm)	
Sinusoid Phase 3	
Sinusoid Maxharm 3	

Delay Time, **Damping Factor**, and **Power Switch** do not apply to additional sinusoids. Instead of **Power Switch**, additional sinusoids are specified in Volts by setting **Sinusoid Ampl N (Vpk)**, or in dbm by setting **Sinusoid Ampl N (dbm)**. You can specify levels in Volts or dbm, but not both.

Maxharm settings are currently ignored.

Although you can treat multi-sinusoids as additions to single- and two-tone specifications (to specify tone 3 and higher), it is recommended to not to mix them together. Instead, use

one form or the other but not both. When using the multi-sinusoid form, the functional form of the waveform is:

$$v(t) = v_0 + \sum_{n=1}^N v_{ampl\ n} \sin (2\pi f_n + \varphi_n)$$

$v_{ampl\ n}$ corresponds to **Sinusoid Ampl n (Vpk)** or **Sinusoid Ampl n (dbm)**, f_n is **Sinusoid Frequency n**, and so forth. v_0 is the **Offset Value** parameter.

Small signal parameters

Small signal parameters set the source amplitudes in AC, small-signal HB (HBAC), and small-signal Shooting Newton (SNAC) analysis. **Power Switch** determines whether the source is specified in open-circuit Volts, available Watts, or available dbm.

Display small signal para...	☒
HBAC Magnitude	
HBAC Phase	
SNAC Magnitude	
SNAC Phase	
AC Magnitude	
AC Phase	

Resistance parameters

The use of resistance parameters is not recommended.

Display resistance param...	☒
Open DC Connection	
DC Resistance	
AC Resistance	
HB Resistance	
HBAC Resistance	
TRAN Resistance	

Noise parameters

Display noise parameters	☒
Generate noise?	no
Noise temperature	

By default, the port's noise voltage contribution is

$$\sqrt{4KTR},$$

where R is port **Impedance** and the default value of T is the circuit temperature. You can turn off noise contribution using **Generate noise**. You can set T to temperature different than circuit using **Noise temperature**.

In noise figure simulations using .LIN, .HBAC, or .SNAC analysis, the input port's temperature is 290 if unspecified, and the output port's contribution is ignored.

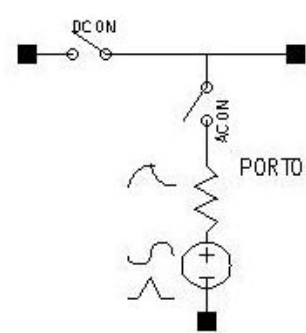
See Also

- [Port Element](#) in the *PrimeSim Continuum User Guide: Elements*

port3t

The port3t element is a port with T-shaped connection of DC (chock inductor) and AC (blocking capacitor) control.

The port element is a voltage source with internal resistance. See [port](#).



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Frequency name 1	fundname	-	-	x	-
Frequency name 2	fundname2	-	-	x	-
Noise file name	noisefile	-	-	x	-
Noise file name is variable	noise_file_as_ 0 var	-	-	X	-
File name	fileName	-	-	x	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Display second sinusoid	numofsines	0	-	x	-
Display modulation params	modulation	0	-	x	-
Display temperature params	tempParam	0	-	x	-
Display small signal params	smallSig	0	-	x	-
Display noise parameters	noiseParam	0	-	x	-
Num. of noise/freq pairs	FNpairs	0	-	x	-
Freq 1-10	F1-10		-	x	-
Noise 1-10	N1-10		-	x	-
Waveform Entry Method	pwlEntryMethod	File	-	x	-
Noise Entry Method	noiseEntryMethod	File	-	x	-
Number of PWL/Time pairs	tvpairs	0	-	x	-
Time 1-50	t1-50		-	x	-
Source type	sourceType	port	-	x	-
Resistance	r	50	-	x	-
Chock ind for net analyser	lchock		-	x	-
Blocking cap for net analyser	cblock		-	x	-
Port number	num		-	x	-
DC voltage	vdc		-	x	-
Delay time	td		-	x	-
Zero value	val0		-	x	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
One value	val1		-	X	-
Period of waveform	per		-	X	-
Rise time	tr		-	X	-
Fall time	tf		-	X	-
Pulse width	pw		-	X	-
Rise time start	td1		-	X	-
Rise time constant	tau1		-	X	-
Fall time start	td2		-	X	-
Fall time constant	tau2		-	X	-
Frequency 1	freq		-	X	-
Amplitude 1 (Vpk)	va		-	X	-
Amplitude 1 (dBm)	vaDBm		-	X	-
Phase for Sinusoid 1	sinephase		-	X	-
Sine DC level	sinedc		-	X	-
Frequency 2	freq2		-	X	-
Amplitude 2 (Vpk)	va2		-	X	-
Amplitude 2 (dBm)	vaDBm2		-	X	-
Phase for Sinusoid 2	sinephase2		-	X	-
FM modulation index	fmmodindex		-	X	-
FM modulation freq	fmmodfreq		-	X	-
AM modulation index	ammodindex		-	X	-
AM modulation freq	ammodfreq		-	X	-
AM modulation phase	ammodphase		-	X	-
Damping factor	theta		-	X	-
Linear temp. coefficient	tc1		-	X	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Quadratic temp. coeff.	tc2		-	x	-
Nominal temperature	tnom		-	x	-
PAC Magnitude (Vpk)	pacm		-	x	-
PAC Magnitude (dBm)	pacDBm		-	x	-
PAC phase	pacp		-	x	-
AC Magnitude (Vpk)	acm		-	x	-
AC phase	acp		-	x	-
XF Magnitude (Vpk)	xfm		-	x	-
Noise temperature	noisetemp		-	x	-
DC offset	offset		-	x	-
Amplitude scale factor	scale		-	x	-
Time scale factor	stretch		-	x	-
Period	pwlperiod		-	x	-
Power of PWL waveform	pwldbm		-	x	-
Transition width	twidth		-	x	-
Multiplier	m		-	x	-
Display multi sinusoid	vecofsines	0	-	x	-
Number of Frequencies	freqvec	0	-	x	-
Sinusoid Frequency 1-9	freqv1-9		-	x	-
Sinusoid Ampl 1-9 (Vpk)	vav1-9		-	x	-
Sinusoid Ampl 1-9 (dbm)	vaDBmv1-9		-	x	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Sinusoid Phase 1-9	phasev1-9		-	x	-
Sinusoid Maxharm 1-9	harmv1-9		-	x	-
Number of FM Files	filenums	none	-	x	-
Name of FM File1	fmmodfile1		-	x	-
Name of FM File2	fmmodfile2		-	x	-
Reactance	x		-	x	-
Source type	srcType	sine	-	x	-
Rise delay	td01		-	x	-
Fall delay	td10		-	x	-
Exp period	expperiod		-	x	-
Desired rms value	rms		-	x	-
Breakpoints	allbrkpts		-	x	-

Note:

- The default of noise_file_as_var parameter is *false*. If the noisefile parameter is set as *NOISEFILE*, the netlisting behavior for noisefile and noise_file_as_var parameters is as follows:

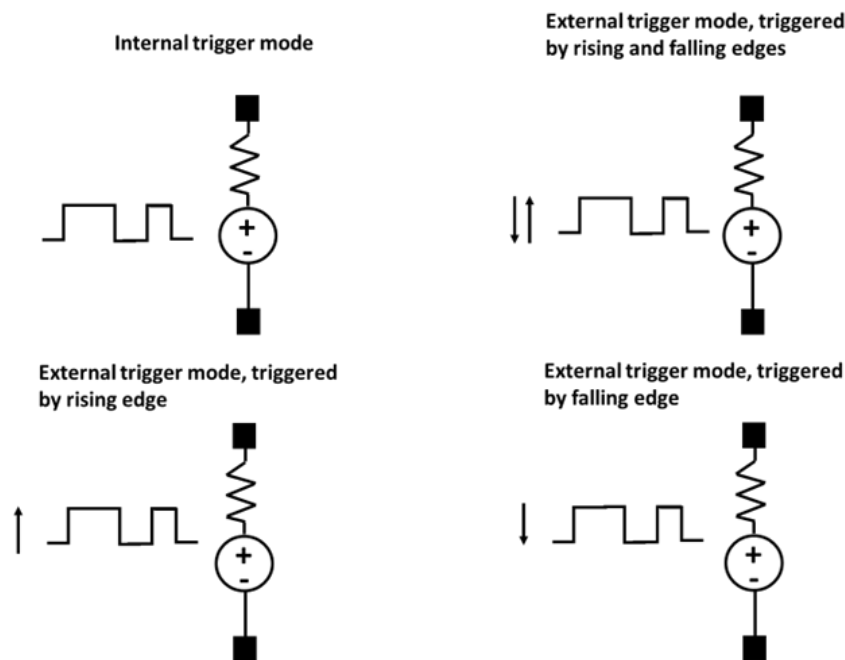
CDF Parameter	hspiceD Output	spectre Output
noise_file_as_var		
false (default)	noisefile='NOISEFILE'	noisefile="NOISEFILE"
true	noisefile=str(NOISEFILE)	noisefile=NOISEFILE

- For Hspice-format netlisting, a pair of single quotes is added to noisefile parameter when noise_file_as_var is *false*. When noise_file_as_var parameter is *true*, str(NOISEFILE) is used as the value format.
- For Spectre-format netlisting, a pair of double quotes is added when noise_file_as_var parameter is *false*. No single/double quotes are added when noise_file_as_var parameter is *true*.

pprbs

Port element with a pseudo random binary sequence function. This element is relevant only for Spectre netlisting and simulation at this moment.

The pprbs device is a PCell. Its PCell behavior is affected by the triggerValue parameter. The following figure shows four vprbs devices that have different trigger modes, which are Internal (on the upper-left), External by both edges (on the upper-right), External by falling edges (on the lower-left), and External by rising edges (on the lower-right) modes. The internal trigger mode has 2 pins (PLUS and MINUS) and other three external trigger modes have 4 pins (P1, M1, P2, and M2). The netlisting procedure writes out their pins accordingly.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Resistance	r	50	-	X	-
Port number	num		-	X	-
triggerValue	triggerValue	1	-	X	-
Delay time	tdPrbs		-	X	-
Zero value	val0		-	X	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
One value	val1		-	X	-
Bit period	perPrbs	1n	-	X	-
Rise time	tr		-	X	-
Fall time	tf		-	X	-
Transition reference	transitionreference		-	X	-
Lower threshold(percent)	lowerthreshold		-	X	-
Upper threshold(percent)	upperthreshold		-	X	-
Edge type	edgetype		-	X	-
Trigger	trigger	Internal	-	X	-
Threshold	triggerthreshold		-	X	-
LFSR Mode	lfsrmode	PN32	-	X	-
Browse and specify bit file	prbsFileBox	0	-	X	-
Bit data file	prbsFile		-	X	-
Seed	lfsrseed	1 3 5 7 9 11 13 15 17 19 21 23 25 27 29 31	-	X	-
Taps	lfsrtaps		-	X	-
RJ(rms)	rjrms		-	X	-
RJ(seed)	rjseed		-	X	-
Number of periodic jitters	nPeriodicJitter	0	-	X	-
PJ1(amplitude)	pjump1		-	X	-
PJ1(frequency)	pjfreq1		-	X	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
PJ1(type)	pjtype1	sine	-	X	-
PJ2(amplitude)	pjamp2		-	X	-
PJ2(frequency)	pjfreq2		-	X	-
PJ2(type)	pjtype2	sine	-	X	-
PJ3(amplitude)	pjamp3		-	X	-
PJ3(frequency)	pjfreq3		-	X	-
PJ3(type)	pjtype3	sine	-	X	-
Pattern Parameter data	data		-	X	-
Reference Value	ref		-	X	-
Random delay time	jitter		-	X	-
Seed	seed		-	X	-
Feedback Shift Register	taps		-	X	-
Source type	srcType	prbs	-	X	-
Multiplier	m		-	X	-
Version	version	2	-	-	-

Example

```
PORT0 (out 0) port r=50 type=prbs period=1n registerlength=32
lfsrseed=[1 3 5 7 \
 9 11 13 15 17 19 21 23 25 27 29 31]
```

Controlling Version Number for Legacy Designs

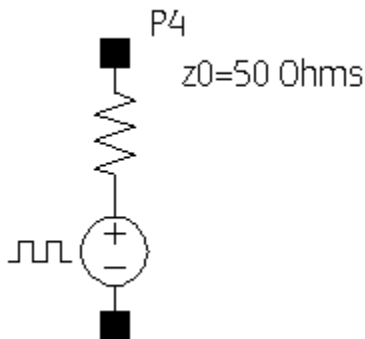
To control the version number for legacy designs, create and use the [dbAnalogLibPprbsVersion](#) Tcl preference before the analogLib library is loaded.

Updating Port Number Automatically

You can create and use the [dbAnalogLibAutoUpdatePortNum](#) Tcl preference to control whether to automatically update the port number of port family devices.

ppulse

Port element with pulse transient function.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Source Type	sourceType	port	-	-	-
Source type	srcType	pulse	-	X	-
Port Number	num		X	X	-
Impedance	z0	50	X	-	-
Resistance	r	50	-	X	-
Power Switch	power		X	-	-
DC Value	dc		X	-	-
DC Voltage	vdc		-	X	-
Value 1	v1	0	X	X	-
Value 2	v2	0	X	X	-
Delay Time	td	0	X	X	-
Rise Time	tr		X	X	-
Fall Time	tf		X	X	-
Pulse Width	pw		X	X	-
Period	per		X	X	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
# HB Tones	hbtones	0	X	-	-
HB Magnitude 1-7	hbmag1-7		X	-	-
HB Phase1-7	hbphase1-7	0	X	-	-
HB Harmonic1-7	hbharm1-7	1	X	-	-
HB Tone1-7	hbtone1-7	1-7	X	-	-
HB ModHarm1-7	hbmodharm1-7		X	-	-
HB ModTone1-7	hbmodtone1-7		X	-	-
Open DC Connection	dcopen		X	-	-
DC Resistance	rdc		X	-	-
AC Resistance	rac		X	-	-
HB Resistance	rhb		X	-	-
HBAC Resistance	rhbac		X	-	-
TRAN Resistance	rtran		X	-	-
HBAC Magnitude	hbacm		X	-	-
HBAC Phase	hbacp		X	-	-
SNAC Magnitude	snacm		X	-	-
SNAC Phase	snacp		X	-	-
Noise file name	noisefile		-	X	-
Noise file name is variable	noise_file_as_var	0	-	X	-
Number of noise/frequency pairs	FNpairs	0	-	X	-
Frequency1-50	F1-50		-	X	-
Noise1-50	N1-50		-	X	-
Frequency name for fundamental frequency	fundname		-	X	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Multiplier	m		-	X	-
AC Magnitude	acm		X	X	-
AC Phase	acp		X	X	-
PAC magnitude	pacm		-	X	-
PAC phase	pacp		-	X	-
XF magnitude	xfm		-	X	-
Temp coefficient 1	tc1		-	X	-
Temp coefficient 2	tc2		-	X	-
Nominal temperature	tnom		-	X	-

Note:

- When frequency/noise pairs have empty values or the number of frequency/noise pairs does not match with the FNpairs parameter, the Spectre netlister shows an error in the console.
- The default of noise_file_as_var parameter is *false*. If the noise file parameter is set as *NOISEFILE*, the netlisting behavior for noise file and noise_file_as_var parameters is as follows:

CDF Parameter	hspiceD Output	spectre Output
noise_file_as_var		
false (default)	noisefile='NOISEFILE'	noisefile="NOISEFILE"
true	noisefile=str(NOISEFILE)	noisefile=NOISEFILE

- For Hspice-format netlisting, a pair of single quotes is added to noise file parameter when noise_file_as_var is *false*. When noise_file_as_var parameter is *true*, `str(NOISEFILE)` is used as the value format.
- For Spectre-format netlisting, a pair of double quotes is added when noise_file_as_var parameter is *false*. No single/double quotes are added when noise_file_as_var parameter is *true*.

Example

```
p4 n1 gnd! port=1 dc=3.3 pulse ( 0 2.5 0 50p 50p 2n 4n ) z0=50
```

See Also

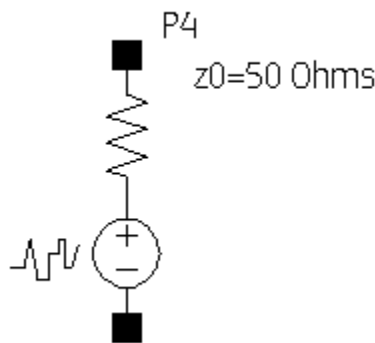
- [Port Element](#) in the *HSPICE® User Guide: Elements*

Updating Port Number Automatically

You can create and use the `dbAnalogLibAutoUpdatePortNum` Tcl preference to control whether to automatically update the port number of port family devices.

ppwl

Port element with PWL transient function.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Source Type	sourceType	port	-	-	-
Source Type	srcType	pwl	-	X	
Port Number	num		X	X	-
Impedance	z0	50	X	-	-
Resistance	r	50	-	X	-
Power Switch	power		X	-	-
DC Value	dc		X	-	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
DC Voltage	vdc		-	X	-
Time Value Pairs	tvpairs	2	-	X	-
Time1-50	t1-50		X	X	-
Value1-50	v1-50	0	X	X	-
Repeat	rpt		-	-	-
Repeat Stop Time	rstop		X	-	-
Repeat Stop Value	stopvalue		X	-	-
Repeat Stop Slope	stopslope		X	-	-
Delay Time	td	0	-	X	-
AC Magnitude	acm		X	X	-
AC Phase	acp		X	X	-
# HB Tones	hbtones	0	X	-	-
HB Magnitude 1-7	hbmag1-7		X	-	-
HB Phase1-7	hbphase1-7	0	X	-	-
HB Harmonic1-7	hbharm1-7	1	X	-	-
HB Tone1-7	hbtone1-7	1-7	X	-	-
HB ModHarm1-7	hbmodharm1-7		X	-	-
HB ModTone1-7	hbmodtone1-7		X	-	-
Open DC Connection	dcopen		X	-	-
DC Resistance	rdc		X	-	-
AC Resistance	rac		X	-	-
HB Resistance	rhb		X	-	-
HBAC Resistance	rhbac		X	-	-
TRAN Resistance	rtran		X	-	-
HBAC Magnitude	hbacm		X	-	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
HBAC Phase	hbacp		X	-	-
SNAC Magnitude	snacm		X	-	-
SNAC Phase	snacp		X	-	-
Number of noise/frequency pairs	FNpairs	0	-	X	-
Noise file name	noisefile		-	X	-
Noise file name is variable	noise_file_as_var	0	-	X	-
Frequency1-50	F1-50		-	X	-
Noise1-50	N1-50		-	X	-
Frequency name for fundamental frequency	fundname		-	X	-
Multiplier	m		-	X	-
Noise temperature	noisetemp		-	X	-
Nominal temperature	tnom		-	X	-
PAC magnitude	pacm		-	X	-
PAC phase	pacp		-	X	-
PWL period	pwlperiod		-	X	-
Amplitude scale factor	scale		-	X	-
Transition width	twidht		-	X	-
Desired rms value	rms		-	X	-
Time scaling factor	stretch		-	X	-
Temp coefficient 1	tc1		-	X	-
Temp coefficient 2	tc2		-	X	-
XF magnitude	xfm		-	X	-
Offset voltage	vo		-	X	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Breakpoints	allbrkpts		-	X	-

Note:

- When frequency/noise pairs have empty values or the number of frequency/noise pairs does not match with the FNpairs parameter, the Spectre netlister shows an error in the console.
- The default of noise_file_as_var parameter is *false*. If the noise file parameter is set as *NOISEFILE*, the netlisting behavior for noise file and noise_file_as_var parameters is as follows:

CDF Parameter	hspiceD Output	spectre Output
noise_file_as_var		
false (default)	noisefile='NOISEFILE'	noisefile="NOISEFILE"
true	noisefile=str(NOISEFILE)	noisefile=NOISEFILE

- For Hspice-format netlisting, a pair of single quotes is added to noise file parameter when noise_file_as_var is *false*. When noise_file_as_var parameter is *true*, str(NOISEFILE) is used as the value format.
- For Spectre-format netlisting, a pair of double quotes is added when noise_file_as_var parameter is *false*. No single/double quotes are added when noise_file_as_var parameter is *true*.

Example

```
p4 n1 gnd! port=1 dc=0 pw1 ( 0 0.0 1n 2.5 td=0 ) z0=50
```

See Also

- [Port Element](#) in the *HSPICE® User Guide: Elements*

Controlling Version Number for Legacy Designs

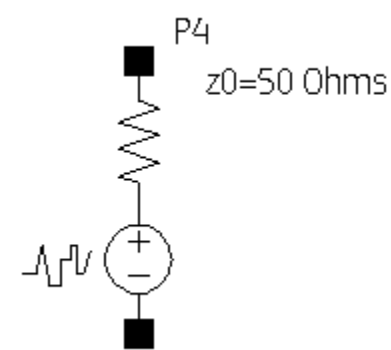
To control the version number for legacy designs, create and use the [dbAnalogLibPpwlVersion](#) Tcl preference before the analogLib library is loaded.

Updating Port Number Automatically

You can create and use the `dbAnalogLibAutoUpdatePortNum` Tcl preference to control whether to automatically update the port number of port family devices.

ppwlf

Port element with PWL transient function specified using a file, containing time voltage value pairs.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Source Type	sourceType	port	-	-	-
Source Type	srcType	pwlf	-	X	-
Port Number	num		X	X	-
Impedance	z0	50	X	-	-
Resistance	r	50	-	X	-
Power Switch	power		X	-	-
DC Value	dc		X	-	-
DC Voltage	vdc		-	X	-
PWL File	fileName		X	X	-
File name is variable	file_as_var	0	X	X	-
Repeat	rpt		X	-	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Repeat Stop Time	rstop		X	-	-
Repeat Stop Value	stopvalue		X	-	-
Repeat Stop Slope	stopslope		X	-	-
Delay Time	td	0	X	X	-
AC Magnitude	acm		X	X	-
AC Phase	acp		X	X	-
# HB Tones	hbtones	0	X	-	-
HB Magnitude 1-7	hbmag1-7		X	-	-
HB Phase1-7	hbphase1-7	0	X	-	-
HB Harmonic1-7	hbharm1-7	1	X	-	-
HB Tone1-7	hbtone1-7	1-7	X	-	-
HB ModHarm1-7	hbmodharm1-7		X	-	-
HB ModTone1-7	hbmodtone1-7		X	-	-
Open DC Connection	dcopen		X	-	-
DC Resistance	rdc		X	-	-
AC Resistance	rac		X	-	-
HB Resistance	rhb		X	-	-
HBAC Resistance	rhbac		X	-	-
TRAN Resistance	rtran		X	-	-
HBAC Magnitude	hbacm		X	-	-
HBAC Phase	hbacp		X	-	-
SNAC Magnitude	snacm		X	-	-
SNAC Phase	snacp		X	-	-
PAC magnitude	pacm		-	X	-
PAC phase	pacp		-	X	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
PWL period	pwlperiod		-	X	-
Amplitude scale factor	scale		-	X	-
Transition width	twidth		-	X	-
Desired rms value	rms		-	X	-
Time scaling factor	stretch		-	X	-
Temp coefficient 1	tc1		-	X	-
Temp coefficient 2	tc2		-	X	-
XF magnitude	xfm		-	X	-
Number of noise/frequency pairs	FNpairs	0	-	X	-
Noise file name	noisefile		-	X	-
Noise file name is variable	noise_file_as_var	0	-	X	-
Frequency1-50	F1-50		-	X	-
Noise1-50	N1-50		-	X	-
Frequency name for fundamental frequency	fundname		-	X	-
Multiplier	m		-	X	-
Nominal temperature	tnom		-	X	-
Offset voltage	vo		-	X	-
Breakpoints	allbrkpts	no	-	X	-

Note:

- When frequency/noise pairs have empty values or the number of frequency/noise pairs does not match with the FNpairs parameter, the Spectre netlister shows an error in the console.
- The default of noise_file_as_var parameter is *false*. If the noise file parameter is set as *NOISEFILE*, the netlisting behavior for noise file and noise_file_as_var parameters is as follows:

CDF Parameter	hspiceD Output	spectre Output
noise_file_as_var		
false (default)	noisefile='NOISEFILE'	noisefile="NOISEFILE"
true	noisefile=str(NOISEFILE)	noisefile=NOISEFILE

- For Hspice-format netlisting, a pair of single quotes is added to noise file parameter when noise_file_as_var is *false*. When noise_file_as_var parameter is *true*, `str(NOISEFILE)` is used as the value format.
- For Spectre-format netlisting, a pair of double quotes is added when noise_file_as_var parameter is *false*. No single/double quotes are added when noise_file_as_var parameter is *true*.

Example

```
p4 n1 gnd! port=1 dc=0 pwl pwlfile='pwl_data.txt' td=0 z0=50
```

See Also

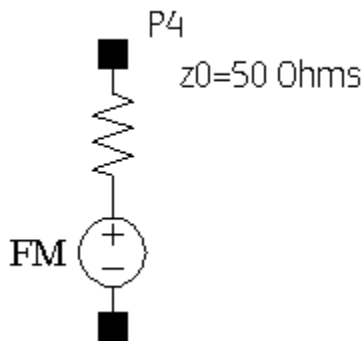
- [Port Element](#) in the *HSPICE® User Guide: Elements*

Updating Port Number Automatically

You can create and use the `dbAnalogLibAutoUpdatePortNum` Tcl preference to control whether to automatically update the port number of port family devices.

psffm

Port element with FM transient function.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Source Type	sourceType	port	-	-	-
Source Type	srcType	sffm	-	-	-
Port Number	num		X	-	-
Impedance	z0	50	X	-	-
Resistance	r	50	-	-	-
Power Switch	power		X	-	-
DC Value	dc		X	-	-
DC Voltage	vdc		-	-	-
Output Amplitude	va	0	X	-	-
Offset Value	vo	0	X	-	-
Carrier Frequency	freq	1	X	-	-
Modulation Index	mdi	0	X	-	-
Signal Frequency	fs		X	-	-
AC Magnitude	acm		X	-	-
AC Phase	acp		X	-	-
# HB Tones	hbtones	0	X	-	-
HB Magnitude 1-7	hbmag1-7		X	-	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
HB Phase1-7	hbphase1-7	0	X	-	-
HB Harmonic1-7	hbharm1-7	1	X	-	-
HB Tone1-7	hbtone1-7	1-7	X	-	-
HB ModHarm1-7	hbmodharm1-7		X	-	-
HB ModTone1-7	hbmodtone1-7		X	-	-
Open DC Connection	dcopen		X	-	-
DC Resistance	rdc		X	-	-
AC Resistance	rac		X	-	-
HB Resistance	rhb		X	-	-
HBAC Resistance	rhbac		X	-	-
TRAN Resistance	rtran		X	-	-
HBAC Magnitude	hbacm		X	-	-
HBAC Phase	hbacp		X	-	-
SNAC Phase	snacp		X	-	-

Example

```
p4 n1 gnd! port=1 dc=0 sffm ( 0.35 500m 400Meg 5 500k ) z0=50
```

See Also

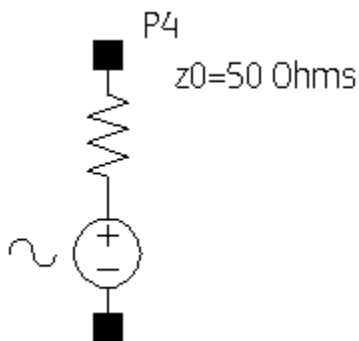
- [Port Element](#) in the *HSPICE® User Guide: Elements*

Updating Port Number Automatically

You can create and use the [dbAnalogLibAutoUpdatePortNum](#) Tcl preference to control whether to automatically update the port number of port family devices.

psin

Port element with sinusoidal transient function.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Source Type	sourceType	port	-	-	-
Source Type	srcType	sine	-	X	-
Port Number	num		X	X	-
Impedance	z0	50	X	-	-
Resistance	r	50	-	X	-
Power Switch	power		X	-	-
DC Value	dc		X	-	-
DC Voltage	vdc		-	X	-
Offset Value	vo	0	X	X	-
Amplitude Value	va		X	X	-
Frequency	freq		X	X	-
Delay Time	td	0	X	X	-
Damping Factor	theta	0	X	X	-
Phase for sinusoid	sinephase	0	X	X	-
AC Magnitude	acm		X	X	-
AC Phase	acp		X	X	-
# HB Tones	hbtones	0	X	-	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
HB Magnitude1-7	hbmag1-7		X	-	-
HB Phase1-7	hbphase1-7	0	X	-	-
HB Harmonic1-7	hbharm1-7	1	X	-	-
HB Tone1-7	hbtone1-7	1-7	X	-	-
HB ModHarm1-7	hbmodharm1-7		X	-	-
HB ModTone1-7	hbmodtone1-7		X	-	-
Open DC Connection	dcopen		X	-	-
DC Resistance	rdc		X	-	-
AC Resistance	rac		X	-	-
HB Resistance	rhb		X	-	-
HBAC Resistance	rhbac		X	-	-
TRAN Resistance	rtran		X	-	-
HBAC Magnitude	hbacm		X	-	-
HBAC Phase	hbacp		X	-	-
SNAC Magnitude	snacm		X	-	-
SNAC Phase	snacp		X	-	-
Amp modulation frequency	ammodfreq		-	X	-
Amp modulation index	ammodindex		-	X	-
Amp modulation phase	ammodphase		-	X	-
Number of FM Files	filenums	none	-	X	-
Name of FM File1	fmmodfile1		-	X	-
FM file1 name is variable	fmmodfile1_as _var	0	-	X	-
Name of FM file 2	fmmodfile2		-	X	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
FM file2 name is variable	fmmodfile2_as_var	0	-	X	-
FM frequency	fmmodfreq		-	X	-
FM index	fmmodindex		-	X	-
Frequency 2	freq2		-	X	-
Number of noise/frequency pairs	FNpairs	0	-	X	-
Noise file name	noisefile		-	X	-
Noise file name is variable	noise_file_as_var	0	-	X	-
Frequency1-50	F1-50		-	X	-
Noise1-50	N1-50		-	X	-
Frequency name for fundamental frequency	fundname		-	X	-
Second frequency name	fundname2		-	X	-
Multiplier	m		-	X	-
Noise temperature	noisetemp		-	X	-
Nominal temperature	tnom		-	X	-
PAC magnitude	pacm		-	X	-
PAC phase	pacp		-	X	-
Phase for sinusoid 2	sinephase2		-	X	-
Number of frequencies	freqvec	0	-	X	-
Sinusoid frequency 1-9	freqv1-9		-	X	-
Sinusoid phase 1-9	phasev1-9		-	X	-
Sinusoid max harmonic 1-9	harmv1-9		-	X	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Voltage level	sinedc	-	-	X	-
Temp coefficient 1	tc1	-	-	X	-
Temp coefficient 2	tc2	-	-	X	-
XF magnitude	xfm	-	-	X	-
Amplitude (dBm)	vaDBm	-	-	X	-
Amplitude 2	va2	-	-	X	-
Amplitude 2 (dBm)	vaDBm2	-	-	X	-
PAC magnitude (dBm)	pacDBm	-	-	X	-
Sinusoid amplitude 1-9 (DBm)	vaDBmv1-9	-	-	X	-

Note:

- When frequency/noise pairs have empty values or the number of frequency/noise pairs does not match with the FNpairs parameter, the Spectre netlister shows an error in the console.
- The default of noise_file_as_var parameter is *false*. If the noisefile parameter is set as *NOISEFILE*, the netlisting behavior for noisefile and noise_file_as_var parameters is as follows:

CDF Parameter	hspiceD Output	spectre Output
noise_file_as_var		
false (default)	noisefile='NOISEFILE'	noisefile="NOISEFILE"
true	noisefile=str(NOISEFILE)	noisefile=NOISEFILE

- For Hspice-format netlisting, a pair of single quotes is added to noisefile parameter when noise_file_as_var is *false*. When noise_file_as_var parameter is *true*, `str(NoiseFile)` is used as the value format.
- For Spectre-format netlisting, a pair of double quotes is added when noise_file_as_var parameter is *false*. No single/double quotes are added when noise_file_as_var parameter is *true*.
- If parameter va is empty, writes out 0 in the Hspice-format netlisting result. For Spectre netlister users, create and use the [dbAnalogLibPsinVersion](#) Tcl preference to get legacy designs.

Example

```
p4 n1 gnd! port=1 dc=0 sin ( 0.35 500m 400Meg 0 0 0 ) z0=50
```

See Also

- [Port Element](#) in the *HSPICE® User Guide: Elements*

Controlling Version Number for Legacy Designs

To control the version number for legacy designs, create and use the [dbAnalogLibPsinVersion](#) Tcl preference before the analogLib library is loaded.

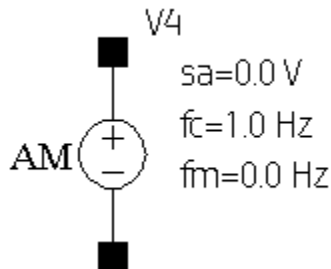
Updating Port Number Automatically

You can create and use the [dbAnalogLibAutoUpdatePortNum](#) Tcl preference to control whether to automatically update the port number of port family devices.

vam

Single-frequency AM voltage source.

$$sourcevalue = sa \cdot \{oc + SIN[2 \cdot \pi \cdot fm \cdot (Time - td)]\} \cdot SIN[2 \cdot \pi \cdot fc \cdot (Time - td)]$$



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
DC Voltage	vdc	0	X	-	-
Source Type	srcType	am	-	-	-
Signal Amplitude	sa	0	X	-	-
Carrier Frequency	fc	1	X	-	-
Modulation Frequency	fm	0	X	-	-
Offset Constant	oc	0	X	-	-
Delay Time	td	0	X	-	-
AC Magnitude	acm		X	-	-
AC Phase	acp		X	-	-
# HB Tones	hbtones	0	X	-	-
HB Magnitude1-7	hbmag1-7		X	-	-
HB Phase1-7	hbphase1-7	0	X	-	-
HB Harmonic1-7	hbharm1-7	1	X	-	-
HB Tone1-7	hbtone1-7	1-7	X	-	-
HB ModHarm1-7	hbmodharm1-7		X	-	-
HB ModTone1-7	hbmodtone1-7		X	-	-
Open DC Connection	dcopen		X	-	-
DC Resistance	rdc		X	-	-
AC Resistance	rac		X	-	-
HB Resistance	rhb		X	-	-
HBAC Resistance	rhbac		X	-	-
TRAN Resistance	rtran		X	-	-
HBAC Magnitude	hbacm		X	-	-
HBAC Phase	hbacp		X	-	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
SNAC Magnitude	snacm		X	-	-
SNAC Phase	snacp		X	-	-
Impedance	z0		-	-	-

Example

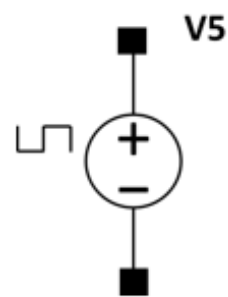
```
vp4 n1 gnd! dc=0 am ( 500m 0.0 200Meg 100Meg 0 )
```

See Also

- [Single-frequency AM Source](#) in the *HSPICE® User Guide: Elements*

vbit

Bit independent voltage source.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Pattern Parameter data	data		X	X	-
Pattern Parameter data is variable	data_as_var		X	X	-
Pattern Parameter rptstart	rptstart	1	X	X	-
Pattern Parameter rpttimes	rpttimes	0	X	X	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Trigger	trigger		-	X	-
RJ(rms)	rjrms		X	X	-
RJ(seed)	rjseed		X	X	-
Number of periodic jitters	nPeriodicJitter	0	X	X	-
PJ%d(amplitude)	pjamp1-3		X	X	-
PJ%d(frequency)	pjfreq1-3		X	X	-
PJ%d(type)	pjtype1-3	sine	X	X	-
One Value	val1		X	X	-
Zero Value	val0		X	X	-
Delay Time	td		X	X	-
Rise Time	tr	10f	X	X	-
Fall Time	tf	10f	X	X	-
Period	per		X	X	-
Source Type	srcType	bit	-	X	-

Example

In the example, the first b-string is 1011, which repeats once and then repeats from the second bit, which is 0. The pattern is repeated 5 times. The final b-string is 1011011011011011011.

```
v4 n1 n2 pat ( 1.2 0.25 0 1n 1n 5n b1011 r=5 rb=2 )
```

Where:

- High voltage level is 1.2 V
- Low voltage level is 0.25 V
- Time delay is 0 n
- Rise time is 1 n

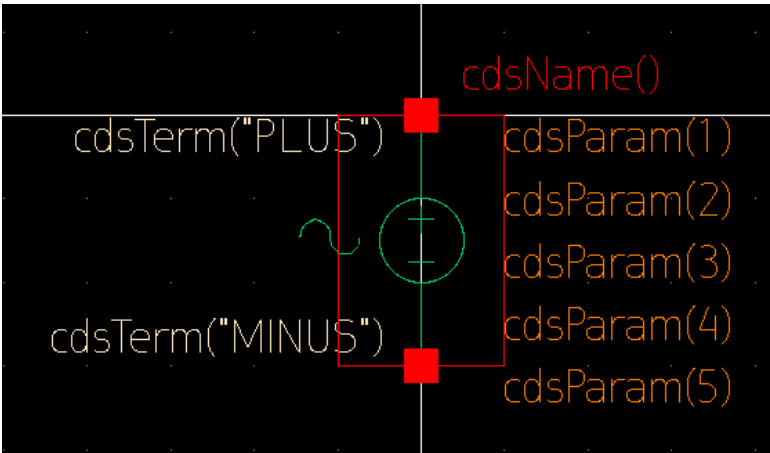
- Fall time is 1 n
- Sample time (period) is 5 n

See Also

- [Pattern Source](#) in the *HSPICE® User Guide: Elements*

VCOS

Voltage source.



CDF Parameters

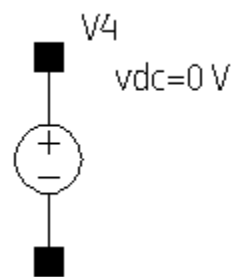
CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Source Type	srcType	cosine	-	-	-
DC Voltage	vdc	0	X	-	-
Voltage Offset	vo	0	X	-	-
Voltage Amplitude	va	0	X	-	-
Frequency	freq		X	-	-
Delay Time	td	0	X	-	-
Damping Factor	q	0	X	-	-
Phase for sinusoid	sinephase	0	X	-	-
Period Jitter	perjitter		X	-	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Seed	seed		X	-	-
AC Magnitude	acm		X	-	-
AC Phase	acp		X	-	-
# HB Tones	hbtones	0	X	-	-
HB Magnitude1-7	hbmag1-7		X	-	-
HB Phase1-7	hbphase1-7	0	X	-	-
HB Harmonic1-7	hbharm1-7	1	X	-	-
HB Tone1-7	hbtone1-7	1-7	X	-	-
HB ModHarm1-7	hbmodharm1-7		X	-	-
HB ModTone1-7	hbmodtone1-7		X	-	-
Open DC Connection	dcopen		-	-	-
DC Resistance	rdc		X	-	-
AC Resistance	rac		X	-	-
HB Resistance	rhb		X	-	-
HBAC Resistance	rhbac		X	-	-
TRAN Resistance	rtran		X	-	-
HBAC Magnitude	hbacm		X	-	-
HBAC Phase	hbacp		X	-	-
SNAC Magnitude	snacm		X	-	-
SNAC Phase	snacp		X	-	-
Amp modulation frequency	ammodfreq		-	-	-
Amp modulation index	ammodindex		-	-	-
Amp modulation phase	ammodphase		-	-	-
Number of FM Files	filenums	none	-	-	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Name of FM File1	fmmodfile1		-	-	-
Name of FM File2	fmmodfile2		-	-	-
FM frequency	fmmodfreq		-	-	-
FM index	fmmodindex		-	-	-
Frequency 2	freq2		-	-	-
Number of noise/frequency pairs	FNpairs	0	-	-	-
Noise file name	noisefile		-	-	-
Frequency 1-10	F1-10		-	-	-
Noise 1-10	N1-10		-	-	-
Frequency name for fundamental frequency	fundname		-	-	-
Second frequency name	fundname2		-	-	-
Noise temperature	noisetemp		-	-	-
Nominal temperature	tnom		-	-	-
PAC magnitude	pacm		-	-	-
PAC phase	pacp		-	-	-
Phase for sinusoid 2	sinephase2		-	-	-
Temp coefficient 1	tc1		-	-	-
Temp coefficient 2	tc2		-	-	-
XF magnitude	xfm		-	-	-
Phase shift	phase		-	-	-
Amplitude 2	va2		-	-	-
Impedance	z0		-	-	-

vdc

DC voltage source.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
DC Voltage	vdc	0	X	X	-
AC Magnitude	acm		X	X	-
AC Phase	acp		X	X	-
# HB Tones	hbtones	0	X	-	-
HB Magnitude1-7	hbmag1-7		X	-	-
HB Phase1-7	hbphase1-7	0	X	-	-
HB Harmonic1-7	hbharm1-7	1	X	-	-
HB Tone1-7	hbtone1-7	1-7	X	-	-
HB ModHarm1-7	hbmodharm1-7		X	-	-
HB ModTone1-7	hbmodtone1-7		X	-	-
Open DC Connection	dcopen		-	-	-
DC Resistance	rdc		X	-	-
AC Resistance	rac		X	-	-
HB Resistance	rhb		X	-	-
HBAC Resistance	rhbac		X	-	-
TRAN Resistance	rtran		X	-	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
HBAC Magnitude	hbacm		X	-	-
HBAC Phase	hbacp		X	-	-
SNAC Magnitude	snacm		X	-	-
SNAC Phase	snacp		X	-	-
Source Type	srcType	dc	-	X	-
Number of noise/frequency pairs	FNpairs	0	-	X	-
Noise file name	noisefile		X	X	-
Frequency1-50	F1-50		-	X	-
Noise1-50	N1-50		-	X	-
Nominal temperature	tnom		X	X	-
PAC magnitude	pacm		-	X	-
PAC phase	pacp		-	X	-
Temp coefficient 1	tc1		X-	X	-
Temp coefficient 2	tc2		X	X	-
XF magnitude	xfm		-	X	-
Impedance	z0		-	-	-

Note:

- When frequency/noise pairs have empty values or the number of frequency/noise pairs does not match with the FNpairs parameter, the Spectre netlister shows an error in the console.
- The default of noise_file_as_var parameter is *false*. If the noisefile parameter is set as *NOISEFILE*, the netlisting behavior for noisefile and noise_file_as_var parameters is as follows:

CDF Parameter	hspiceD Output	spectre Output
noise_file_as_var		

CDF Parameter	hspiceD Output	spectre Output
false (default)	noisefile='NOISEFILE'	noisefile="NOISEFILE"
true	noisefile=str(NOISEFILE)	noisefile=NOISEFILE

- For Hspice-format netlisting, a pair of single quotes is added to noisefile parameter when noise_file_as_var is *false*. When noise_file_as_var parameter is *true*, str(NOISEFILE) is used as the value format.
- For Spectre-format netlisting, a pair of double quotes is added when noise_file_as_var parameter is *false*. No single/double quotes are added when noise_file_as_var parameter is *true*.

Example

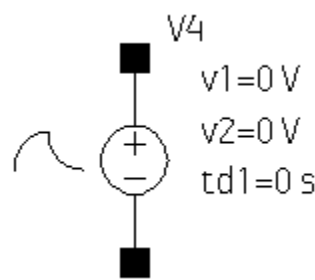
```
v4 n1 gnd! dc=2.5 ac=1
```

See Also

- [DC Sources](#) in the *HSPICE® User Guide: Elements*

vexp

Exponential voltage source.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
DC Voltage	vdc	0	X	X	-
Source Type	srcType	exp	-	X	-
Voltage1	v1	0	X	X	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Voltage2	v2	0	X	X	-
Rise Delay Time	td1	0	X	X	-
Fall Delay Time	td2		X	X	-
Rise Time Constant	tau1		X	X	-
Fall Time Constant	tau2	0	X	X	-
AC Magnitude	acm		X	X	-
AC Phase	acp		X	X	-
# HB Tones	hbtones	0	X	-	-
HB Magnitude1-7	hbmag1-7		X	-	-
HB Phase1-7	hbphase1-7	0	X	-	-
HB Harmonic1-7	hbharm1-7	1	X	-	-
HB Tone1-7	hbtone1-7	1-7	X	-	-
HB ModHarm1-7	hbmodharm1-7		X	-	-
HB ModTone1-7	hbmodtone1-7		X	-	-
Open DC Connection	dcopen		-	-	-
DC Resistance	rdc		X	-	-
AC Resistance	rac		X	-	-
HB Resistance	rhb		X	-	-
HBAC Resistance	rhbac		X	-	-
TRAN Resistance	rtran		X	-	-
HBAC Magnitude	hbacm		X	-	-
HBAC Phase	hbacp		X	-	-
SNAC Magnitude	snacm		X	-	-
SNAC Phase	snacp		X	-	-
Number of noise/frequency pairs	FNpairs	0	-	X	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Noise file name	noisefile	-	-	X	-
Noise file name is variable	noise_file_as_var	0	-	X	-
Frequency1-50	F1-50	-	-	X	-
Noise1-50	N1-50	-	-	X	-
Nominal temperature	tnom	-	X	X	-
PAC magnitude	pacm	-	-	X	-
PAC phase	pacp	-	-	X	-
Temp coefficient 1	tc1	-	X	X	-
Temp coefficient 2	tc2	-	X	X	-
XF magnitude	xfm	-	-	X	-
Delay time	td	-	-	X	-
Impedance	z0	-	-	-	-

Note:

- When frequency/noise pairs have empty values or the number of frequency/noise pairs does not match with the FNpairs parameter, the Spectre netlister shows an error in the console.
- The default of noise_file_as_var parameter is *false*. If the noisefile parameter is set as *NOISEFILE*, the netlisting behavior for noisefile and noise_file_as_var parameters is as follows:

CDF Parameter	hspiceD Output	spectre Output
noise_file_as_var		
false (default)	noisefile='NOISEFILE'	noisefile="NOISEFILE"
true	noisefile=str(NOISEFILE)	noisefile=NOISEFILE

- For Hspice-format netlisting, a pair of single quotes is added to noisefile parameter when noise_file_as_var is *false*. When noise_file_as_var parameter is *true*, `str(NOISEFILE)` is used as the value format.
- For Spectre-format netlisting, a pair of double quotes is added when noise_file_as_var parameter is *false*. No single/double quotes are added when noise_file_as_var parameter is *true*.

Example

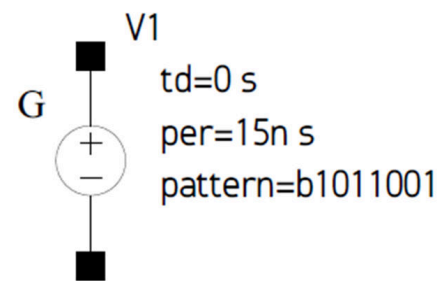
```
v4 n1 gnd! dc=2.5 ac=1 exp ( 0 5 5n 3n 0.5n 15n )
```

See Also

- [Exponential Source Function](#) in the *HSPICE® User Guide: Elements*

vgpulse

Gaussian voltage pulse.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
DC Voltage	vdc	0	X	-	-
Voltage1	v1		X	-	-
Voltage2	v2		X	-	-
Delay time	td		X	-	-
Pulse Width	Pw		X	-	-
Period	per		X	-	-

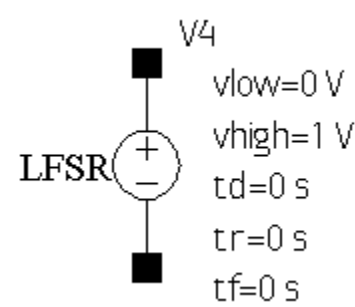
CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Pattern	pattern		X	-	-
Repeat	r		X	-	-
Repeat Starts	rb		X	-	-
Pseudo Random Seq.	bprbs		X	-	-

Example

```
v9 a gnd! dc=0 gpulse (0 1 0 5n 15n b1011001 bprbs=12 r=100n)
```

vlfsr

Pseudo Random Bit Generator Source (PRBS) function as an independent voltage source. The PRBS source uses a Linear Feedback Shift Register (LFSR) to generate a pseudo random bit sequence. HSPICE supports eight-bit data byte conversion to a 10-bit transmission character-8B/10B encoding for the LFSR keyword.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
DC Voltage	vdc	0	X	-	-
Source Type	srcType	lfsr	-	-	-
Voltage Low	vlow	0	X	-	-
Voltage High	vhigh	1	X	-	-
Delay Time	td	0	X	-	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Rise Time	tr	0	X	-	-
Fall Time	tf	0	X	-	-
Bit Rate	rate	1M	X	-	-
Seed	seed	1	X	-	-
Taps	taps	[1 2]	X	-	-
Output Resistance	rout		X	-	-
AC Magnitude	acm		X	-	-
AC Phase	acp		X	-	-
# HB Tones	hbtones	0	X	-	-
HB Magnitude1-7	hbmag1-7		X	-	-
HB Phase1-7	hbphase1-7	0	X	-	-
HB Harmonic1-7	hbharm1-7	1	X	-	-
HB Tone1-7	hbtone1-7	1-7	X	-	-
HB ModHarm1-7	hbmodharm1-7		X	-	-
HB ModTone1-7	hbmodtone1-7		X	-	-
Open DC Connection	dcopen		-	-	-
DC Resistance	rdc		X	-	-
AC Resistance	rac		X	-	-
HB Resistance	rhb		X	-	-
HBAC Resistance	rbac		X	-	-
TRAN Resistance	rtran		X	-	-
HBAC Magnitude	hbacm		X	-	-
HBAC Phase	hbacp		X	-	-
SNAC Magnitude	snacm		X	-	-
SNAC Phase	snacp		X	-	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
PAM4 modulation	pam4_modulation	none	X	-	-
PAM4 mapping	pam4_mapping		X	-	-
Impedance	z0		-	-	-

Example

The following example shows the pattern voltage source connected between nodes n1 and n2.

Example of LFSR, output is 1100011111001101:

```
v4 n1 n2 dc=0 lfsr ( 0 50u 0 1n 1n 10meg 3 [2,5] rout=10 )
```

Where:

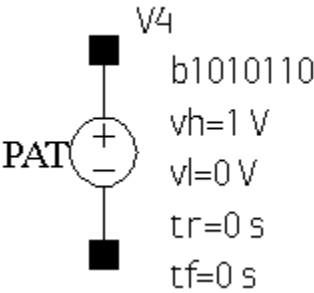
- The output low voltage is 0, and the output high voltage is 1 v.
- The delay time is 0 ms.
- The rise and fall times are each 1 ns.
- The bit rate is 10 meg bits/s.
- The seed is 3 (bits: 00011).
- The taps are at the output of the 2nd and 5th shift registers and are xor'd together as the input to the first shift register.
- The output resistance is 10 ohms.

See Also

- [Pseudo Random Bit Generator Source](#) in the *HSPICE® User Guide: Elements*

vpat

Pattern voltage source function. It uses four states, '1','0','m', and 'z', which represent the high, low, middle voltage, or current and high impedance state respectively. The series of these four states is a "b-string." HSPICE supports 8-bit data byte conversion to a 10-bit transmission character-8B/10B encoding, 64-bit data conversion to a 66-bit transmission character-64B/66B encoding, 128-bit data byte conversion to a 130-bit or 132-bit transmission character-128B/130B or 128B/132B encoding, for the PAT sources.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
DC Voltage	vdc	0	X	-	-
Source Type	srcType	pat	-	-	-
Voltage High	vh	1	X	-	-
Voltage Low	vl	0	X	-	-
Delay Time	td	0	X	-	-
Rise Time	tr	0	X	-	-
Fall Time	tf	0	X	-	-
Sample Time	tsample	1n	X	-	-
Data	data	b1010 110	X	-	-
Repeat	r		X	-	-
Repeat Bit	rb		X	-	-
AC Magnitude	acm		X	-	-
AC Phase	acp		X	-	-
RJ(rms)	rjrms		X	X	-
RJ(seed)	rjseed		X	X	-
Number of periodic jitters	nPeriodicJitter	0	X	X	-
PJ%d(amplitude)	pjump1-3		X	X	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
PJ%d(frequency)	pjfreq1-3		X	X	-
PJ%d(type)	pjtype1-3	sine	X	X	-
# HB Tones	hbtones	0	X	-	-
HB Magnitude1-7	hbmag1-7		X	-	-
HB Phase1-7	hbphase1-7	0	X	-	-
HB Harmonic1-7	hbharm1-7	1	X	-	-
HB Tone1-7	hbtone1-7	1-7	X	-	-
HB ModHarm1-7	hbmodharm1-7		X	-	-
HB ModTone1-7	hbmodtone1-7		X	-	-
Open DC Connection	dcopen		X	-	-
DC Resistance	rdc		X	-	-
AC Resistance	rac		X	-	-
HB Resistance	rhb		X	-	-
HBAC Resistance	rhbac		X	-	-
TRAN Resistance	rtran		X	-	-
HBAC Magnitude	hbacm		X	-	-
HBAC Phase	hbacp		X	-	-
SNAC Magnitude	snacm		X	-	-
SNAC Phase	snacp		X	-	-
PAM4 modulation	pam4_modulation	none	X	-	-
PAM4 mapping	pam4_mapping		X	-	-
Impedance	z0		-	-	-

Example

```
v4 n1 n2 pat ( 1.2 0.25 0 1n 1n 5n b1011 r=5 rb=2 )
```

In this example:

- High voltage level is 1.2 V
- Low voltage level is 0.25 V
- Time delay is 0 n
- Rise time is 1 n
- Fall time is 1 n
- Sample time is 5 n

The first b-string is 1011, which repeats once and then repeats from the second bit, which is 0. The pattern is repeated 5 times.

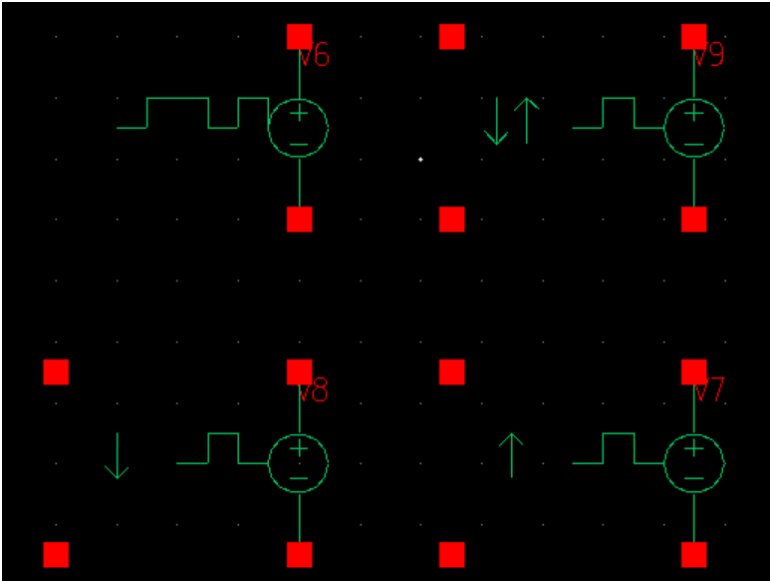
See Also

- [Pattern Source](#) in the *HSPICE® User Guide: Elements*

vprbs

Voltage source with pseudo random bit generator source transient function source.

The vprbs device is a PCell controlled by the trigger parameter. The following figure shows four vprbs devices that have different trigger modes, which are Internal (on the upper left), External by both edges (on the upper right), External by falling edges (on the lower left), and External by rising edges (on the lower right) modes. The internal trigger mode has two pins (PLUS and MINUS) and other 3 external trigger modes have 4 pins (P1 M1 P2 M2). The netlisting procedure writes out their pins accordingly.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
triggerValue	triggerValue	1	-	X	-
Delay time	tdPrbs		-	X	-
Zero value	val0		-	X	-
One value	val1		-	X	-
Bit period	perPrbs	1n	-	X	-
Rise Time	tr		-	X	-
Fall Time	tf		-	X	-
Transition reference	transitionreference		-	X	-
Lower threshold(percent)	lowerthreshold		-	X	-
Upper threshold(percent)	upperthreshold		-	X	-
Edge type	edgetype		-	X	-
Trigger	trigger	Internal	-	X	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Threshold	triggerthresh old		-	X	-
LFSR Mode	lfsrmode	PN32	-	X	-
Browse and specify bit file	prbsFileBox	0	-	-	-
Bit data file	prbsFile		-	X	-
Seed	lfsrseed	1 3 5 7 9 11 13 15 17 19 21 23 25 27 29 31	-	X	-
Taps	lfsrtaps		-	X	-
RJ(rms)	rjrms		-	X	-
RJ(seed)	rjseed		-	X	-
Number of periodic jitters	nPeriodicJitter	0	-	X	-
PJ%d(amplitude)	pjamp1-3		-	X	-
PJ%d(frequency)	pjfreq1-3		-	X	-
PJ%d(type)	pjtype1-3	sine	-	X	-
Pattern Parameter data	data		-	X	-
Reference Value	ref		-	X	-
Random delay time	jitter		-	X	-
Seed	seed		-	X	-
Feedback Shift Register	taps		-	X	-
Source Type	srcType	prbs	-	X	-
Version	version	2	-	-	-

Note:

The nPeriodicJitter value is limited to the range of 0 and 3. If the input value is smaller than 0, it is reset to 0. If the input value is greater than 3, it is reset to 3. No message is displayed for resetting the limit value.

Controlling Version Number for Legacy Designs

To control the version number for legacy designs, create and use the [dbAnalogLibVprbsVersion](#) Tcl preference before the analogLib library is loaded.

vpulse

This symbol specifies an independent source element and modes pulse generator.

CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
DC Current	vdc	0	X	X	-
Source Type	srcType	pulse	X	X	-
Voltage1	v1	0	X	X	-
Voltage2	v2	0	X	X	-
Delay Time	td	0	X	X	-
Rise Time	tr		X	X	-
Fall Time	tf		X	X	-
Pulse Width	pw		X	X	-
Period	per		X	X	-
Period Jitter	perjitter		X	-	-
Seed	seed		X	-	-
AC Magnitude	acm		X	X	-
AC Phase	acp		X	X	-
# HB Tones	hbtones	0	X	-	-
HB Magnitude1-7	hbmag1-7		X	-	-
HB Phase1-7	hbphase1-7	0	X	-	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
HB Harmonic1-7	hbharm1-7	1	X	-	-
HB Tone1-7	hbtone1-7	1-7	X	-	-
HB ModHarm1-7	hbmodharm1-7		X	-	-
HB ModTone1-7	hbmodtone1-7		X	-	-
Open DC Connection	dcopen		X	-	-
DC Resistance	rdc		X	-	-
AC Resistance	rac		X	-	-
HB Resistance	rhb		X	-	-
HBAC Resistance	rhbac		X	-	-
TRAN Resistance	rtran		X	-	-
HBAC Magnitude	hbacm		X	-	-
HBAC Phase	hbacp		X	-	-
SNAC Magnitude	snacm		X	-	-
SNAC Phase	snacp		X	-	-
Number of noise/frequency pairs	FNpairs	0	-	X	-
Noise file name	noisefile		-	X	-
Noise file name is variable	noise_file_as_var	0	-	X	-
Frequency1-50	F1-50		-	X	-
Noise1-50	N1-50		-	X	-
Frequency name for fundamental frequency	fundname		-	X	-
Nominal temperature	tnom		X	X	-
PAC magnitude	pacm		-	X	-
PAC phase	pacp		-	X	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Edge type	edgetype		X	X	-
Temp coefficient 1	tc1		X	X	-
Temp coefficient 2	tc2		X	X	-
XF magnitude	xfm		-	X	-
Impedance	z0		-	-	-

Note:

- When frequency/noise pairs have empty values or the number of frequency/noise pairs does not match with the FNpairs parameter, the Spectre netlister shows an error in the console.
- The default of noise_file_as_var parameter is *false*. If the noise file parameter is set as *NOISEFILE*, the netlisting behavior for noise file and noise_file_as_var parameters is as follows:

CDF Parameter	hspiceD Output	spectre Output
noise_file_as_var		
false (default)	noisefile='NOISEFILE'	noisefile="NOISEFILE"
true	noisefile=str(NOISEFILE)	noisefile=NOISEFILE

- For Hspice-format netlisting, a pair of single quotes is added to noise file parameter when noise_file_as_var is *false*. When noise_file_as_var parameter is *true*, `str(NOISEFILE)` is used as the value format.
- For Spectre-format netlisting, a pair of double quotes is added when noise_file_as_var parameter is *false*. No single/double quotes are added when noise_file_as_var parameter is *true*.

Example

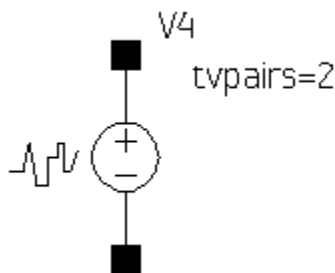
```
v1 plus minus dc=0 pulse ( 0 0 0 )
```

See Also

- [Independent Source Elements](#) in the *HSPICE® User Guide: Elements*

vpwl

PWL independent voltage source.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
DC Voltage	vdc	0	X	X	-
Source Type	srcType	pwl	-	X	-
Time Value Pairs	tvpairs	2	X	X	-
Display Pairs Count	dpcount	5	X	-	-
Time1-50	t1-50		X	X	-
Voltage1-50	v1-50	0	X	X	-
Repeat	r		X	-	-
Repeat Stop Time	rstop		X	-	-
Repeat Stop Value	stopvalue		X	-	-
Repeat Stop Slope	stopslope		X	-	-
Delay Time	td	0	X	X	-
AC Magnitude	acm		X	X	-
AC Phase	acp		X	X	-
# HB Tones	hbtone	0	X	-	-
HB Magnitude1-7	hbmag1-7		X	-	-
HB Phase1-7	hbphase1-7	0	X	-	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
HB Harmonic1-7	hbharm1-7	1	X	-	-
HB Tone1-7	hbtone1-7	1-7	X	-	-
HB ModHarm1-7	hbmodharm1-7		X	-	-
HB ModTone1-7	hbmodtone1-7		X	-	-
Open DC Connection	dcopen		-	-	-
DC Resistance	rdc		X	-	-
AC Resistance	rac		X	-	-
HB Resistance	rhb		X	-	-
HBAC Resistance	rhbac		X	-	-
TRAN Resistance	rtran		X	-	-
HBAC Magnitude	hbacm		X	-	-
HBAC Phase	hbacp		X	-	-
SNAC Magnitude	snacm		X	-	-
SNAC Phase	snacp		X	-	-
Number of noise/frequency pairs	FNpairs	0	-	X	-
Noise file name	noisefile		-	X	-
Noise file name is variable	noise_file_as_var	0	-	X	-
Frequency1-50	F1-50		-	X	-
Noise1-50	N1-50		-	X	-
Frequency name for fundamental frequency	fundname		-	X	-
Nominal temperature	tnom		X	X	-
PAC magnitude	pacm		-	X	-
PAC phase	pacp		-	X	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
PWL period	pwlperiod		-	X	-
Amplitude scale factor	scale		-	X	-
Edge type	edgetype	linear	-	X	-
Transition width	twidth		-	X	-
Time scaling factor	stretch		-	X	-
Temp coefficient 1	tc1		X	X	-
Temp coefficient 2	tc2		X	X	-
XF magnitude	xfm		-	X	-
Delay time	delay		-	X	-
Offset voltage	vo		-	X	-
Impedance	z0		-	-	-
PWL	pwl		-	-	-

Note:

- When frequency/noise pairs have empty values or the number of frequency/noise pairs does not match with the FNpairs parameter, the Spectre netlister shows an error in the console.
- The default of noise_file_as_var parameter is *false*. If the noise file parameter is set as *NOISEFILE*, the netlisting behavior for noise file and noise_file_as_var parameters is as follows:

CDF Parameter	hspiceD Output	spectre Output
noise_file_as_var		
false (default)	noisefile='NOISEFILE'	noisefile="NOISEFILE"
true	noisefile=str(NOISEFILE)	noisefile=NOISEFILE

- For Hspice-format netlisting, a pair of single quotes is added to noisefile parameter when noise_file_as_var is *false*. When noise_file_as_var parameter is *true*, `str(NoiseFile)` is used as the value format.
- For Spectre-format netlisting, a pair of double quotes is added when noise_file_as_var parameter is *false*. No single/double quotes are added when noise_file_as_var parameter is *true*.

Example

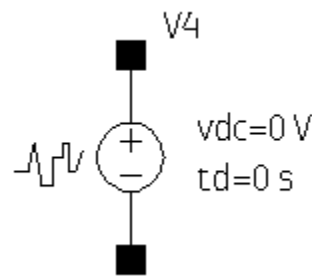
```
v4 n1 n2 dc=0 pwlf ( 0 0.0 1n 0.02 1.2n 0.5 2n 1.2 )
```

See Also

- [PWL Source](#) in the *HSPICE® User Guide: Elements*

vpwlf

PWL source, that accepts a text file containing time and voltage pairs. The data in the text file can be either a table or a list.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Source Type	srcType	pwlf	-	X	-
PWL File	fileName		X	X	-
File name is variable	file_as_var	0	X	X	-
DC Voltage	vdc	0	X	X	-
Repeat	r		X	-	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Repeat Stop Time	rstop		X	-	-
Repeat Stop Value	stopvalue		X	-	-
Repeat Stop Slope	stopslope		X	-	-
Delay Time	td	0	X	X	-
AC Magnitude	acm		X	X	-
AC Phase	acp		X	X	-
# HB Tones	hbtones	0	X	-	-
HB Magnitude1-7	hbmag1-7		X	-	-
HB Phase1-7	hbphase1-7	0	X	-	-
HB Harmonic1-7	hbharm1-7	1	X	-	-
HB Tone1-7	hbtone1-7	1-7	X	-	-
HB ModHarm1-7	hbmodharm1-7		X	-	-
HB ModTone1-7	hbmodtone1-7		X	-	-
Open DC Connection	dcopen		-	-	-
DC Resistance	rdc		X	-	-
AC Resistance	rac		X	-	-
HB Resistance	rhb		X	-	-
HBAC Resistance	rhbac		X	-	-
TRAN Resistance	rtran		X	-	-
HBAC Magnitude	hbacm		X	-	-
HBAC Phase	hbacp		X	-	-
SNAC Magnitude	snacm		X	-	-
SNAC Phase	snacp		X	-	-
Desired rms value	rms		-	X	-
Filter for pwl source	pwlfilter		-	X	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Rolloff factor	rolloff		-	X	-
Bandwidth of the pwl signal	pwlbandwidth		-	X	-
PAC magnitude	pacm		-	X	-
PAC phase	pacp		-	X	-
PWL period	pwlperiod		-	X	-
Amplitude scale factor	scale		-	X	-
Edge type	edgetype	linear	-	X	-
Transition width	twidth		-	X	-
Time scaling factor	stretch		-	X	-
Temp coefficient 1	tc1		X	X	-
Temp coefficient 2	tc2		X	X	-
XF magnitude	xfm		-	X	-
Number of noise/frequency pairs	FNpairs	0	-	X	-
Noise file name	noisefile		-	X	-
Noise file name is variable	noise_file_as_var	0	-	X	-
Frequency1-50	F1-50		-	X	-
Noise1-50	N1-50		-	X	-
Frequency name for fundamental frequency	fundname		-	X	-
Nominal temperature	tnom		X	X	-
Offset voltage	vo		-	X	-
Impedance	z0		-	-	-

Note:

- When frequency/noise pairs have empty values or the number of frequency/noise pairs does not match with the FNpairs parameter, the Spectre netlister shows an error in the console.
- The default of noise_file_as_var parameter is *false*. If the noise file parameter is set as *NOISEFILE*, the netlisting behavior for noise file and noise_file_as_var parameters is as follows:

CDF Parameter	hspiceD Output	spectre Output
noise_file_as_var		
false (default)	noisefile='NOISEFILE'	noisefile="NOISEFILE"
true	noisefile=str(NOISEFILE)	noisefile=NOISEFILE

- For Hspice-format netlisting, a pair of single quotes is added to noise file parameter when noise_file_as_var is *false*. When noise_file_as_var parameter is *true*, `str(NOISEFILE)` is used as the value format.
- For Spectre-format netlisting, a pair of double quotes is added when noise_file_as_var parameter is *false*. No single/double quotes are added when noise_file_as_var parameter is *true*.

Example

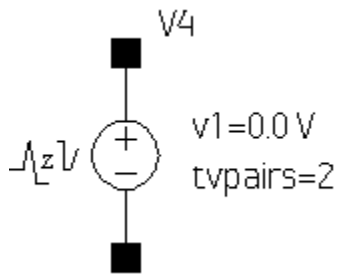
```
v4 n1 n2 PWL PWLFILE='Vmod.dat'
```

See Also

- [File-driven PWL Source](#) in the *HSPICE® User Guide: Elements*

vpwlz

The high Z state of the PWLZ source adds the capability to disconnect the PWL source for time periods marked with the keyword Z.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
DC Voltage	vdc	0	X	X	-
Source Type	srcType	pwlz	-	X	-
Time Value Pairs	tvpairs	2	X	X	-
Time1-50	t1-50		X	X	-
Voltage1-50	v1-50	0	X	X	-
Z State1-50	Z1-50	0	X	X	-
Repeat	r		X	-	-
Delay Time	td	0	X	X	-
AC Magnitude	acm		X	X	-
AC Phase	acp		X	X	-
# HB Tones	hbtones	0	X	-	-
HB Magnitude1-7	hbmag1-7		X	-	-
HB Phase1-7	hbphase1-7	0	X	-	-
HB Harmonic1-7	hbharm1-7	1	X	-	-
HB Tone1-7	hbtone1-7	1-7	X	-	-
HB ModHarm1-7	hbmodharm1-7		X	-	-
HB ModTone1-7	hbmodtone1-7		X	-	-
Open DC Connection	dcopen		-	-	-
DC Resistance	rdc		X	-	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
AC Resistance	rac		X	-	-
HB Resistance	rhb		X	-	-
HBAC Resistance	rhbac		X	-	-
TRAN Resistance	rtran		X	-	-
HBAC Magnitude	hbacm		X	-	-
HBAC Phase	hbacp		X	-	-
SNAC Magnitude	snacm		X	-	-
SNAC Phase	snacp		X	-	-
Number of noise/frequency pairs	FNpairs	0	-	X	-
Noise file name	noisefile		-	X	-
Noise file name is variable	noise_file_as_var	0	-	X	-
Frequency1-50	F1-50		-	X	-
Noise1-50	N1-50		-	X	-
Frequency name for fundamental frequency	fundname		-	X	-
Nominal temperature	tnom		X	X	-
PAC magnitude	pacm		-	X	-
PAC phase	pacp		-	X	-
PWL period	pwlperiod		-	X	-
Amplitude scale factor	scale		-	X	-
Edge type	edgetype	linear	-	X	-
Transition width	twidth		-	X	-
Time scaling factor	stretch		-	X	-
Temp coefficient 1	tc1		X	X	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Temp coefficient 2	tc2		X	X	-
XF magnitude	xfm		-	X	-
Delay time	delay		-	X	-
Offset voltage	vo		-	X	-
Impedance	z0		-	-	-

Note:

- When frequency/noise pairs have empty values or the number of frequency/noise pairs does not match with the FNpairs parameter, the Spectre netlister shows an error in the console.
- The default of noise_file_as_var parameter is *false*. If the noise file parameter is set as *NOISEFILE*, the netlisting behavior for noise file and noise_file_as_var parameters is as follows:

CDF Parameter	hspiceD Output	spectre Output
noise_file_as_var		
false (default)	noisefile='NOISEFILE'	noisefile="NOISEFILE"
true	noisefile=str(NOISEFILE)	noisefile=NOISEFILE

- For Hspice-format netlisting, a pair of single quotes is added to noise file parameter when noise_file_as_var is *false*. When noise_file_as_var parameter is *true*, *str(NOISEFILE)* is used as the value format.
- For Spectre-format netlisting, a pair of double quotes is added when noise_file_as_var parameter is *false*. No single/double quotes are added when noise_file_as_var parameter is *true*.

Example

```
v4 n1 n2 dc=0 pwlz ( 0 0.0 1n z 1.2n 0.5 2n 1.2 )
```

See Also

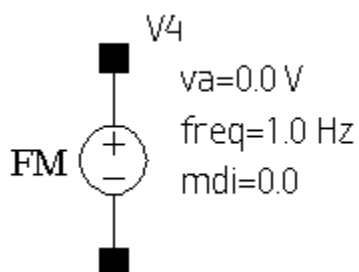
- [High Impedance State PWL \(PWLZ\)](#) in the *HSPICE® User Guide: Elements*

vsffm

Single-frequency FM voltage source.

The following expression defines the waveform shape:

$$sourcevalue = vo + va \cdot SIN[2 \cdot \pi \cdot fc \cdot Time + mdi \cdot SIN(2 \cdot \pi \cdot fs \cdot Time)]$$



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
DC Current	idc	0	-	-	-
Source Type	srcType	sffm	-	-	-
Output Amplitude	va	0	X	-	-
Offset Voltage	vo	0	X	-	-
Carrier Frequency	freq	1	X	-	-
Modulation Index	mdi	0	X	-	-
Signal Frequency	fs		X	-	-
AC Magnitude	acm		X	-	-
AC Phase	acp		X	-	-
# HB Tones	hbtones	0	X	-	-
HB Magnitude1-7	hbmag1-7		X	-	-
HB Phase1-7	hbphase1-7	0	X	-	-
HB Harmonic1-7	hbharm1-7	1	X	-	-
HB Tone1-7	hbtone1-7	1-7	X	-	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
HB ModHarm1-7	hbmodharm1-7	X	-	-	-
HB ModTone1-7	hbmodtone1-7	X	-	-	-
Open DC Connection	dcopen	-	-	-	-
DC Resistance	rdc	X	-	-	-
AC Resistance	rac	X	-	-	-
HB Resistance	rhb	X	-	-	-
HBAC Resistance	rhbac	X	-	-	-
TRAN Resistance	rtran	X	-	-	-
HBAC Magnitude	hbacm	X	-	-	-
HBAC Phase	hbacp	X	-	-	-
SNAC Magnitude	snacm	X	-	-	-
SNAC Phase	snacp	X	-	-	-
Impedance	z0	-	-	-	-

Example

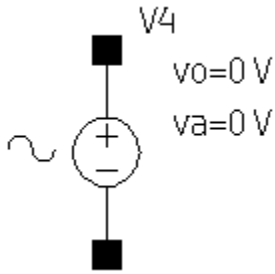
```
v4 n1 gnd! dc=0.5 sffm ( 0.0 1.1 20k 10 5k )
```

See Also

- [Single-frequency FM Source](#) in the *HSPICE® User Guide: Elements*

vsin

Sinusoidal voltage source that can be either a constant sinusoid or a damped, that produces a dying exponential with a sine wave.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Source Type	srcType	sine	-	X	-
DC Voltage	vdc	0	X	X	-
Voltage Offset	vo	0	X	X	-
Voltage Amplitude	va	0	X	X	-
Frequency	freq		X	X	-
Delay Time	td	0	X	X	-
Damping Factor	theta	0	X	X	-
Phase for sinusoid	sinephase	0	X	X	-
Period Jitter	perjitter		X	-	-
Seed	seed		X	-	-
AC Magnitude	acm		X	X	-
AC Phase	acp		X	X	-
# HB Tones	hbtones	0	X	-	-
HB Magnitude1-7	hbmag1-7		X	-	-
HB Phase1-7	hbphase1-7	0	X	-	-
HB Harmonic1-7	hbharm1-7	1	X	-	-
HB Tone1-7	hbtone1-7	1-7	X	-	-
HB ModHarm1-7	hbmodharm1-7		X	-	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
HB ModTone1-7	hbmodtone1-7		X	-	-
Open DC Connection	dcopen		-	-	-
DC Resistance	rdc		X	-	-
AC Resistance	rac		X	-	-
HB Resistance	rhb		X	-	-
HBAC Resistance	rhbac		X	-	-
TRAN Resistance	rtran		X	-	-
HBAC Magnitude	hbacm		X	-	-
HBAC Phase	hbacp		X	-	-
SNAC Magnitude	snacm		X	-	-
SNAC Phase	snacp		X	-	-
Amp modulation frequency	ammodfreq		-	X	-
Amp modulation index	ammodindex		-	X	-
Amp modulation phase	ammodphase		-	X	-
Number of FM Files	filenums	none	-	X	-
Name of FM File1	fmmodfile1		-	X	-
FM file1 name is variable	fmmodfile1_as 0 _var		-	X	-
Name of FM file 2	fmmodfile2		-	X	-
FM file2 name is variable	fmmodfile2_as 0 _var		-	X	-
FM frequency	fmmodfreq		-	X	-
FM index	fmmodindex		-	X	-
Frequency 2	freq2		-	X	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Number of noise/frequency pairs	FNpairs	0	-	X	-
Noise file name	noisefile		-	X	-
Noise file name is variable	noise_file_as_var	0	-	X	-
Frequency1-50	F1-50		-	X	-
Noise1-50	N1-50		-	X	-
Frequency name for fundamental frequency	fundname		-	X	-
Second frequency name	fundname2		-	X	-
Noise temperature	noisetemp		-	X	-
Nominal temperature	tnom		X	X	-
PAC magnitude	pacm		-	X	-
PAC phase	pacp		-	X	-
Phase for sinusoid 2	sinephase2		-	X	-
Temp coefficient 1	tc1		X	X	-
Temp coefficient 2	tc2		X	X	-
XF magnitude	xfm		-	X	-
Phase shift	phase		-	X	-
Amplitude 2	va2		-	X	-
Impedance	z0		-	-	-

Note:

- When frequency/noise pairs have empty values or the number of frequency/noise pairs does not match with the FNpairs parameter, the Spectre netlister shows an error in the console.
- The default of noise_file_as_var parameter is *false*. If the noise file parameter is set as *NOISEFILE*, the netlisting behavior for noise file and noise_file_as_var parameters is as follows:

CDF Parameter	hspiceD Output	spectre Output
noise_file_as_var		
false (default)	noisefile='NOISEFILE'	noisefile="NOISEFILE"
true	noisefile=str(NOISEFILE)	noisefile=NOISEFILE

- For Hspice-format netlisting, a pair of single quotes is added to noise file parameter when noise_file_as_var is *false*. When noise_file_as_var parameter is *true*, `str(NOISEFILE)` is used as the value format.
- For Spectre-format netlisting, a pair of double quotes is added when noise_file_as_var parameter is *false*. No single/double quotes are added when noise_file_as_var parameter is *true*.

Example

```
v4 n1 0 dc=0 sin ( 0 1v 100Meg 1n 1e+6 0 )
```

This damped sinusoidal source connects between nodes n1 and 0. In this waveform:

- Peak value is 1V
- Offset is 0 V
- Frequency is 100 MHz
- Time delay is 1 ns
- Damping factor is 1e+6
- Phase delay is 0 deg

See Also

- [Sinusoidal Source Function](#) in the *HSPICE® User Guide: Elements*

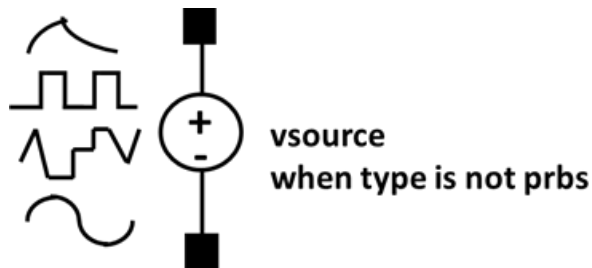
Controlling Version Number for Legacy Designs

To control the version number for legacy designs, create and use the `dbAnalogLibVsinVersion` Tcl preference before the analogLib library is loaded.

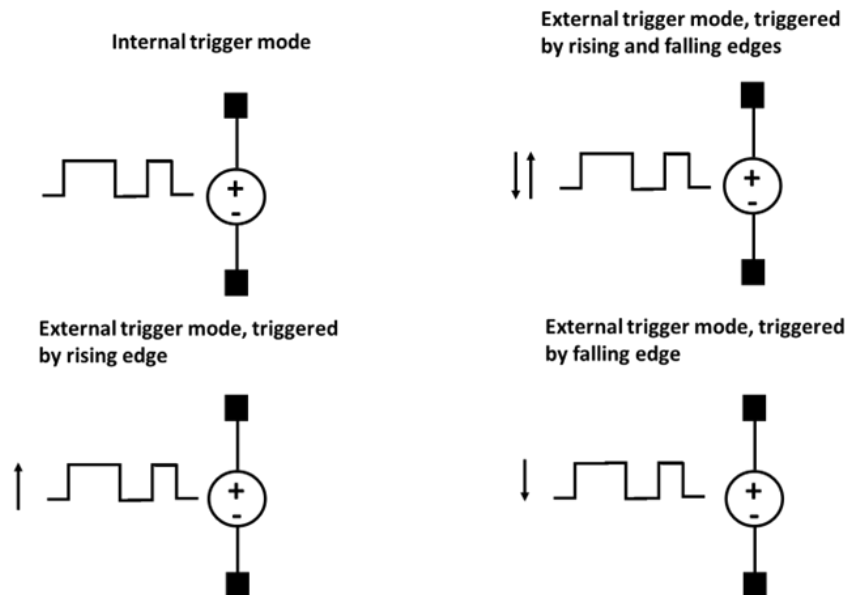
vsources

Voltage source element with various parameters settings.

The vsources device is a PCell controlled by its source type. For source type other than prbs, it is a 2-pin device and its symbol looks as follows:



For prbs source type, its PCell behavior is affected by the trigger parameter. The following figure shows four vsources devices that have different trigger modes, which are Internal (on the upper-left), External by rising edges (on the upper-right), External by falling edges (on the lower-left), and External by both edges (on the lower-right) modes. The internal trigger mode has 2 pins (PLUS and MINUS) and other three external trigger modes have 4 pins (P1, M1, P2, and M2). The netlisting procedure writes out their pins accordingly.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
DC Voltage	vdc	0	X	X	-
srcTypeValue	srcTypeValue	1	-	-	-
Source Type	srcType	dc	X	X	-
Frequency name 1	fundname		-	X	-
Waveform entry method	pwlEntryMethod	File	X	X	-
File name	fileName		X	X	-
File name is variable	file_as_var	0	X	X	-
Time Value Pairs	tvpairs	0	X	X	-
Time1-50	t1-50		X	X	-
Z State1-50	Z1-50	0	X	X	-
Voltage1-50	v1-50		X	X	-
Zero value	val0		-	X	-
One value	val1		-	X	-
Rise Delay Time	td1	0	X	X	-
Fall Delay Time	td2	0	X	X	-
Rise Time Constant	tau1	0	X	X	-
Fall Time Constant	tau2	0	X	X	-
Exp period	expperiod		-	X	-
Rise Time	tr	0	X	X	-
Fall Time	tf	0	X	X	-
Pulse Width	pw	0	X	X	-
Period	per	0	X	X	-
Repeat	r		X	-	-
Repeat Stop Time	rstop		X	-	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Repeat Stop Value	stopvalue		X	-	-
Repeat Stop Slope	stopslope		X	-	-
Voltage Offset	vo	0	X	-	-
Frequency 1	freq		X	X	-
Voltage Amplitude	va	0	X	X	-
Phase for sinusoid	sinephase	0	X	X	-
Voltage level	sinedc		-	X	-
Delay Time	td	0	X	X	-
Damping Factor	theta		X	X	-
DC offset	offset		-	X	-
Desired rms value	rms		-	X	-
Time scaling factor	stretch		-	X	-
PWL period	pwlperiod		-	X	-
Edge type	edgetype		X	X	-
Width of transition	twidth		-	X	-
Breakpoints	allbrkpts	no	-	X	-
Period start time	pwlperiodstart		-	X	-
# HB Tones	hbtones	0	X	-	-
HB Magnitude1-7	hbmag1-7		X	-	-
HB Phase1-7	hbphase1-7	0	X	-	-
HB Harmonic1-7	hbharm1-7	1	X	-	-
HB Tone1-7	hbtone1-7	1-7	X	-	-
HB ModHarm1-7	hbmodharm1-7		X	-	-
HB ModTone1-7	hbmodtone1-7		X	-	-
Power Switch	power	V	X	-	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Impedance	z0		-	-	-
Open DC Connection	dcopen		X	-	-
Display second sinusoid	numofsines	0	-	X	-
Frequency name 2	fundname2		-	X	-
Frequency 2	freq2		-	X	-
Amplitude 2 (Vpk)	va2		-	X	-
Phase for sinusoid 2	sinephase2		-	X	-
Display multi sinusoid	vecofsines	0	-	X	-
Number of frequencies	freqvecs	0	-	X	-
Sinusoid frequency 1-9	freqv1-9		-	X	-
Sinusoid Ampl 1-9 (Vpk)	vav1-9		-	x	-
Sinusoid phase 1-9	phasev1-9		X	X	-
Sinusoid max harmonic 1-9	harmv1-9		X	X	-
Display modulation params	modulation	0	-	X	-
Amp modulation frequency 1	ammodfreq		-	X	-
Amp modulation index 1	ammodindex		-	X	-
Amp modulation phase 1	ammodphase		-	X	-
Number of FM files	filenums	none	-	X	-
Name of FM file 1	fmmodfile1		-	X	-
FM file1 name is variable	fmmodfile1_as_var	0	-	X	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Name of FM file 2	fmmodfile2		-	X	-
FM file2 name is variable	fmmodfile2_as_var	0	-	X	-
FM index 1	fmmodindex		-	X	-
FM frequency 1	fmmodfreq		-	X	-
IQ modulation I File	imodfile		-	X	-
I File name is variable	imodfile_as_var	0	-	X	-
IQ modulation Q File	qmodfile		-	X	-
Q File name is variable	qmodfile_as_var	0	-	X	-
Delay time	tdPrbs		-	X	-
Bit period	perPrbs	1n	-	X	-
Transition reference	transitionreference		-	X	-
Lower threshold(percent)	lowerthreshold		-	X	-
Upper threshold(percent)	upperthreshold		-	X	-
Edge type	edgetypePrbs		-	X	-
Rise delay	td01		-	X	-
Fall delay	td10		-	X	-
PAM4 modulation	pam4_modulation	none	-	X	-
PAM4 mapping	pam4_mapping		-	X	-
triggerValue	triggerValue	1	-	X	-
Trigger	trigger	Internal	-	X	-
Threshold	triggerthreshold		-	X	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
LFSR Mode	lfsrmode	PN32	-	X	-
Browse and specify bit file	prbsFileBox	0			
Bit data file	prbsFile		-	X	-
Bit data file name is variable	prbsFile_as_var	0	-	X	-
Seed	lfsrseed	1 3 5 7 9 11 13 15 17 19 21 23 25 27 29 31	-	X	-
Taps	lfsrtaps		-	X	-
RJ(rms)	rjrms		-	X	-
RJ(seed)	rjseed		-	X	-
Number of periodic jitters	nPeriodicJitter	0	-	X	-
PJ%d(amplitude)	pjamp1-3		-	X	-
PJ%d(frequency)	pjfreq1-3		-	X	-
PJ%d(type)	pjtype1-3		-	X	-
Pattern Parameter data	data		-	X	-
Pattern Parameter data is variable	data_as_var	0	-	X	-
Pattern Parameter rptstart	rptstart	1	-	X	-
Pattern Parameter rpttimes	rpttimes	0	-	X	-
Amplitude scale factor	scale		-	X	-
Filter for pwl source	pwlfilter		-	X	-
Rolloff factor	rolloff		-	X	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Bandwidth of the pwl signal	pwlbandwidth		-	X	-
Display small signal params	smallSig	0	X	X	-
HBAC Magnitude	hbacm		X	X	-
HBAC Phase	hbacp		X	X	-
SNAC Magnitude	snacm		X	-	-
SNAC Phase	snacp		X	-	-
AC Magnitude	acm		X	X	-
AC Phase	acp		X	X	-
PAC magnitude	pacm		-	X	-
PAC phase	pacp		-	X	-
XF magnitude	xfm		-	X	-
Display resistance parameters	resistanceParam	0	X	-	-
DC Resistance	rdc		X	-	-
AC Resistance	rac		X	-	-
HB Resistance	rhb		X	-	-
HBAC Resistance	rhbac		X	-	-
TRAN Resistance	rtran		X	-	-
Display temperature params	tempParam	0	-	X	-
Nominal temperature	tnom		X	X	-
Linear temperature coefficient	tc1		X	X	-
Quadratic temperature coefficient	tc2		X	X	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Display noise parameters	noiseParam	0	-	X	-
Generate noise?	isnoisyNoiseModel		-	X	-
Noise entry method	noiseEntryMethod	File	-	X	-
Noise file name	noisefile		-	X	-
Noise file name is variable	noise_file_as_var	0	X	X	-
Noise type	noiseModelType	Noise Voltage(V ² /Hz)	-	X	-
Interpolation method	noiseModelInterMethod		-	X	-
Number of noise/freq pairs	FNpairs	0	-	X	-
Frequency1-50	F1-50		-	X	-
Noise1-50	N1-50		-	X	-
Noise temperature	noisetemp		-	X	-
This dummy parameter exists only for using its default value	paramForDefault	evalDefaultZeroOrEmpty("")			
High-Z impedance	highz		-	X	-
Min high-Z transition width	min_z_transition_width		-	X	-
Multiplier	m		-	X	-
Version	version	2	-	-	-

Note:

- When frequency/noise pairs have empty values or the number of frequency/noise pairs does not match with the FNpairs parameter, the Spectre netlister shows an error in the console.
- The default of noise_file_as_var parameter is *false*. If the noise_file parameter is set as *NOISEFILE*, the netlisting behavior for noise_file and noise_file_as_var parameters is as follows:

CDF Parameter	hspiceD Output	spectre Output
noise_file_as_var		
false (default)	noisefile='NOISEFILE'	noisefile="NOISEFILE"
true	noisefile=str(NOISEFILE)	noisefile=NOISEFILE

- For Hspice-format netlisting, a pair of single quotes is added to noise_file parameter when noise_file_as_var is *false*. When noise_file_as_var parameter is *true*, `str(NOISEFILE)` is used as the value format.
- For Spectre-format netlisting, a pair of double quotes is added when noise_file_as_var parameter is *false*. No single/double quotes are added when noise_file_as_var parameter is *true*.

Example

```
v4 n1 n2 dc=1 power=0
v4 n1 n2 dc=0 exp ( 0 1 0 20n 0 30n )
v4 n1 n2 dc=0.35 pulse ( 0 0.75 1n 1n 1n 10n 20n )
```

See Also

- [Independent Sources and Stimuli](#) in the *PrimeSim Continuum User Guide: Elements*

Controlling Version Number for Legacy Designs

To control the version number for legacy designs, create and use the `dbAnalogLibVsourceVersion` Tcl preference before the analogLib library is loaded.

Using vsource to Set up Harmonic Balance or Shooting PSS Simulations

vsource can be used as stimulus for one-, two-, and N-tone Harmonic Balance and Shooting Newton simulations. Refer to [port](#) for details, because the setup is similar and most parameters are shared.

Note:

vsources **Resistance parameters**, **#HB Tones parameters**, and **Power Switch** are kept for reasons of compatibility with PrimeSim HSPICE, but their ongoing use is not recommended.

vvmrf

Vector Modulated RF voltage source. It allows digital modulation of an RF carrier using in-phase and quadrature components created from a binary data stream.

CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
DC Voltage	vdc	0	X	-	-
Source Type	srcType	vmrf	-	-	-
Voltage Amplitude	amp		X	-	-
Carrier Frequency	freq		X	-	-
Carrier Phase	phase		X	-	-
Modulation Method	mod	bpsk	X	-	-
Filter Method	filter	rect	X	-	-
Filter Parameter	filcoef		X	-	-
Bit Rate	rate		X	-	-
Bitstream	bitstream		X	-	-
AC Magnitude	acm		X	-	-
AC Phase	acp		X	-	-
# HB Tones	hbtones	0	X	-	-
HB Magnitude1-7	hbmag1-7		X	-	-
HB Phase1-7	hbphase1-7	0	X	-	-
HB Harmonic1-7	hbharm1-7	1	X	-	-
HB Tone1-7	hbtone1-7	1-7	X	-	-
HB ModHarm1-7	hbmodharm1-7		X	-	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
HB ModTone1-7	hbmodtone1-7		X	-	-
Open DC Connection	dcopen		-	-	-
DC Resistance	rdc		X	-	-
AC Resistance	rac		X	-	-
HB Resistance	rhb		X	-	-
HBAC Resistance	rhbac		X	-	-
TRAN Resistance	rtran		X	-	-
HBAC Magnitude	hbacm		X	-	-
HBAC Phase	hbacp		X	-	-
SNAC Magnitude	snacm		X	-	-
SNAC Phase	snacp		X	-	-
Multiplier	m		X	-	-
Impedance	z0		X	-	-

Example

```
v4 n1 n2 dc=0 vmrf amp=0.85 freq=10k phase=0 rate=100Meg
+ mod=bpsk filter=rect bitstream=01010010011100b
```

See Also

- [Vector-modulated RF \(VMRF\) Source](#) in the *HSPICE® User Guide: Elements*

Dependent Sources

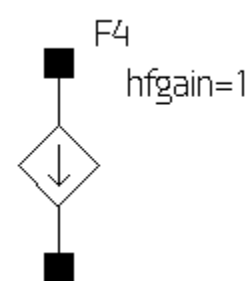
The following topics describe:

- [cccs](#)
- [ccvs](#)
- [pcccs](#)
- [pccvs](#)

- [pvccs](#)
- [pvcvs](#)
- [pvcvs2](#)
- [svccs](#)
- [svcvs](#)
- [vccs](#)
- [vcvs](#)
- [zvccs](#)
- [zvcvs](#)

CCCS

Current-controlled current source. Can be of *linear*, *pwl*, and *delay* types.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Name of Voltage Source	vref		X	X	-
Current Gain (Spectre)	fgain	1	-	X	-
Current Gain (HSPICE)	hfgain	1	X	-	-
Type	csType	linear	X	-	-
Controlling Pairs	xypairs	2	X	-	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Delta	delta		-	X	-
Control Current 1-50	x1-50		X	-	-
Corresp Element1-50	y1-50		X	-	-
Maximum Output Current	maxi		X	-	-
Minimum Output Current	mini		X	-	-
Maximum Output Current	maxm		-	X	-
Minimum Output Current	minm		-	X	-
Delay Time	htd	0	X	-	-
Scale Factor	scale		X	X	-
Temperature Coefficient 1	tc1		X	X	-
Temperature Coefficient 2	tc2		X	X	-
Absolute Output Current	absol	0	-	X	-
Absolute Output Current	habs	0	X	-	-
Initial Output Current	hic		X	-	-
Probe 1-20	p1-20		-	X	-
Port 1-20	port1-20		-	X	-
Probe count	probeCnt	4	-	X	-
Port	port		-	X	-
PWL vector source	iVectSpec	PwlVect	-	X	-
File containing PWL Vectors	pwlFile		-	X	-
Scale for input	stretchpwl		-	X	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Scale for output	scalepwl		-	X	-
Smoothing factor	smoothing		-	X	-
Type of source	typesrc	cccs	-	X	-
Type of transfer char	trfType	Linear	-	X	-
Number of PWL Vectors	pwlCnt	4	-	X	-
PWL Input	pwi1-20		-	X	-
PWL Output	pwo1-20		-	X	-
Multiplier	m		-	X	-
Time Delay	td		-	X	-

Example

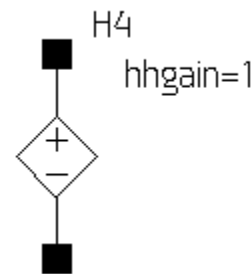
```
f4 n1 gnd! cccs vcntrl 0.45 abs=0
```

See Also

- [Current-dependent Current Sources—F-elements](#) in the *HSPICE® User Guide: Elements*

CCVS

Current-controlled voltage source. Can be of *linear*, *pwl*, and *delay* types.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Name of Voltage Source	vref		X	X	-
Transresistance (Spectre)	hgain	1	-	X	-
Transresistance (HSPICE)	hhgain	1	X	-	-
Type	csType	linear	X	-	-
Controlling Pairs	xypairs	2	X	-	-
Delta	delta		X	X	-
Control Current 1-50	x1-50		X	X	-
Corresp Element1-50	y1-50		X	-	-
Maximum Output Voltage	maxv		X	-	-
Minimum Output Voltage	minv		X	-	-
Maximum Output Voltage	maxm		-	X	-
Minimum Output Voltage	minm		-	X	-
Delay Time	htd	0	X	-	-
Scale Factor	scale		X	X	-
Temperature Coefficient 1	tc1		X	X	-
Temperature Coefficient 2	tc2		X	X	-
Absolute Output Voltage	absol	0	-	X	-
Absolute Output Voltage	habs	0	X	-	-
Initial Output Voltage	hic		X	-	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Probe 1-20	p1-20		-	X	-
Port 1-20	port1-20		-	X	-
Probe count	probeCnt	4	-	X	-
Port	port		-	X	-
PWL vector source	iVectSpec	PwlVect	-	X	-
File containing PWL Vectors	pwlFile		-	X	-
Scale for input	stretchpwl		-	X	-
Scale for output	scalepwl		-	X	-
Smoothing factor	smoothing		-	X	-
Type of Source	typesrc		-	X	-
Type of transfer char	trfType	Linear	-	X	-
Number of PWL Vectors	pwlCnt	4	-	X	-
PWL Input	pwi1-20		-	X	-
PWL Output	pwo1-20		-	X	-
Multiplier	m		-	X	-

Example

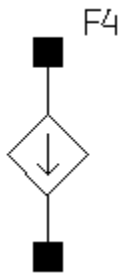
```
h4 n1 gnd! ccvs vcntrl 1 max='vmax' min='vmin' abs=0
```

See Also

- [Current-dependent Voltage Sources—H-elements](#) in the *HSPICE® User Guide: Elements*

pcccs

Polynomial current-controlled current source.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Dimension (Number of Voltage Sources)	dim	1	X	X	-
Name of Voltage Source 1	vref1		X	X	-
Name of Voltage Source 2	vref2		X	X	-
Name of Voltage Source 3	vref3		X	X	-
Type	csType	poly	X	-	-
Initial Controlling Current	hic		X	-	-
Scale Factor	scale		X	-	-
Absolute output current	absol	off	X	X	-
Maximum number of coefficients	polyCoef	4	X	X	-
Poly coefficient 0-20	c0-20	0	X	X	-
Coefficients to be specified	coefSpec	form	-	X	-
File containing coefficients	filecoef		-	X	-
Gain	gain	1	-	X	-
Smoothing factor	smoothing		-	X	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Temp coefficient 1	tc1		-	X	-
Temp coefficient 2	tc2		-	X	-
Probe 1-20	p1-20		-	X	-
Port 1-20	port1-20		-	X	-
Probe count	probeCnt	4	-	X	-
Min output current	minm		X	X	-
Max output current	maxm		X	X	-
Multiplier	m		X	X	-

Example

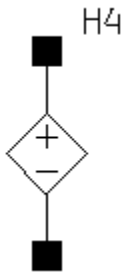
```
f4 n1 gnd! cccs poly(2) v1 v2 ic=1u 1 1 3 1 6
```

See Also

- [Polynomial \(POLY\)](#) in the *HSPICE® User Guide: Elements*
- [Polynomial Functions](#) in the *HSPICE® User Guide: Elements*

pccvs

Polynomial current-controlled voltage source.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Dimension (Number of Voltage Sources)	dim	1	X	X	-
Name of Voltage Source 1	vref1		X	X	-
Name of Voltage Source 2	vref2		X	X	-
Name of Voltage Source 3	vref3		X	X	-
Type	csType	poly	X	-	-
Initial Controlling Current	hic		X	-	-
Scale Factor	scale		X	-	-
Absolute output voltage	absol	off	X	X	-
Maximum number of coefficients	polyCoef	4	X	X	-
Poly coefficient 0-20	c0-20	0	X	X	-
Coefficients to be specified	coefSpec	form	-	X	-
File containing coefficients	filecoef		-	X	-
Gain	gain	1	-	X	-
Smoothing factor	smoothing		-	X	-
Temp coefficient 1	tc1		X	X	-
Temp coefficient 2	tc2		X	X	-
Probe 1-20	p1-20		-	X	-
Port 1-20	port1-20		-	X	-
Probe count	probeCnt	4	-	X	-
Min output voltage	minm		X	X	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Max output voltage	maxm		X	X	-
Multiplier	m		-	X	-

Example

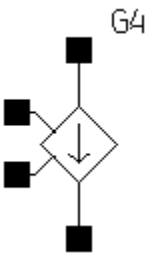
```
h4 n1 gnd! ccvs poly(2) v1 v2 1 1 3 1 6 ic=1u
```

See Also

- [Polynomial \(POLY\)](#) in the *HSPICE® User Guide: Elements*
- [Polynomial Functions](#) in the *HSPICE® User Guide: Elements*

pvccs

Polynomial voltage-controlled current source.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Dimension (Number of Voltage Sources)	dim	1	X	X	-
Type	csType	poly	X	-	-
Initial Controlling Voltage	hic		X	-	-
Scale Factor	scale		X	-	-
Absolute output current	absol	off	X	X	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Maximum number of coefficients	polyCoef	4	X	X	-
Poly coefficient 0-20	c0-20	0	X	X	-
Coefficients to be specified	coefSpec	form	-	X	-
File containing coefficients	filecoef		-	X	-
Gain	gain	1	-	X	-
Smoothing factor	smoothing		-	X	-
Temp coefficient 1	tc1		X	X	-
Temp coefficient 2	tc2		X	X	-
Min output current	minm		X	X	-
Max output current	maxm		X	X	-
Multiplier	m		X	X	-

Example

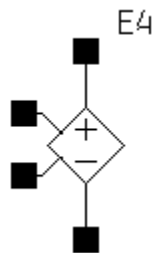
```
g5 n1 gnd! vccs poly(1) n2 gnd! ic=0.5 3 1 5
```

See Also

- [Polynomial \(POLY\)](#) in the *HSPICE® User Guide: Elements*

pvcvs

Polynomial voltage-controlled voltage source.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Dimension (Number of Voltage Sources)	dim	1	X	X	-
Type	csType	poly	X	-	-
Initial Controlling Voltage	hic		X	-	-
Scale Factor	scale		X	-	-
Maximum number of coefficients	polyCoef	4	X	X	-
Poly coefficient 0-20	c0-20	0	X	X	-
Coefficients to be specified	coefSpec	form	-	X	-
File containing coefficients	filecoef		-	X	-
Gain	gain	1	-	X	-
Smoothing factor	smoothing		-	X	-
Temp coefficient 1	tc1		X	X	-
Temp coefficient 2	tc2		X	X	-
Min output voltage	minm		X	X	-
Max output voltage	maxm		X	X	-
Multiplier	m		-	X	-
Absolute output voltage	absol	off	X	X	-

Example

```
e5 n1 gnd! vcvs poly(1) n2 gnd! 3 1 5 ic=0.5
```

See Also

- [Polynomial \(POLY\)](#) in the *HSPICE® User Guide: Elements*

pvcvs2

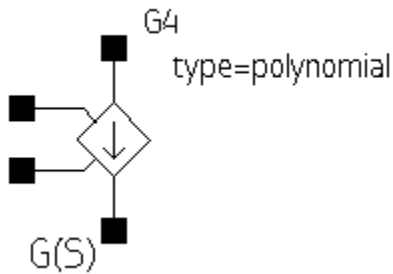
Polynomial voltage controlled voltage sources, which is controlled by two voltage sources.

CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Max Coefficient Number	polyCoef	-	-	X	-
Poly Coeff 0-20	c0 – c20	-	-	X	-
Initial condition	ic	-	-	X	-
Scale factor	scale	-	-	X	-
Absolute value	abs	-	-	X	-
Gain	gain	-	-	X	-
Multiplier	m	-	-	X	-
Minimum Output Voltage	minm	-	-	X	-
Maximum Output Voltage	maxm	-	-	X	-
Absolute Output Voltage	absol	-	-	X	-
Smoothing Factor	smoothing	-	-	X	-
Temperature coefficient 1	tc1	-	-	X	-
Temperature coefficient 2	tc2	-	-	-	-

SVCCS

Voltage-controlled current source with Laplace transform and Pole-Zero function.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Laplace Specification Type	type	polynomial	X	-	-
Order of Transfer Function	order	1	X	-	-
Coefficient of Num. Const Term	k0		X	-	-
Coefficient of Num. Term 1	k1		X	-	-
Coefficient of Num. Term 2	k2		X	-	-
Coefficient of Num. Term 3	k3		X	-	-
Coefficient of Num. Term 4	k4		X	-	-
Coefficient of Num. Term 5	k5		X	-	-
Coefficient of Num. Term 6	k6		X	-	-
Coefficient of Num. Term 7	k7		X	-	-
Coefficient of Num. Term 8	k8		X	-	-
Coefficient of Num. Term 9	k9		X	-	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Coefficient of Num. Term 10	k10		X	-	-
Coefficient of Den. Const Term	d0		X	-	-
Coefficient of Den. Term 1	d1		X	-	-
Coefficient of Den. Term 2	d2		X	-	-
Coefficient of Den. Term 3	d3		X	-	-
Coefficient of Den. Term 4	d4		X	-	-
Coefficient of Den. Term 5	d5		X	-	-
Coefficient of Den. Term 6	d6		X	-	-
Coefficient of Den. Term 7	d7		X	-	-
Coefficient of Den. Term 8	d8		X	-	-
Coefficient of Den. Term 9	d9		X	-	-
Coefficient of Den. Term 10	d10		X	-	-
Zero Multiplier Gain	a		X	-	-
Zero Real Part 1	az1		X	-	-
Zero Imaginary Part 1	fz1		X	-	-
Zero Real Part 2	az2		X	-	-
Zero Imaginary Part 2	fz2		X	-	-
Zero Real Part 3	az3		X	-	-
Zero Imaginary Part 3	fz3		X	-	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Zero Real Part 4	az4		X	-	-
Zero Imaginary Part 4	fz4		X	-	-
Zero Real Part 5	az5		X	-	-
Zero Imaginary Part 5	fz5		X	-	-
Zero Real Part 6	az6		X	-	-
Zero Imaginary Part 6	fz6		X	-	-
Zero Real Part 7	az7		X	-	-
Zero Imaginary Part 7	fz7		X	-	-
Zero Real Part 8	az8		X	-	-
Zero Imaginary Part 8	fz8		X	-	-
Zero Real Part 9	az9		X	-	-
Zero Imaginary Part 9	fz9		X	-	-
Zero Real Part 10	az10		X	-	-
Zero Imaginary Part 10	fz10		X	-	-
Pole Multiplier Gain	b		X	-	-
Pole Real Part 1	ap1		X	-	-
Pole Imaginary Part 1	fp1		X	-	-
Pole Real Part 2	ap2		X	-	-
Pole Imaginary Part 2	fp2		X	-	-
Pole Real Part 3	ap3		X	-	-
Pole Imaginary Part 3	fp3		X	-	-
Pole Real Part 4	ap4		X	-	-
Pole Imaginary Part 4	fp4		X	-	-
Pole Real Part 5	ap5		X	-	-
Pole Imaginary Part 5	fp5		X	-	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Pole Real Part 6	ap6		X	-	-
Pole Imaginary Part 6	fp6		X	-	-
Pole Real Part 7	ap7		X	-	-
Pole Imaginary Part 7	fp7		X	-	-
Pole Real Part 8	ap8		X	-	-
Pole Imaginary Part 8	fp8		X	-	-
Pole Real Part 9	ap9		X	-	-
Pole Imaginary Part 9	fp9		X	-	-
Pole Real Part 10	ap10		X	-	-
Pole Imaginary Part 10	fp10		X	-	-
Scale Factor	scale		X	-	-
Temperature Coefficient 1	tc1		X	-	-
Temperature Coefficient 2	tc2		X	-	-
Multiplier	m		X	-	-

Example

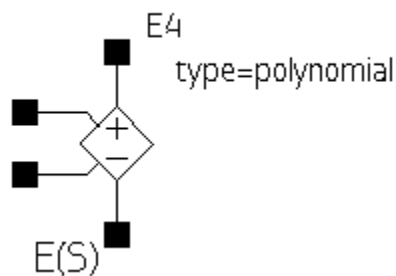
```
glowpass out gnd! laplace in gnd! 1.0 / 1.0, 2.0, 2.0, 1.0
ghigh_pass out gnd! pole in gnd! 1 / 1 0.001,0.0
```

See Also

- [Laplace Transform](#) in the *HSPICE® User Guide: Elements*
- [Pole-Zero Function](#) in the *HSPICE® User Guide: Elements*

SVCVS

Voltage-controlled voltage source with Laplace transform and Pole-Zero function.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Laplace Specification Type	type	polynomial	X	-	-
Order of Transfer Function	order	1	X	-	-
Coefficient of Num. Const Term	k0		X	-	-
Coefficient of Num. Term 1	k1		X	-	-
Coefficient of Num. Term 2	k2		X	-	-
Coefficient of Num. Term 3	k3		X	-	-
Coefficient of Num. Term 4	k4		X	-	-
Coefficient of Num. Term 5	k5		X	-	-
Coefficient of Num. Term 6	k6		X	-	-
Coefficient of Num. Term 7	k7		X	-	-
Coefficient of Num. Term 8	k8		X	-	-
Coefficient of Num. Term 9	k9		X	-	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Coefficient of Num. Term 10	k10		X	-	-
Coefficient of Den. Const Term	d0		X	-	-
Coefficient of Den. Term 1	d1		X	-	-
Coefficient of Den. Term 2	d2		X	-	-
Coefficient of Den. Term 3	d3		X	-	-
Coefficient of Den. Term 4	d4		X	-	-
Coefficient of Den. Term 5	d5		X	-	-
Coefficient of Den. Term 6	d6		X	-	-
Coefficient of Den. Term 7	d7		X	-	-
Coefficient of Den. Term 8	d8		X	-	-
Coefficient of Den. Term 9	d9		X	-	-
Coefficient of Den. Term 10	d10		X	-	-
Zero Multiplier Gain	a		X	-	-
Zero Real Part 1	az1		X	-	-
Zero Imaginary Part 1	fz1		X	-	-
Zero Real Part 2	az2		X	-	-
Zero Imaginary Part 2	fz2		X	-	-
Zero Real Part 3	az3		X	-	-
Zero Imaginary Part 3	fz3		X	-	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Zero Real Part 4	az4		X	-	-
Zero Imaginary Part 4	fz4		X	-	-
Zero Real Part 5	az5		X	-	-
Zero Imaginary Part 5	fz5		X	-	-
Zero Real Part 6	az6		X	-	-
Zero Imaginary Part 6	fz6		X	-	-
Zero Real Part 7	az7		X	-	-
Zero Imaginary Part 7	fz7		X	-	-
Zero Real Part 8	az8		X	-	-
Zero Imaginary Part 8	fz8		X	-	-
Zero Real Part 9	az9		X	-	-
Zero Imaginary Part 9	fz9		X	-	-
Zero Real Part 10	az10		X	-	-
Zero Imaginary Part 10	fz10		X	-	-
Pole Multiplier Gain	b		X	-	-
Pole Real Part 1	ap1		X	-	-
Pole Imaginary Part 1	fp1		X	-	-
Pole Real Part 2	ap2		X	-	-
Pole Imaginary Part 2	fp2		X	-	-
Pole Real Part 3	ap3		X	-	-
Pole Imaginary Part 3	fp3		X	-	-
Pole Real Part 4	ap4		X	-	-
Pole Imaginary Part 4	fp4		X	-	-
Pole Real Part 5	ap5		X	-	-
Pole Imaginary Part 5	fp5		X	-	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Pole Real Part 6	ap6		X	-	-
Pole Imaginary Part 6	fp6		X	-	-
Pole Real Part 7	ap7		X	-	-
Pole Imaginary Part 7	fp7		X	-	-
Pole Real Part 8	ap8		X	-	-
Pole Imaginary Part 8	fp8		X	-	-
Pole Real Part 9	ap9		X	-	-
Pole Imaginary Part 9	fp9		X	-	-
Pole Real Part 10	ap10		X	-	-
Pole Imaginary Part 10	fp10		X	-	-
Scale Factor	scale		X	-	-
Temperature Coefficient 1	tc1		X	-	-
Temperature Coefficient 2	tc2		X	-	-

Example

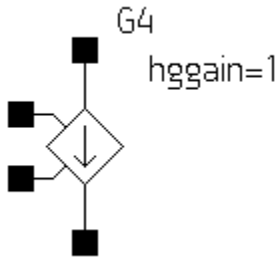
```
ehipass out 0 LAPLACE in 0 0.0,0.0,0.0,1.0 / 1.0,2.0,2.0,1.0
elow_pass out 0 POLE in 0 1.0 / 1.0, 1.0,0.0 0.5,0.1379
```

See Also

- [Laplace Transform](#) in the *HSPICE® User Guide: Elements*
- [Pole-Zero Function](#) in the *HSPICE® User Guide: Elements*

VCCS

Voltage-controlled current source with *linear*, *pwl*, and *delay* functions.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Transconductance (Spectre)	ggain	1	-	X	-
Transconductance (HSPICE)	hggain	1	X	-	-
Type	csType	linear	X	X	-
Controlling Pairs	xypairs	2	X	-	-
Delta	Delta		X	X	-
Control Volt 1-50	x1-50		X	-	-
Corresp Element1-50	y1-50		X	-	-
Maximum Output Current	maxi		X	-	-
Minimum Output Current	mini		X	-	-
Delay Time	htd	0	X	-	-
Scale Factor	scale		X	X	-
Temperature Coefficient 1	tc1		X	X	-
Temperature Coefficient 2	tc2		X	X	-
Absolute Output Current	absol	0	-	X	-
Absolute Output Current	habs	0	X	-	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Initial Output Current	hic		X	-	-
Multiplier	hm		X	-	-
Multiplier	m		-	X	-
PWL vector source	iVectSpec	PwlVect	-	X	-
File containing PWL Vectors	pwlFile		-	X	-
Scale for input	stretchpwl		-	X	-
Scale for output	scalepwl		-	X	-
Smoothing factor	smoothing		-	X	-
Type of source	typesrc	vccs	X	X	-
Capacitance Calculation Mode	ctype	0	X	-	-
Type of input of source	inputtype		-	X	-
Type of transfer char	trfType	Linear	-	X	-
Number of PWL Vectors	pwlCnt	4	-	X	-
PWL Input	pwi1-20		-	X	-
PWL Output	pwo1-20		-	X	-
Time Delay	td	0	X	-	-
Multiplier	m		-	X	-

Example

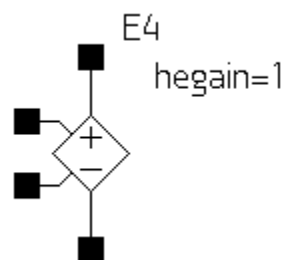
```
G4 out gnd! vccs cntrl gnd! 1 max=100u min=10u abs=0
```

See Also

- [Voltage-dependent Current Sources—G-elements](#) in the *HSPICE® User Guide: Elements*

VCVS

Voltage-controlled voltage source with *linear*, *pwl*, and *delay* functions.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Voltage Gain (Spectre)	egain	1	-	X	-
Voltage Gain (HSPICE)	hegain	1	X	-	-
Type	csType	linear	X	X	-
Controlling Pairs	xypairs	2	X	X	-
Delta	delta		X	X	-
Control Volt 1-50	x1-50		X	X	-
Corresp Element1-50	y1-50		X	X	-
Maximum Output Voltage	maxv		X	-	-
Minimum Output Voltage	minv		X	-	-
Maximum Output Voltage	maxm		-	X	-
Minimum Output Voltage	minm		-	X	-
Delay Time	htd	0	X	X	-
Scale Factor	scale		X	X	-

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Temperature Coefficient 1	tc1		X	X	-
Temperature Coefficient 2	tc2		X	X	-
Absolute Output Voltage	absol	0	-	X	-
Absolute Output Voltage	habs	0	X	-	-
Initial Output Voltage	hic		X	-	-
PWL vector source	iVectSpec	PwlVect	-	X	-
File containing PWL Vectors	pwlFile		-	X	-
Scale for input	stretchpwl		-	X	-
Scale for output	scalepwl		-	X	-
Smoothing factor	smoothing		-	X	-
Type of source	typesrc	vcvs	-	X	-
Multiplier	m		-	X	-

Example

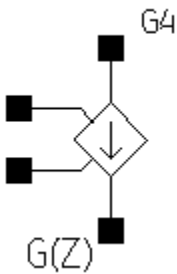
```
E4 out gnd! vcvs cntrl gnd! 1 max=1 min=0.25 abs=0
```

See Also

- [Voltage-Dependent Voltage Sources—E-elements](#) in the *HSPICE® User Guide: Elements*

ZVCCS

Voltage-controlled current source with Z transform function.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Order of Transfer Function	order	1	X	-	-
Coefficient of Num. Const Term	k0		X	-	-
Coefficient of Num. Term 1-10	k1-10		X	-	-
Coefficient of Den. Const Term	d0		X	-	-
Coefficient of Den. Term 1-10	d1-10		X	-	-
Nyquist Critical Frequency	maxf		X	-	-
Scale Factor	scale		X	-	-
Temperature Coefficient 1	tc1		X	-	-
Temperature Coefficient 2	tc2		X	-	-
Multiplier	m		X	-	-

Example

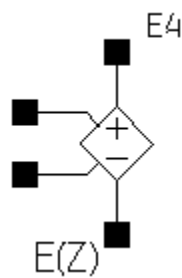
```
glow_pass 0 out ZTRANS in 0 0.0317 0.0951
+ 0.0951 0.0317 / 1.0 -1.459 0.9104 -0.1978
```

See Also

- [Z Transform Function Cell](#) in the *HSPICE® User Guide: Elements*

ZVCVS

Voltage-controlled voltage source with Z transform function.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default Value	hspiceD	spectre	auCdl
Order of Transfer Function	order	1	X	-	-
Coefficient of Num. Const Term	k0		X	-	-
Coefficient of Num. Term 1-10	k1-10		X	-	-
Coefficient of Den. Const Term	d0		X	-	-
Coefficient of Den. Term 1-10	d1-10		X	-	-
Nyquist Critical Frequency	maxf		X	-	-
Scale Factor	scale		X	-	-
Temperature Coefficient 1	tc1		X	-	-
Temperature Coefficient 2	tc2		X	-	-

Example

```
ehigh_pass out 0 ZTRANS in 0 -0.0082 -0.1793  
+ 0.6579 -0.1793 -0.0082 / 1
```

See Also

- [Z Transform Function Cell](#) in the *HSPICE® User Guide: Elements*

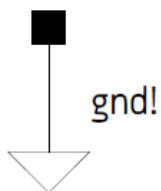
Global Nets

The following topics describe:

- [gnd](#)
- [gnd_connDef](#)
- [gndd!](#)
- [vcc](#)
- [vccd!](#)
- [vdd](#)
- [vdd_connDef](#)
- [vdda](#)
- [vddd](#)
- [vee!](#)
- [veed](#)
- [vssa!](#)
- [vssd!](#)

gnd

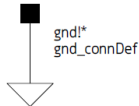
The symbol specifies a gnd! global ground net. The cell has no CDF parameters.



gnd_connDef

The symbol specifies a ground net with a connection definition and gnd! default net name. This symbol is similar to *gnd!* and has no CDF parameters.

The cell has one connection assignment with the name “gnd”. With this connection assignment it is possible to create a new global power net and pass it to a higher hierarchy level of the design.



Example

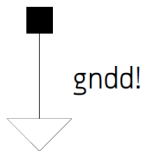
Connection Assignments	
Name	Net
gnd	vss_net!
Click to add	

gnd = vss_net!



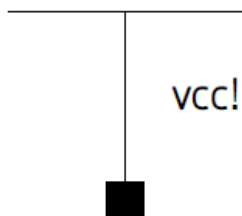
gndd!

The symbol specifies a gndd! global ground net. The cell has no CDF parameters.



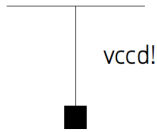
vcc

The symbol specifies vcc! power global net. The cell has no CDF parameters.



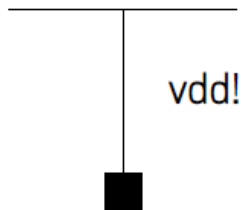
vccd!

The symbol specifies a power connection with vccd! global net. The cell has no CDF parameters.



vdd

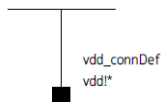
The symbol specifies a power connection with vdd! global net. The cell has no CDF parameters.



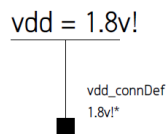
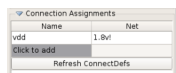
vdd_connDef

The symbol specifies a power net with a connection definition and the default net name vdd!. The cell has no CDF parameters.

The cell has one connection assignment with the name “vdd”. With this connection assignment it is possible to create a new global power net and pass it to a higher hierarchy level of the design.

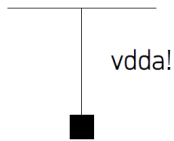


Example



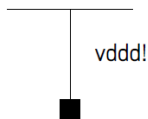
vdda

The symbol specifies a vdda! global power net. The cell has no CDF parameters.



vddd

The symbol specifies a vddd! global power net. The cell has no CDF parameter.



vee!

The symbol specifies a ground connection with vee! global power net. The cell has no CDF parameters.



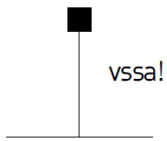
veed

The symbol specifies a veed! global power net. The cell has no CDF parameters.



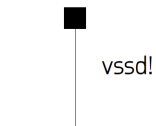
vssa!

The symbol specifies a vssa! global ground net. It is similar to vss! and gnd! symbols and the cell has no CDF parameters.



vssd!

The symbol specifies a vssd! global power net. The cell has no CDF parameters.



2

analogLib Preferences

Summarizes the supported Tcl preferences of analogLib.

Tcl preferences of analogLib are used for version control and parameter settings. Each preference has:

- Preference purpose and name in the page's title
- A table lists:
 - The Custom Compiler build with a new version of related element.
 - The description of that version.
 - The preference value which points to that version.
- Sometimes the preference values are enabled or redefined in a newer build than the element version.
 - The build information related to preference is in the note under the table. The preference values are valid with that build.
- Check [dbAnalogLibSwitchVersion](#) for an example.

Syntax

The syntax to use an analogLib preference:

```
db::createPref <preference_name> -value <preference_value>
```

Example

```
db::createPref dbAnalogLibIsinVersion -value 1
```


Guidelines

Guidelines to use analogLib preferences:

- If need a third-party compatible element: don't set version control preferences. The default version is third-party compatible.
- If there is any back-compatible issue in legacy schematics: try a version control preference. But whenever it is possible, it is recommended to migrate to the latest analogLib versions.
- Always set analogLib preferences right after launching Custom Compiler, to make sure the preferences work as expected.

The following sections describe the preferences in details.

- [setAnalogLibCompatibleVersion](#)
- [dbAnalogLibAutoUpdatePortNum](#)
- [dbAnalogLibCccsVersion](#)
- [dbAnalogLibCcvsvVersion](#)
- [dbAnalogLibIndVersion](#)
- [dbAnalogLibIndqVersion](#)
- [dbAnalogLibIprbsVersion](#)
- [dbAnalogLibIsinVersion](#)
- [dbAnalogLibIsourceVersion](#)
- [dbAnalogLibNcsVersion](#)
- [dbAnalogLibNportVersion](#)
- [dbAnalogLibNvsVersion](#)
- [dbAnalogLibNxPortVersion](#)
- [dbAnalogLibMTlineVersion](#)
- [dbAnalogLibPcapacitorVersion](#)
- [dbAnalogLibPdiodeVersion](#)
- [dbAnalogLibPinductorVersion](#)
- [dbAnalogLibPortVersion](#)
- [dbAnalogLibPprbsVersion](#)

- [dbAnalogLibPpwlVersion](#)
- [dbAnalogLibPsinVersion](#)
- [dbAnalogLibPulseExpV1V2Remap](#)
- [dbAnalogLibSp1tswitchVersion](#)
- [dbAnalogLibSp1t2tswitchEmptyOffset](#)
- [dbAnalogLibSwitchVersion](#)
- [dbAnalogLibTlineVersion](#)
- [dbAnalogLibVccsVersion](#)
- [dbAnalogLibVcresVersion](#)
- [dbAnalogLibVcvsVersion](#)
- [dbAnalogLibVprbsVersion](#)
- [dbAnalogLibVsinVersion](#)
- [dbAnalogLibVsourceVersion](#)
- [dbAnalogLibWindingVersion](#)
- [nlHSPICESkipEmptyDC](#)

setAnalogLibCompatibleVersion

Sets all analogLib preferences to the default value in specified version.

Syntax

```
db::setAnalogLibCompatibleVersion <release version number>
```

Examples

```
db::setAnalogLibCompatibleVersion U-2023.03-SP1
```

Note:

- Use this command line before opening any libraries or library manager.
- This preference sets all analogLib preferences to specified release version. After closing Custom Compiler, all analogLib preferences are written to the file: `<work_dir>/synopsys_custom/prefs.xml`

Next time when launching Custom Compiler, these analogLib preferences are loaded automatically.

- After using the preference `setAnalogLibCompatibleVersion`, the analogLib preferences are not changed with newer CC versions. If want to use the latest analogLib, manually delete all analogLib preferences in the file, `<work_dir>/synopsys_custom/prefs.xml.`,
- When using top level preference and individual preference together, individual preference wins.

For example:

```
db::setAnalogLibCompatibleVersion U-2023.03
db::setAnalogLibCompatibleVersion U-2023.03
```

In this case, the individual preference wins. The `lprbs` version is set to 2, all other elements are set to the default version in U-2023.02. However, the individual preference set in console won't be remembered next time you open Custom Compiler

The permanent solution is to open `<work_dir>/synopsys_custom/prefs.xml`, change the value of the `dbAnalogLibIprbsVersion` preference to 2 directly

dbAnalogLibAutoUpdatePortNum

Disables auto-assignment of port number.

First Build to Enable the Enhancements	Description
Before S-2021.09-SP2	Automatically assign port number if the parameter “num” is not specified This applies when creating an instance, performing Copy action or Property Editor display on a selected device that has an empty port number. Port family devices include port, port_ref, mport, pdc, pexp, ppulse, ppwl, ppwlf, plfsr, pp sin, pam, and psffm.
N.A.	Turn off the auto-assignment of the port number.

Note:

Preference values true and false are enabled in U-2023.03-SP2. The default value is true.

dbAnalogLibCccsVersion

Controls cccs versions.

First Build to Enable the Enhancements	Description
--	-------------

Before V-2023.12

Legacy cccs

V-2023.12

More enhancements for third-party compatibility

Spectre-format: changed several CDF parameter definitions. Now it is compatible with third-party's spectre-format

Hspice-format: no change

Note:

Preference values 1-2 are redefined or enabled in V-2023.12. The default is 2.

dbAnalogLibCcvsvVersion

Controls ccvs versions.

First Build to Enable the Enhancements	Description
--	-------------

Before V-2023.12

Legacy ccvs

V-2023.12

More enhancements for third-party compatibility

Spectre-format: changed several CDF parameter definitions. Now it is compatible with third-party's spectre-format

Hspice-format: no change

Note:

Preference values 1-2 are redefined or enabled in V-2023.12. The default is 2.

dbAnalogLibIndVersion

Controls ind versions.

First Build to Enable the Enhancements	Description
--	-------------

Before V-2023.12-SP1

Legacy ind

V-2023.12-SP1

Use c1 ~ c20 instead of c0 ~c20 for poly coefficients in spectre-format display and netlis

Note:

Preference values 1-2 are redefined or enabled in V-2023.12-SP1. The default is 2.

dbAnalogLibIndqVersion

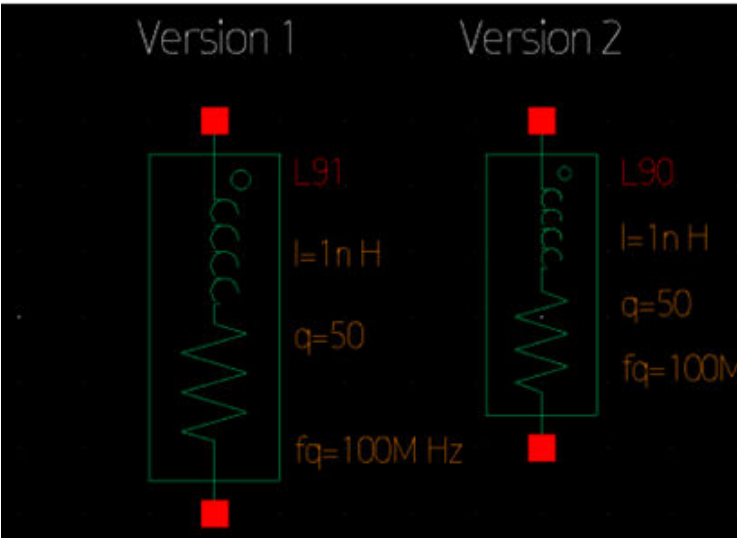
Controls `ind` versions.

First Build to Enable the Enhancements	Description
Before V-2023.12-SP1	Legacy pinductor
V-2023.12-SP1	Use new symbol, for third-party compatibility.

Note:

Preference values 1-2 are redefined or enabled in V-2023.12-SP1. The default value is 2.

The comparison of symbols in version 1 and version 2 (The height is changed):



dbAnalogLiblprbsVersion

Controls `lprbs` versions.

First Build to Enable the Enhancements	Description
U-2023.03-SP2	Add iprbs in analogLib Symbol: is a PCell

Note:
Preference value 1 is enabled in U-2023.03-SP2. The default value is 1.

dbAnalogLibIsinVersion

Controls isin versions.

First Build to Enable the Enhancements	Description
Before U-2023.03	Legacy isin CDF parameters: ia default is 0, io default is 0
U-2023.03	Remove default value of io Spectre-format: netlist is changed, simulation result is not changed Hspice-format: netlist is not changed Remove default value of ia Spectre-format: netlist and simulation result are changed Hspice-format: netlist is not changed

Note:
Preference values 1-2 are redefined and enabled in U-2023.03-SP2. The default value is 2.

dbAnalogLibIsinSourceVersion

Controls isin source versions.

First Build to Enable the Enhancements	Enhancements Description
Before U-2023.03	Legacy isource Symbol: is not a PCell CDF parameters: tvpairs default is 2, ia default is 0 Netlisting behavior: Noise related parameters (noisefile, F/N pairs) are only netlisted when they are displayed in the Property Editor (parameter noiseParam is selected)
U-2023.03	Remove default value of ia Spectre-format: netlist and simulation result are changed Hspice-format: netlist is not changed Inherited from last version: tvpair default is 2, netlisting behavior of noisefile and F/N pairs is not a PCell
U-2023.03-SP2	Symbol is changed to PCell (for “type=prbs” & “external trigger modes”) No impact on legacy designs if not in this type and mode Change tvpairs default value to 0 Both of Spectre-format and Hspice-format: netlist and simulation result are changed Netlist noise-related parameters (noisefile, F/N pairs) when they are hidden in PE Spectre-format: netlist and simulation result are changed Hspice-format: netlist is not changed Inherited from last version: ia has no default value

Note:

Preference values 1-3 are in U-2023.03-SP2. The default value is 3.

dbAnalogLibNcsVersion

Controls ncs versions.

First Build to Enable the Enhancements	Enhancements Description
Before U-2023.03-SP2	Legacy ncs Property Editor displays noise, noisefile, and noisefile_as_var at the same time Hspice netlister writes noise and noisefile at the same time if they both have non-empty

First Build to Enable the Enhancements	Description
U-2023.03-SP2	Add new CDF parameter noiseEntryMethod to control the display and netlisting of and noise file Choices are: file and expression Default value is: file Spectre-format: no impact. (ncs is for Hspice-format only) Hspice-format: legacy schematic is impacted if the ncs element sets Noise Expression a The noise expression is not netlisted because it is hidden by default noiseEntryMethod.

Note:

Preference values 1-2 are enabled in U-2023.03-SP2. The default value is 2.

dbAnalogLibNportVersion

Controls nport versions.

First Build to Enable the Enhancements	Description
Before S-2021.09	Legacy nport
Before S-2021.09	Redefine nport device to make it more compatible with third party

Note:

Preference values 1-2 are enabled before S-2021.09. The default value is 2.

dbAnalogLibNvsVersion

Controls nvs versions.

First Build to Enable the Enhancements	Description
Before U-2023.03-SP2	Legacy nvs Property Editor displays noise, noise file, and noise file _as_ var at the same time Hspice netlister writes noise and noise file at the same time if they both have non-empty

First Build to Enable the Enhancements	Description
U-2023.03-SP2	Add new CDF parameter noiseEntryMethod to control the display and netlisting of and noise file Choices are: file and expression Default value is: file Spectre-format: no impact. (nvs is for Hspice-format only) Hspice-format: legacy schematic is impacted if the ncs element sets Noise Expression a The noise expression is not netlisted because it is hidden by default noiseEntryMethod.

Note:

Preference values 1-2 are enabled in U-2023.03-SP2. The default value is 2.

dbAnalogLibNxPortVersion

Controls n1port, n2port, n3port, and n4port versions.

First Build to Enable the Enhancements	Description
Before U-2023.03	Legacy n1port CDF parameter: interp default value is default
U-2023.03	Change default value of interp to linear Spectre-format: netlist and simulation result are changed Hspice-format: netlist is not changed. (interp is for Spectre-format only)

Note:

Preference values 1-2 are enabled in U-2023.03-SP1-1. The default value is 2.

dbAnalogLibMTlineVersion

Controls mtline versions.

First Build to Enable the Enhancements	Description
Before S-2021.09	Legacy mtline
Before S-2021.09	Redefine mtline to make it more compatible with third party

First Build to Enable the Enhancements	Description
V-2023.12-SP1	Use CDF parameter <code>file</code> instead of <code>rlgcfile</code> in spectre-format display and netlisting. spectre-format: Set this preference to value 2 to use <code>rlgcfile</code> in spectre-format display netlisting. Hspice-format: Use <code>rlgcfile</code> only for Hspice-format.

Note:
Preference values 1-2 are enabled before S-2021.09.
Preference value 3 is enabled in V-2023.12-SP1. The default is 3.

dbAnalogLibPcapacitorVersion

Controls pcapacitor versions.

First Build to Enable the Enhancements	Description
Before V-2023.12	Legacy pcapacitor
V-2023.12	More enhancements for third-party compatibility Spectre-format: use CDF parameter “trise” instead of “dtemp” Hspice-format: no change

Note:
Preference values 1-2 are redefined or enabled in V-2023.12. The default is 2.

dbAnalogLibPdiodeVersion

Controls pdiode versions.

First Build to Enable the Enhancements	Description
Before V-2023.12	Legacy pdiode
V-2023.12	More enhancements for third-party compatibility Spectre-format: the CDF parameter definition and netlist are third-party compatible. Hspice-format: no change

Note:

Preference values 1-2 are redefined or enabled in V-2023.12. The default is 2.

dbAnalogLibPinductorVersion

Controls `pinductor` versions.

First Build to Enable the Enhancements	Description
Before V-2023.12-SP1	Legacy <code>pinductor</code>
V-2023.12-SP1	Use CDF parameter <code>trise</code> instead of <code>dtemp</code> in spectre-format display and netlisting

Note:

Preference values 1-2 are redefined or enabled in V-2023.12-SP1. The default value is 2.

dbAnalogLibPortVersion

Controls port versions.

First Build to Enable the Enhancements	Description
Before U-2023.03-SP2	Legacy port Symbol: is not a PCell CDF parameter: <code>tvpairs</code> default is 2 Netlisting behavior: - Noise related parameters (<code>noisefile</code>, F/N pairs) are only netlisted when they are displayed in the Property Editor (parameter <code>noiseParam</code> is selected) - Modulation parameter (<code>type=sine</code>) <code>fmmodfiles</code> and <code>scale</code> are only netlisted when they are displayed in the Property Editor (parameter <code>modulation</code> is selected)

First Build to Enable the Enhancements	Description
U-2023.03-SP2	Symbol is changed to PCell (for “type=prbs” & “external trigger modes”) No impact on legacy designs if not in this type and mode Change tvpairs default value to 0 Both of spectre-format and hspice-format: netlist and simulation result are changed Netlist noise-related parameters (noisefile, F/N pairs) when they are hidden in the Property Editor by parameter noiseParam Spectre-format: netlist and simulation result are changed Hspice-format: netlist is not changed Netlist (type=sine) fmmodfiles and scale when they are hidden in the Property Editor by parameter modulation Spectre-format: netlist and simulation result are changed Hspice-format: netlist is not changed

Note:

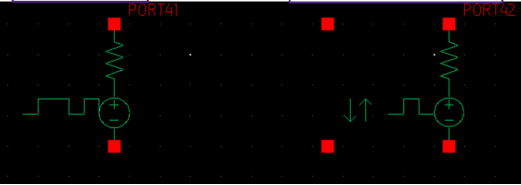
Preference values 1-2 are enabled in U-2023.03-SP2. The default value is 2.

dbAnalogLibPprbsVersion

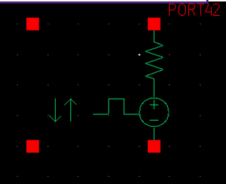
Controls pprbs versions.

First Build to Enable the Enhancements	Description
Before U-2023.03-SP2	Legacy pprbs Symbol: is not a PCell When the trigger mode is external, because there is no pin to connect the external trigger signal, the netlist is incorrect
U-2023.03-SP2	Change the symbol to PCell When the trigger mode is external, adds pins to connect the external signals No impact on legacy designs if the trigger mode is internal

old symbol



new pcell symbol



Transition reference	
Edge type	
Trigger	External both edges
Threshold	
LFSR Mode	PN32
Seed	1 3 5 7 9 11 13 15 17 ...

Note:

Preference values 1-2 are enabled in U-2023.03-SP2. The default value is 2.

dbAnalogLibPpwlVersion

Controls ppwl versions.

First Build to Enable the Enhancements	Enhancements Description
Before U-2023.03-SP2	Legacy ppwl CDF parameter: allbrkpts default is no
U-2023.03-SP2	Remove default value of allbrkpts Spectre-format: netlist is changed. Simulation result may be changed depending on simulator's default values. Hspice-format: netlist is not changed

Note:

Preference values 1-2 are enabled in U-2023.03-SP2. The default value is 2.

dbAnalogLibPsinVersion

Controls psin versions.

First Build to Enable the Enhancements	Enhancements Description
Before U-2023.03-SP2	Legacy psin CDF parameter: va default is 0
U-2023.03-SP2	Remove default value of va Spectre-format: netlist and simulation result are changed Hspice-format: netlist is not changed

Note:

Preference values 1-2 are enabled in U-2023.03-SP2. The default value is 2.

dbAnalogLibPulseExpV1V2Remap

Disables using val0 and val1 in vsource, isource, or port devices (type=pulse/exp) in Hspice-format.

First Build to Enable the Enhancements	Description
Before U-2023.03	Legacy vsource, isource, or port devices (type=pulse/exp) in Hspice-format CDF parameter for high/low values: v1 and v2, prompts are “Voltage1” and “Voltage 2”
U-2023.03	Use CDF parameters val0 and val1 for high/low values, prompts are “Zero value” and “One value” Hspice-format: netlist is not changed Spectre-format: no change. This is only for Hspice-format.

Note:

Preference values true and false are enabled in U-2023.03. The default value is true.

dbAnalogLibSp1tswitchVersion

Controls sp1tswitch versions.

First Build to Enable the Enhancements	Description
Before S-2021.09	Legacy sp1tswitch
Before S-2021.09	Redefine sp1tswitch to make it more compatible with third party

Note:

Preference values 1-2 are enabled before S-2021.09. The default value is 2.

dbAnalogLibSp1t2tswitchEmptyOffset

Write offset=0 in netlist by default.

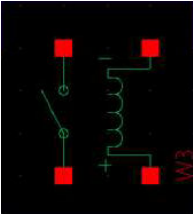
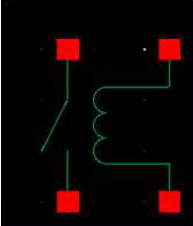
First Build to Enable the Enhancements	Description
Before V-2023.12	Write offset=0 by default in netlist CDF parameter transfactor has the default 1.
V-2023.12	Do not write the offset=0 by default in netlist The simulation result is not changed.

Note:

Preference values true and false are redefined or enabled in V-2023.12. The default is true.

dbAnalogLibSwitchVersion

Controls switch versions.

First Build to Enable the Enhancements	Description
Before T-2022.06	Legacy switch Symbol: version 1  CDF parameters: vt1 default value is 0.1, vt2 default value is 0
T-2022.06	Change the symbol to version 2 Impact both Hspice-format design and Spectre-format schematic  Inherited from last version: vt1 default value is 0.1, vt2 default value is 0

First Build to Enable the Enhancements	Enhancements Description
U-2023.03	Remove default values of vt1 and vt2 Spectre-format: netlist and simulation result are changed Hspice-format: netlist is not changed
V-2023.12-SP2	Change the hspice-format netlist syntax Spectre-format netlist: no change hspice-format netlist: • Previous syntax: <code>gw0 n+ n- vcr pw1(1) nc+ nc- vt2 r vt1 ropen</code> • New syntax: <code>gw0 n+ n- vcr relay nc+ nc- vt1 vt2 ropen r</code>

Note:

Preference values 1-3 are redefined or enabled in U-2023.03-SP1. Preference value 4 is enabled in V-2023.12-SP2. The default value is 4.

Examples

A legacy schematic was created with Custom Compiler T-2022.06-SP1. The switch in the design should have symbol version 2, vt1 default value is 0.1, vt2 default value is 0. Suppose the legacy schematic used these default values.

- If opening the legacy schematic with U-2023.03-SP1, set the preference value to 2 for back-compatibility.
- If opening the example schematic with version earlier than U-2023.03-SP1, the preference value 2 may not work as the table shown.
- If the switch does not use the default vt1 and vt2, no need to set the preference.

dbAnalogLibTlineVersion

Controls tline versions.

First Build to Enable the Enhancements	Enhancements Description
Before T-2022.06-SP2	Legacy tline CDF parameter "td" has default as 0 CDF parameter "rs" has default as 1K CDF parameter "nl" has no default Use CDF parameter "z0" and "l" in spectre-formant netlisting procedure

First Build to Enable the Enhancements	Description
--	-------------

T-2022.06-SP2	Remove td default value Spectre-format: netlist is changed, simulation result is not changed Hspice-format: no change
V-2023.12	More enhancements for third-party compatibility Remove CDF parameter “rs” default Set CDF parameter “nl” default to 0.25 Use CDF parameter “zo” instead of “z0” in spectre-format Use CDF parameter “len” instead of “l” in spectre-format Spectre-format: netlist is changed. Simulation result is changed Hspice-format: netlist is changed. Simulation result is changed

Note:

Preference values 1-2 are redefined or enabled in V-2023.12. The default is 3.

dbAnalogLibVccsVersion

Controls vccs versions.

First Build to Enable the Enhancements	Description
--	-------------

Before V-2023.12	Legacy vccs
V-2023.12	More enhancements for third-party compatibility Spectre-format: changed several CDF parameter definitions. Now it is compatible with third-party's spectre-format Hspice-format: no change

Note:

Preference values 1-2 are redefined or enabled in V-2023.12. The default is 2.

dbAnalogLibVcresVersion

Controls writing hm=1 in spectre-format netlist by default.

First Build to Enable the Enhancements	Description
Before V-2023.12	Legacy vres CDF parameter transfactor has the default 1.
V-2023.12	Remove the default value of transfactor Spectre-format: netlist does not have gm=1 by default. Simulation result changed Hspice-format: netlist has gm=2 by default. No change The simulation result is not changed

Note:

Preference values 1-2 are redefined or enabled in V-2023.12. The default is 2.

dbAnalogLibVcvsVersion

Controls vcvs versions.

First Build to Enable the Enhancements	Description
Before T-2022.06	Legacy vcvs
T-2022.06	Add CDF parameter trfType for spectre-format Spectre-format: replace CDF parameter “csType” with “trfType” Hspice-format: not changed
V-2023.12	More enhancements for third-party compatibility Spectre-format: changed several CDF parameter definitions. Now it is compatible with third-party's spectre-format Hspice-format: no change

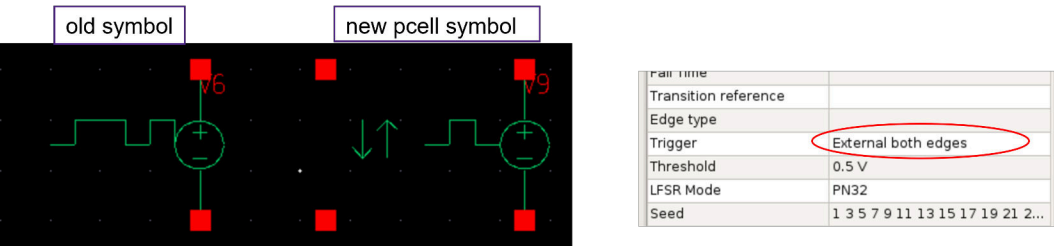
Note:

Preference values 1-3 are redefined or enabled in V-2023.12. The default is 3.

dbAnalogLibVprbsVersion

Controls vprbs versions.

First Build to Enable the Enhancements	Description
Before U-2023.03-SP1	Legacy vprbs Symbol: is not a PCell When the trigger mode is external, because there is no pin to connect the external trigger signal, the netlist is incorrect
U-2023.03-SP1	Change the symbol to PCell When the trigger mode is external, adds pins to connect the external signals No impact on legacy designs if the trigger mode is internal



Note:
Preference values 1-2 are enabled in U-2023.03-SP1. The default value is 2.

dbAnalogLibVsinVersion

Controls vsin versions.

First Build to Enable the Enhancements	Description
Before S-2021.09-SP2-1	Legacy vsin CDF parameters: va default is 0, vo default is 0
S-2021.09-SP2-1	Remove default value of vo Spectre-format: netlist is changed, simulation result is not changed Hspice-format: netlist is not changed Inherited from last version: va default is 0
U-2023.03	Remove default value of va Spectre-format: netlist and simulation result are changed Hspice-format: netlist is not changed Inherited from last version: vo has no default value

Note:

Preference values 1-3 are redefined or enabled in U-2023.03-SP2. The default value is 3.

dbAnalogLibVsourceVersion

Controls vsource versions.

First Build to Enable the Enhancements	Description
Before U-2023.03	Legacy vsource Symbol: is not a PCell CDF parameters: tvpairs default is 2, va default is 0
U-2023.03	Remove default value of va Spectre-format: netlist and simulation result are changed Hspice-format: netlist is not changed Inherited from last version: tvpairs default is 2, symbol is not a PCell
U-2023.03-SP1	Symbol is changed to PCell (for “type=prbs” & “external trigger modes”) No impact on legacy designs if not in this type and mode Change tvpairs default value to 0 Both of spectre-format and hspice-format: netlist and simulation result are changed Inherited from last version: va has no default value

Note:

Preference values 1-3 are redefined or enabled in U-2023.03-SP2. The default value is 3.

dbAnalogLibWindingVersion

Version control for winding devices.

First Build to Enable the Enhancements	Description
V-2023.12-SP1	Use parameter <code>r</code> and map it as <code>resis</code> in Spectre flow.
V-2023.12-SP2	Use parameter <code>resis</code> and remove the mapping of <code>r</code> to <code>resis</code> .

Note:

Preference values 1-2 are redefined or enabled in V-2023.12-SP2. The default value is 2.

nIHSPICESkipEmptyDC

Controls whether to write dc=0 by default in hspice-format netlist.

First Build to Enable the Enhancements		Enhancements Description
Before V-2023.12		Write dc=0 by default for independent sources.
V-2023.12		Do not write dc=0 by default for independent sources.

Note:

Preference values true and false are redefined or enabled in V-2023.12. The default is true.

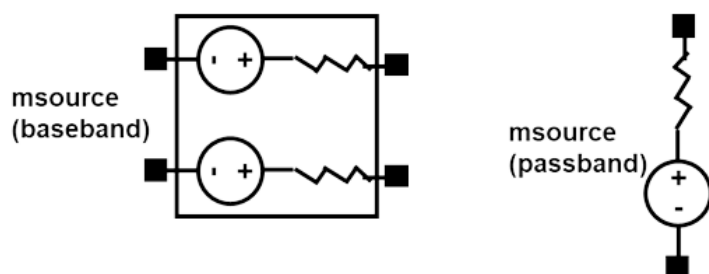
3

rfLib

Support for third-party compatible RF sample library (rfLib) for migration of user's design created by third-party tools. Also, this library can be used by designers to create RF designs.

msource

IQ modulated source with various parameters settings. The msource device is a PCell. Its PCell behavior is controlled by its signal type. When the signal type is `passband`, the symbol is a two-terminal device. When the signal type is `baseband`, the symbol is a four-terminal device.



CDF Parameters

CDF Parameter Prompt	CDF Parameter	Default	hspiceD	spectre	auCdl
Signal type	wtype	passband	X	X	—
Output type	output_type	port	X	—	—
Source type	wsource_type	txInput	—	X	—
Carrier harmonic	wsource_carrier_harm	1	—	X	—

CDF Parameter Prompt	CDF Parameter	Default	hspiceD	spectre	auCdl
LO source type	local_osc_type	I&Q	–	X	–
LO/I source name	local_osc_name_i		–	X	–
LO/Q source name	local_osc_name_q		–	X	–
LO source name	local_osc_name		–	X	–
Average Power	dbm	6.99	X	X	–
Amplitude of signal	amp		X	–	–
Resistance	r	50	X	X	–
Number of frames	framenum	1	–	X	–
Oversample ratio	oversample	4	–	X	–
SNR (dB)	snr		–	X	–
Channel frequency	MSource_Frequency	5G	X	X	–
Data rate (bps)	MSource_DataRate	48M	X	X	–
Modulation type	MSource_ModType	QAM16	–	X	–
Modulation type	mod	QAM16	X	–	–
Modulation order	order		X	–	–
Number of symbols	MSource_SymbolNum	512	–	X	–
Symbols per Impulse	filter_symbols	20	X	–	–
artParameterInToolDisplay('filter_symbols')	filter_samples	4	X	–	–
PN code	MSource_DataType	PN15	X	X	–
OFDM type	MSource_OFDMType	No	X	X	–

CDF Parameter Prompt	CDF Parameter	Default	hspiceD	spectre	auCdl
Pulse shaping filter type	MSource_FilterType	None	–	X	–
Roll-off factor	MSource_Alpha	0.2	–	X	–
Pulse shaping filter type	filter	None	X	–	–
Roll-off factor	filter_coef	0.2	X	–	–
IFFT/FFT size	MSource_FFT Length	64	X	X	–
Number of Data carriers	MSource_Data Carriers	52	–	X	–
Cyclic prefix	MSource_CP	1/4	–	X	–
Guard band	ofdm_gband	6.5	X	–	–
Prefix length	ofdm_prefix	16	X	–	–
Number of DC carriers	MSource_DC Num	1	–	X	–
Number of carriers in CA	MSource_CA Num	1	–	X	–

Examples

Passband netlist in spectre-format:

```
W0_pass (net1 net2) wsource wsource_type=txInput wsource_carrier_harm=1 \
    standard="MSource" dbm=6.99 r=50 framenum=10 oversample=4
    snr=3 \
    MSource_Frequency=5G MSource_DataRate=48M MSource_ModType=QAM16 \
    MSource_SymbolNum=512 MSource_DataType=PN15 \
    MSource_FilterType=None MSource_OFDMType=No
```

Baseband netlist in spectre-format:

```
W1_base (net4 net3 net6 net5) wsource wsource_type=txInput \
    wsource_carrier_harm=1 local_osc_name_i=test_I \
    local_osc_name_q=test_Q standard="MSource" dbm=6.99 r=50 \
    framenum=10 oversample=4 snr=3 MSource_Frequency=5G \
    MSource_DataRate=48M MSource_ModType=QAM16 MSource_SymbolNum=512
    \
    MSource_DataType=PN15 MSource_FilterType=None MSource_OFDMType=No
```


Passband netlist in hspice-format:

```
PW_pass_ofdm net3 0 iqsrc ofdm_fft=64 ofdm_prefix=16 ofdm_gband=6,5
freq=5e+09 mod=QAM16
+      bit_rate=4.8e+07 filter_symbols=20 data=WPN15 axis=iq
filter_samples=4 z0=50
+      dbm=6.99
```

Baseband netlist in hspice-format:

```
PW5 net24 net23 iqsrc ofdm_fft=64 ofdm_prefix=16 ofdm_gband=6,5 mod=QAM16
+      bit_rate=4.8e+07 filter_symbols=20 data=WPN15 axis=i
filter_samples=4 z0=50 dbm=6.99
PW5@1 net26 net25 iqsrc ofdm_fft=64 ofdm_prefix=16 ofdm_gband=6,5
mod=QAM16
+      bit_rate=4.8e+07 filter_symbols=20 data=WPN15 axis=q
filter_samples=4 z0=50 dbm=6.99
```